Who Gains from Maternal Employment? Self-Employment, Childcare, and Child Development*

Hyun Soo Suh†

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Abstract

Whether a mother's employment helps or harms her child depends on how well formal childcare substitutes for her time. I estimate a household life-cycle model of work arrangements, time allocation, and child skill formation on UK panel data, using the paid-versus-self-employment margin to separate flexibility from earnings level and volatility. With a young child, the wife sorts into a more flexible work arrangement while the husband's remains unchanged. The mother specializes in child-rearing because her comparative advantage over the father comes from productivity in child skills, not in the labor market. Formal care substitutes well for a non-college mother's time but poorly for a college mother's, whose time is the child's most productive input. Full-time paid employment therefore lowers skills by 0.14 to 0.20 SD for children of college mothers and has little effect otherwise. Flexible hours without a pay penalty raise welfare and maternal employment, by about 4 percentage points, but leave child skills unchanged. Childcare raises skills mainly for children of non-college mothers, so tagging it by education targets those who gain the most. Pairing the two policies benefits both groups and raises the projected lifetime earnings of children of non-college mothers by about 2.5%.

Keywords: Child development, maternal employment, work flexibility, childcare

JEL Classification: D13, J13, J16, J22, J24

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†Washington University in St. Louis, Department of Economics, hyunsoo@wustl.edu. www.hyunsoosuh.com

1 Introduction

Whether a mother's employment helps or harms her child is debated (Hsin and Felfe, 2014). Some studies find gains, especially for disadvantaged children, while others find losses.[1] The net effect matters because early skills are persistent and yield high returns.[2] The disagreement stems from a trade-off between the two primary inputs into child development, parental time and money, and from how a couple divides the labor. Each parent's work arrangement – whether and how much they work – drives this division and how much of each the child receives. Its effect depends on how spouses allocate time and how well formal childcare substitutes for a parent's time. This allocation embeds comparative advantage, each parent's relative productivity in the labor market and in producing child skills. The choice is particularly salient for the mother, whose time may vary in productivity with her education.

This heterogeneity drives the payoff to family policy. Governments across developed countries increasingly subsidize early childcare and mandate flexible work to ease the work-family trade-off. Because early skills are among the strongest predictors of adult earnings and mobility (Heckman, Stixrud, and Urzua, 2006), which children gain and lose determines how far family spending narrows the gap in skills and earnings children inherit, and at what cost. This bears on policy design: Untargeted childcare expansion subsidizes children who would have formed the same skills at home, and can even lower them. Which policy helps which children turns on this substitutability differential. Yet prior work has not shown how flexibility and childcare each affect children, nor has it isolated flexibility from pay.

In this paper, I develop a framework in which each parent's work arrangement shapes the household's division of labor, which determines the quantity and composition of the inputs into child development. I allow these inputs to differ in skill productivity across each parent's time and formal care. Three forces underlie this division: selection into work arrangements, productivity differences between spouses in the market and at home, and the nonlinear accumulation of human capital over the life cycle. By embedding both the labor-market determinants of earnings (Goldin, 2014; Bertrand, Goldin, and Katz, 2010) and the child-skill returns to parental time and care (Del Boca, Flinn, and Wiswall, 2014; Caucutt et al., 2026), I add a child-production margin to the household's specialization.

Within this framework, I study flexibility which is valued highly by mothers with young children (Mas and Pallais, 2020), through the margin between paid employment and self-employment. Rather than treating self-employment as flexibility per se, I use the two arrangements to decompose a work arrangement into three attributes a household faces. The first, flexibility, defined by the hours menu and the earnings curvature, governs how parents allocate time between market work, home production, and leisure. The other two, the earnings level and its volatility, govern labor-force entry and investment in the child through formal childcare. While paid employment (with its

[1] For gains, see, for example, Vandell and Ramanan, 1992; Dunifon et al., 2013. For losses, see, for example, Bernal, 2008; Bernal and Keane, 2011; Ruhm, 2004.

[2] See, for example, Cunha and Heckman, 2007; Attanasio et al., 2020; Attanasio, 2026.

rigid schedules) offers little control over hours, self-employment offers a wide, nearly continuous menu but at lower and more volatile pay.[3] The decomposition lets me pose a counterfactual the self-employment margin alone cannot: what would occur if flexible hours were extended to paid employees without imposing the pay penalty.

I estimate a dynamic household labor supply model with child skill formation, in which forward-looking spouses make their work arrangement, saving, and child-investment decisions over the life cycle. The model nests two trade-offs. The first weighs parental-child time against household income. Market hours buy income and formal childcare at the expense of the parent's own time, which care substitutes for imperfectly. The second involves parental labor supply choice. Employment raises household resources and builds parental human capital but costs home production time and leisure, and the arrangement supplying the hours differs in flexibility, pay, and volatility, which households smooth through savings. Moreover, human capital depreciates during employment gaps and early child skills are self-productive (Cunha, Heckman, and Schennach, 2010). Thus, early investment choices carry lasting returns not only for the child but also the mother's career.

The model captures each trade-off through distinct features. The child skill technology governs the first. Cognitive and non-cognitive skills are persistent and accumulate from parental time and formal childcare through a separate technology for each domain. In addition, a supervision constraint requires the two main inputs to jointly cover a young child's needs, tying formal childcare to the hours the arrangement supplies. The work arrangement governs the second. Each period, spouses choose an arrangement and, conditional on it, market hours, home production, consumption, and savings. When there is a young child, they additionally choose a formal childcare arrangement at part-time or full-time intensity. Hours options within each arrangement come from weighted k-means clustering, which captures the differences in flexibility across arrangements. The price of that flexibility varies by gender through the hours menu and the hours-earnings curvature, which together set how much pay a parent forgoes to control their hours. The variance of earnings prices each arrangement's volatility (the income risk the household bears), while the tax-and-transfer system sets the net return each household faces.

I use UK panel data, primarily the British Household Panel Survey (BHPS) and the Millennium Cohort Study (MCS). The BHPS provides annual waves on a nationally representative sample of UK households from 1991 to 2008, covering labor supply, earnings, and household wealth; I restrict the sample to married and cohabiting couples with biological children and working husbands. The MCS follows about 19,000 children born in 2000-2002 with age-appropriate cognitive and non-cognitive assessments. I focus on birth to age 7, spanning the preschool investment period and the early school years in which formal childcare remains a parental choice. I supplement the analysis with the UK Time Use Survey, Labour Force Survey, Family Resources Survey, and

[3] Flexibility here is the within-week hours margin, not weeks worked per year. In the BHPS, paid and self-employed parents work a similar number of weeks (49.5 and 50.8 on average, with a median of 52 for both), and the self-employed are if anything less likely to work part-year (6 versus 10%). The arrangement difference operates through weekly hours, not intermittency.

O*Net to document time use, earnings, and job characteristics by work arrangement. The UK setting is well suited to this study for two main reasons. First, health insurance is universal and not tied to employment. Insurance access therefore does not distort the choice between paid and self-employment, unlike in the United States, where job-based coverage can deter it (Fairlie, Kapur, and Gates, 2011). Second, I simulate the tax-and-transfer system using the microsimulation library FORTAX (Shephard, 2009). It allows me to compute household net income under policies such as working tax credits, child tax credits, and childcare subsidies, which shift work arrangements and childcare use.

I first document how spouses' labor supply and time allocations change at childbirth. Wives move toward a more flexible arrangement conditional on working, despite lower average earnings, while husbands' arrangements remain unchanged. Within a given work arrangement, wives with flexibility reallocate more toward home production than those facing rigid schedules, while husbands use it to work longer market hours. Childcare use differs by maternal education. Because these choices shape the inputs into child development, they translate into distinct trajectories in children's cognitive and non-cognitive skills. Identifying the effect, however, poses several challenges. Parents sort into work arrangements on unobserved preferences that also shape their time and money investments, which are themselves simultaneously determined. They also decide as a household, so one spouse's choice affects the other through shared constraints. These challenges compound over the life cycle, shaping future earnings and the household's capacity to invest.

Based on the empirical patterns, I estimate the model in two steps. First, I estimate the child production function correcting for two sources of endogeneity. Measurement error in the skill assessments biases the estimated technology, so I recover the latent skills using lagged cross-domain instruments following the approach of Agostinelli and Wiswall (2025). Parental inputs are chosen jointly with unobserved determinants of child skills, so I correct their endogeneity with a control function approach. The tax-and-transfer system shifts the price of childcare and the returns to each arrangement, separately identifying the productivity of parental time and formal childcare by skill domain and parental education. Second, I recover the remaining structural parameters, governing household preferences, parental income processes, and human capital dynamics, by simulated method of moments (McFadden, 1989). I identify them from how labor supply, time use, and childcare respond to the budget and to the arrival of a young child, and from how wages and childcare use differ by maternal education.

The structural estimates reveal three mechanisms: (i) how households value flexibility against pay, (ii) the productivity of parental inputs, and (iii) the constraints that bind them. First, both spouses value the flexibility of self-employment, revealed by its lower disutility of work than paid employment. They use it in opposite directions: the wife toward home production and shorter hours, the husband toward longer market hours. Second, parental inputs differ in productivity. Maternal time is the most productive, especially for cognitive skills and college mothers, while fathers' home time is much less productive and raises both skill domains only through skill-biased parenting. Formal childcare sits between college and non-college maternal time, which generates

the education gradient. Formal care is an inferior substitute for a college mother's time but a superior one for a non-college mother's. Third, a supervision constraint requires parental time and formal care to jointly cover a young child. Because the father barely adjusts, the mother is the margin of adjustment.

The model prices the household's division of labor. The mother specializes in child-rearing because her comparative advantage over the father is in producing child skills, not in the labor market. A weekly hour with the child forgoes £1,300 to £2,200 a year for the father against £400 to £700 for the mother. I quantify what an equal division would cost the child and find a loss of 0.1 to 0.6 SD of skills, or 2 to 9% of lifetime consumption, which is largest when the mother is college educated and the father is not. The same mechanism shapes maternal employment: full-time work with childcare replaces the mother's time, and the resulting child-skill loss is consistent with Bernal (2008). However, the pooled effect masks heterogeneity by maternal education. Full-time paid employment entry lowers a child's cognitive and non-cognitive skills by 0.20 and 0.14 SD for college-educated mothers but has essentially no effect for non-college-educated mothers. I then ask what flexibility, offering paid employees flexible hours without lower, more volatile pay, does on its own. Since the mother is now able to choose intermediate hours, she is drawn into work below full-time and uses less formal care. Yet she barely cuts her time with the child. The mothers drawn in by flexibility work part-time rather than full-time, and incumbents at full-time do not cut their hours. Among those who do work more, mothers protect their time with the child and substitute leisure for work.

This education gradient motivates the design of family policies. Family-friendly flexibility, which offers reduced and intermediate hours, prevents the skill loss for children of college-educated mothers while raising employment. Education-targeted free childcare provides formal care to children of non-college mothers, from which they gain skills. Because college children gain little from care, concentrating it on non-college children narrows the skill gap across maternal education at lower fiscal cost than universal provision. Education makes a better tag than income (Akerlof, 1978) on efficiency grounds because it is fixed, tracks the productivity gap, and, unlike an income test, invites neither a participation tax nor gaming. Flexibility is thus an employment lever, while childcare policy directly affects child skills.

Pairing the two policies makes both groups better off. The welfare gains are larger from flexibility than from childcare because flexibility raises parental welfare through control over hours while childcare works through child skills. For college households, almost all the gain runs through the parental channel. Family-friendly flexibility is worth about £5,400 a year, with little effect on child skills. For non-college households, flexibility alone raises parental welfare but slightly lowers child skills, so pairing it with targeted childcare delivers both: about £3,800 in parental welfare, mainly from flexibility, and the child-skill gains that flexibility alone does not deliver. The returns also have intergenerational effects. The combined package raises the expected lifetime earnings of children of non-college mothers by about 2.5%, almost entirely from childcare, while family-friendly flexibility raises that of children of college-educated mothers by about 0.4%.

This paper contributes to three strands of the literature. The first studies parental investment and child development. Early skills predict long-run outcomes (Heckman, Stixrud, and Urzua, 2006; Chetty et al., 2011a), and they are formed through parental time and money in a dynamic, self-productive technology (Cunha and Heckman, 2007; Todd and Wolpin, 2007; Cunha, Heckman, and Schennach, 2010; Del Boca, Flinn, and Wiswall, 2014; Attanasio et al., 2020; Agostinelli and Sorrenti, 2021). Building on this, a policy literature evaluates the levers that shape these investments and, through them, child skills: transfer programs (Mullins, 2025), parenting and early childhood interventions (García et al., 2020), childcare subsidies (Baker, Gruber, and Milligan, 2008; Havnes and Mogstad, 2015; Kottelenberg and Lehrer, 2017; Cornelissen et al., 2018; Felfe and Lalive, 2018; Attanasio et al., 2022; Moschini, 2023), childcare markets (Berlinski et al., 2024), and childcare regulations (Garcia-Vazquez, 2023). I add a policy lever this literature has overlooked – work flexibility – by treating the work arrangement as a determinant of both the quantity and composition of the inputs the child receives.

The second strand studies how maternal employment shapes child development. Early maternal employment, especially in the first few years, is often found to lower children's cognitive skills (Waldfogel, Han, and Brooks-Gunn, 2002; Baum, 2003; Ruhm, 2004; James-Burdumy, 2005; Brooks-Gunn, Han, and Waldfogel, 2010), a finding reinforced by structural estimates that treat employment and childcare as endogenous (Bernal, 2008; Bernal and Keane, 2011). Some papers report null or positive effects (Vandell and Ramanan, 1992; Dunifon et al., 2013; Suh, 2026). The sign hinges on what replaces the mother's time: Universal childcare has harmed children in some settings (Baker, Gruber, and Milligan, 2008, 2019; Fort, Ichino, and Zanella, 2020) yet helped disadvantaged children in others (Cornelissen et al., 2018; Felfe and Lalive, 2018; Sylva et al., 2004). The closest work is Verriest (2024), which places household labor supply and childcare policy in a child development model, and Brilli (2022), which estimates the education gradient in maternal and non-parental time. I add the margin between paid employment and self-employment, its decomposition into flexibility, earnings level, and earnings volatility, and the joint choices of both spouses.

I also connect this evidence to a literature on schedule flexibility and maternal labor supply that has traced flexibility only to labor-market outcomes. Most work focuses on paid employment, distinguishing *greedy jobs* that reward long hours (Goldin, 2014, 2021) from family-friendly professions (Goldin and Katz, 2016). In the UK, flexible arrangements strengthen mothers' attachment (Chung and Van der Horst, 2018), with gains that are not gender-neutral (Arntz, Yahmed, and Berlingieri, 2022). This connects to the gender pay gap, largely within occupations (Blau and Kahn, 2017). My paper measures selection into work arrangement and measures how workplace flexibility shapes child development, in addition to maternal labor supply.

The third strand relates to self-employment and household labor supply. The choice of self-employment is often gendered: Men value autonomy despite lower earnings (Hamilton, 2000; Taylor, 1996), while women enter self-employment for work-family balance and flexible schedules (Boden Jr, 1996, 1999; Marlow, 1997; Lombard, 2001; Mattis, 2004; Georgellis and Wall, 2005; Budig, 2006; Wellington, 2006; Joona, 2017; Lim, 2019; Berniell et al., 2025). That work-family motivation

is the flexibility my model prices. At the household level, spouses' self-employment decisions are interdependent: The spouses coordinate work schedules (Devine, 2001; Bryan and Sevilla, 2017), and one spouse's resources and experience shape the other's entry (Bruce, 1999; Parker, 2008). Occupational flexibility also shapes how couples adjust at childbirth (Bang, 2022). This literature traces self-employment to gendered labor-market outcomes surrounding motherhood: wages (Harrison, Kray, and Sethi, 2026), firm performance (Bonney, Pistaferri, and Voena, 2025), and entrepreneurship rates (Ferrando et al., 2025; Rutigliano, 2025). The closest antecedent is Lim (2017), who incorporates self-employment into a life-cycle model of married women's fertility and employment on the NLSY79 and values its flexible amenities as a single package. I instead separate the work arrangement into its three attributes and embed self-employment in a joint model of both spouses, leveraging the work-family flexibility that draws mothers to it into a measurable input into their children's skills, not the mother's labor supply alone.

The remainder of the paper is organized as follows. Section 2 describes the data and documents the empirical patterns motivating the structural model. Section 3 presents the household life-cycle model, and Section 4 outlines the identification and estimation strategy. Section 5 shows the estimation results, and Section 6 covers comparative statics and policy experiments. Section 7 concludes.

2 Parental Work Arrangement and Child Investment

Each work arrangement trades off time and money, shaping how parents invest in their children. Paid employment offers a rigid menu of part-time and full-time schedules, self-employment lets hours adjust nearly continuously, and not working frees all time for home production at the cost of earnings. Self-employment's schedule control, though, comes with lower and more volatile pay. A work arrangement therefore bundles three attributes: flexibility (control over when and how many hours to work), earnings level, and earnings volatility. I document three patterns that motivate the model: at childbirth the mother re-sorts across work arrangements, the arrangements trade flexibility against the level and the risk of pay, and these choices shift the time and childcare inputs the child receives.

2.1 Data

British Household Panel Survey The BHPS (University of Essex, 2018) is an annual longitudinal household panel survey that began in 1991, sampling approximately 5,500 households in Great Britain. The survey ran until 2008, providing 18 waves of detailed information on family structure, labor market outcomes, and income dynamics. The survey also contains rich information on household finances and assets, including data on home ownership, house values, and mortgage status and values. More detailed asset information, including financial assets, savings, and debts, is available every five years to study longer-term wealth.

For the main analysis, I focus on married and cohabiting couples[4] with no changes in family structure, as well as families with biological children.[5] I condition on husbands always working,[6] restrict the sample to women aged 21–51 (covering early to mid-career stages), and drop couples missing key demographic and labor market information. The final sample consists of 49,958 observations of 9,619 individuals and 4,871 households. Wages are computed as hourly earnings by dividing weekly earnings by weekly hours in the main job; self-employment income uses reported gross monthly pay and profit converted to a weekly measure.[7] Table A.4 shows the summary statistics, and variable construction is detailed in Appendix A.1.

Millennium Cohort Study For child development outcomes, I use the MCS (Centre for Longitudinal Studies, 2023), a nationally representative longitudinal study of approximately 19,000 children born in the UK between September 2000 and January 2002. I focus on early childhood – from birth to age 7 – when skill investments are most productive and their effects most persistent (Cunha and Heckman, 2007; Heckman, Stixrud, and Urzua, 2006). Cognitive skills are measured by the Denver Development Screening Test at 9 months and the British Ability Scales (BAS) at ages 3, 5, and 7 (Frankenburg and Dodds, 1967; Elliott, Smith, and McCulloch, 1997); non-cognitive skills by the Carey Infant Temperament Scale (Carey and McDevitt, 1978) at nine months and the Strengths and Difficulties Questionnaire (SDQ) Total Difficulties Score (Goodman, 1997) at ages 3, 5, and 7. This age window also covers the period in which parents are the primary decision-makers over formal childcare arrangements, before school entry substantially reshapes the childcare environment.[8] I apply the same restrictions as the BHPS and further restrict to singleton births and mothers aged 21–45 at birth.[9] The final sample consists of 5,219 children and 28,802 parent-wave observations. Further details are in Appendix A.2.

Table A.6 shows that the dominant pattern in both the MCS and BHPS is a large reallocation of wives across work arrangements as children age, with no comparable shift for husbands. In both

[4]I pool married and cohabiting couples because the UK tax-and-transfer system treats them identically over the sample period. Income tax is assessed individually, and tax credits and means-tested benefits are assessed on joint couple income under the same rules whether the couples are married or cohabiting.

[5]Since changes in family composition can affect couples' behavior and child outcomes, I exclude divorced households, which account for around 8% of the sample. This restriction aligns the empirical analysis with the model environment and avoids confounding effects arising from step-parenting, non-resident parents, or complex household arrangements (Case, Lin, and McLanahan, 2000; Ginther and Pollak, 2004; Francesconi, Jenkins, and Siedler, 2010).

[6]The key margin in the data and model is the work arrangement, particularly the wife's choice across paid employment (part-time and full-time), self-employment, and non-employment. Conditioning on the husband working (7% loss) focuses the analysis on this margin and is consistent with the evidence that the husband's labor supply is largely invariant to childbirth (Kleven, Landais, and Søgaard, 2019).

[7]Self-employment income is bottom-coded at zero in the BHPS, so self-employed individuals reporting losses are recorded as having zero income. The left tail of self-employment income is therefore truncated, and the estimated self-employment earnings dispersion understates true downside unexplained volatility. I verify in Section 6 that the flexibility results are insensitive to this measurement by re-solving the model over a range of self-employment dispersions.

[8]Over 98% of MCS children enter school by age five. By focusing on ages up to seven, I capture both the pre-school investment period and the early school years in which formal childcare remains a meaningful parental choice, while avoiding confounding from school-quality differences.

[9]I keep biological children only and track households with no change in family composition since separation may affect child development outcomes (Amato and Keith, 1991; Cherlin, Kiernan, and Chase-Lansdale, 1995; Kim, 2011). Twins account for 1.3% of the sample; I exclude them for consistency with the singleton model assumption.

datasets, the non-working share among wives falls from close to half at nine months to roughly one in four by age seven, with most of the reallocation flowing into part-time paid employment and, to a lesser extent, self-employment, and the concentration in part-time easing as children age. Husbands, by contrast, remain predominantly in paid employment throughout. This asymmetry between wife and husband work-arrangement dynamics motivates the model's focus on work-arrangement choice as the key margin of the wife's adjustment.

Parental time inputs entering the child skill production function are not consistently observed in the MCS.[10] I therefore construct them by statistically matching BHPS time-use data to MCS households based on parent's gender, work arrangement, and education, as well as household characteristics such as the number of children and child age.[11] The identifying assumption is conditional exchangeability, such that within these matched cells, BHPS time allocations are representative of what MCS households would report.[12]

I also use the UK Time Use Survey (Office for National Statistics, 2003) to validate the imputed time inputs (Appendix C) and document time-use patterns by work arrangement (Appendix A.4), and I supplement the analysis with the Labour Force Survey (Office for National Statistics, 2022b), Family Resources Survey (Department for Work and Pensions, 2005), and O*Net (O*NET Resource Center, 2019) (Appendix A.5).

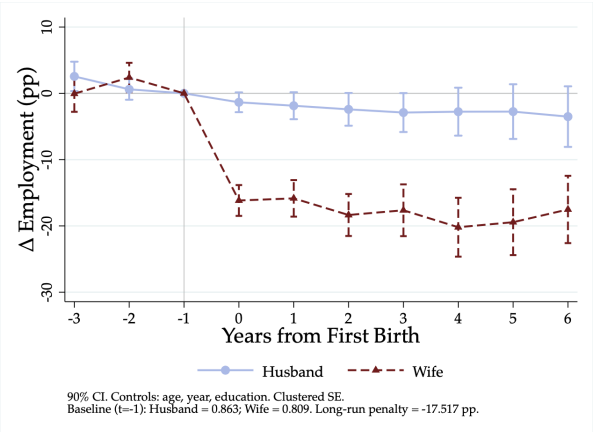
2.2 Work Arrangement at Childbirth

Figure 1 plots the event-study analysis of work arrangements around the first birth. Controlling for individual, age, and year fixed effects, the estimates compare each parent to the path they would have followed without the birth (Appendix B.1). Panel (a) shows that women's employment falls sharply at first birth, by about 17 percentage points, and remains 15–20 percentage points below the pre-birth trend through the sixth year, consistent with the findings of Kleven, Landais, and Søgaard (2019). Work arrangements shift too, but only for mothers. Among mothers who remain employed, full-time paid employment contracts markedly and part-time paid employment expands. Self-employment also rises in the years following birth. This within-employment shift

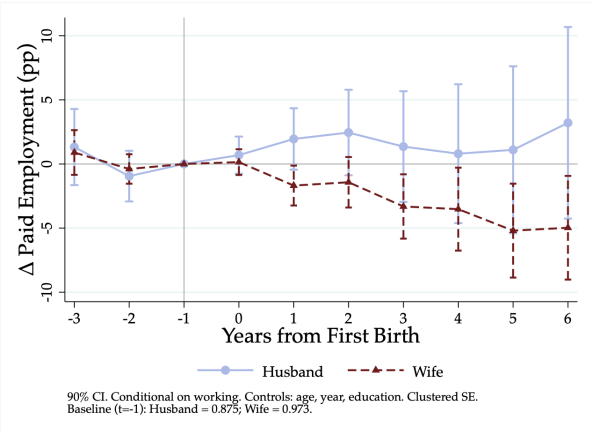
[10] The MCS records parental investment by frequency (for example, how many times per week parents read to the child), which does not capture intensity. Self-employed mothers report higher frequencies than not working mothers. The MCS includes a time-with-child measure in some waves, but as qualitative bands (nowhere near, not quite, just enough, plenty, more) rather than in hours; I show that the measure is highly correlated with the BHPS time-use data (Appendix C).

[11] The match uses five cell categories (parent gender, work arrangement, education, number of children, and child age group), which gives imputed maternal and paternal time inputs of 72 and 53 distinct cell values in the estimation sample, so the identifying variation is between-cell by construction.

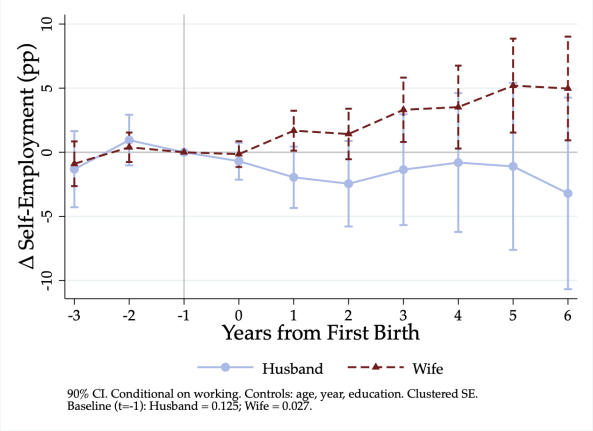
[12] Differences in the marginal distributions across datasets do not threaten validity because matching is done within each matched cell, so what matters is within-cell comparability, not equal marginal shares. The BHPS confirms that conditional on the five matching variables, time allocation varies little with income. The matched values are also consistent with UK Time Use Survey diary estimates within matched cells (Appendix C). Because the procedure assigns conditional means rather than reconstructing the joint distribution of time inputs and child outcomes, it does not require the conditional independence assumption of classical statistical matching (Ridder and Moffitt, 2007). Because the time input is defined at the arrangement level, the coefficient identifies the effect of moving between arrangements, and the instruments, which shift a household's arrangement, need only be excluded across arrangements rather than within them (Corollary 2, Appendix E).



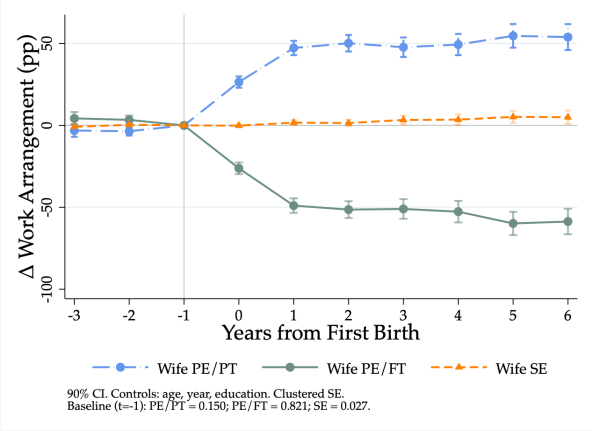
(a) Employment



(b) Paid Employment (Conditional)



(c) Self-Employment (Conditional)



(d) Wife's Work Arrangements (Conditional)

Note: The event is the birth of the first child. The y-axis shows percentage point changes relative to $t = -1$. Panels (b)–(d) are conditional on being employed. Controls include age, year, education, and marital status; standard errors are clustered at the individual level. Confidence intervals are at 90% level.
Source: British Household Panel Study.

Figure 1: Event Study: Parental Work Arrangements

toward self-employment is consistent with the idea that mothers seek work arrangements that offer greater scheduling control (Harkness, Borkowska, and Pelikh, 2019).[13] The rise concentrates in unincorporated self-employment – sole traders and freelancers – rather than incorporated firms.[14] Self-employed women also sort into services and personal care rather than the schedule-rigid administrative roles that dominate paid employment (Appendix B.4). Men, by contrast, show no comparable adjustment in employment or work arrangement across all panels.

These shifts in work arrangement are accompanied by changes in hours worked, income, and home production time (Appendix B.2). Wives reduce market hours and increase home production, which comes with a fall in weekly income from paid employment, and self-employed wives in particular cut hours more steeply but see little change in hourly wages. Husbands' hours, income, and home production again adjust little. Household assets also smooth consumption, which matters given the income-volatility differences across work arrangements that the next subsection documents (Crossley and O'Dea, 2010; Adda, Dustmann, and Stevens, 2017).[15]

2.3 Flexibility, Earnings, and Childcare

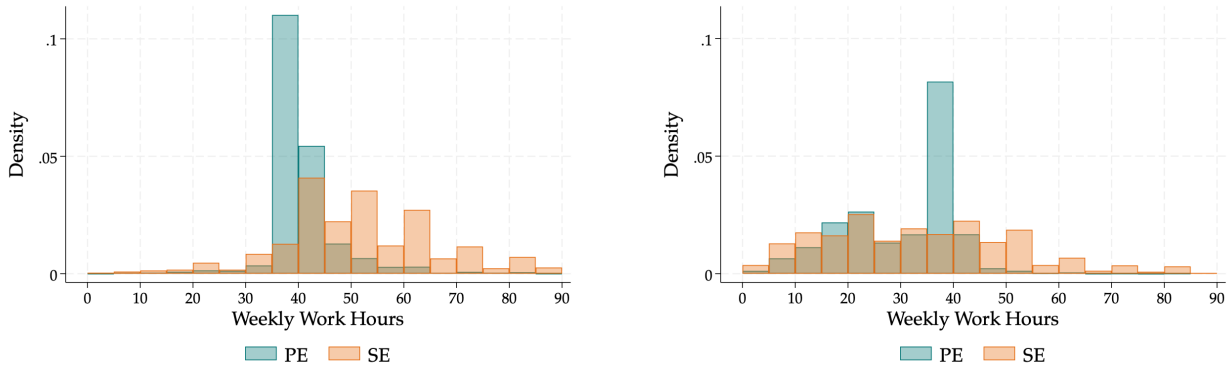
If parents sort into different work arrangements, how do their labor supply choices affect the household's time-money trade-off, especially when children are young? Beginning with time, Figure 2 shows that hours worked differ across arrangements. Paid employed workers cluster tightly and bunch around the standard full-time hours, with husbands concentrated near 40 hours and wives showing a bimodal distribution at part-time and full-time. Self-employed workers, by contrast, spread broadly across the hours range, with substantial masses at both shorter and longer hours. This dispersion reflects the absence of the rigid wage-hours package that characterizes paid employment (Rosen, 1976; Altonji and Paxson, 1988): Self-employed workers set their own schedules and adjust hours in response to changing household needs. They also adjust hours more across years, with higher within-person standard deviation (Appendix B.2).

How parents use this flexibility differs by gender. Figure 3 shows mean hours worked and home production by work arrangement and the presence of a young child (age five or below). Self-employed husbands work about 10 hours more per week but spend similar time in home production compared to paid employed husbands, and both adjust little with a young child. Self-employed wives with a young child work 8.3 fewer market hours than those without, while for full-time paid employed wives the difference is only 0.2 hours (Gimenez-Nadal, Molina, and Ortega, 2012). Among wives with a young child, self-employed wives average 18.6 hours of home production per week compared to 12.6 hours for full-time paid employed wives, a 48% difference

[13] Using sequence analysis on UK Understanding Society data, Harkness, Borkowska, and Pelikh (2019) document the same divergence. In the three years after birth, mothers disperse across eight distinct employment clusters, with 7% sorting into self-employment, while fathers vary little, with 62% remaining consistently in full-time employment.

[14] Rutigliano (2025) and Bonney, Pistaferri, and Voena (2025) use administrative firm records to show large post-birth declines in the performance of female-owned incorporated businesses. Figure B.8 in Appendix B documents the unincorporated concentration in my sample.

[15] The full asset event-study plots are in Appendix B.5.

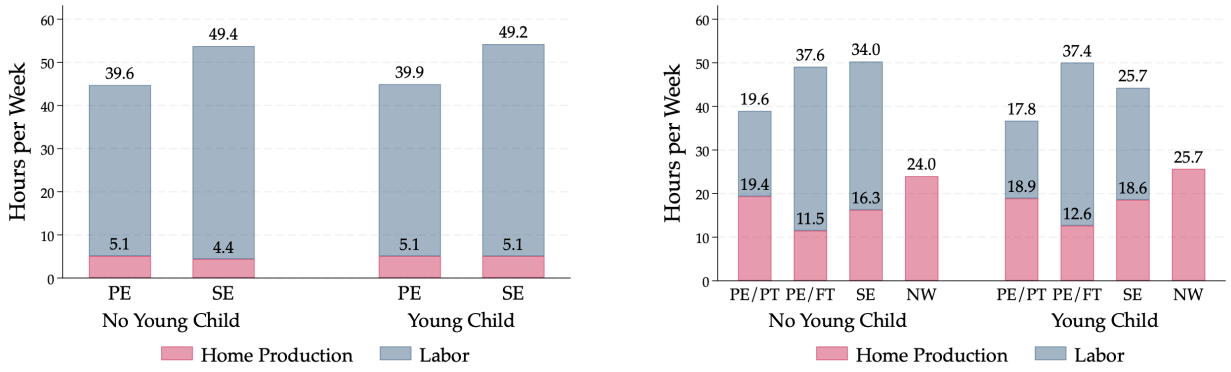


(a) Husband (b) Wife

Note: Bars report the share of person-year observations in five-hour bins, conditional on working. PE = paid employment; SE = self-employment.
Source: British Household Panel Study.

Figure 2: Distribution of Weekly Work Hours by Work Arrangement

in weekly home production time, including childcare time (Appendix C).[16] Self-employed workers, especially those with young children, also work from home more often, take more breaks during the day, and keep more irregular hours (Appendix B.3).



(a) Husband (b) Wife

Note: Bars show mean hours per week in market work and home production by work arrangement and young-child (age five or below) status. PE = paid employment; SE = self-employment.
Source: British Household Panel Study.

Figure 3: Time Allocation by Parental Work Arrangement

Turning to money, flexibility comes at the cost of lower and more volatile income. Self-employed workers have a wider and flatter weekly earnings distribution than paid employees, with thicker tails on both sides (Appendix B.2 Figure B.2). Table 1 shows the mean and standard deviation of

[16]These home production levels are consistent with diary-based UK estimates. Gimenez-Nadal, Molina, and Zhu (2014) report that fathers and mothers devote 0.91 and 1.56 hours per day to housework (about 6 and 11 hours per week), close to the figures here. The UK Time Use Survey confirms the levels (Appendix A.4), though it covers too few self-employed parents to validate the cross-arrangement gap directly. Decompositions by self-employment incorporation status are in Appendix B.

weekly earnings, hours, and hourly wages by work arrangement and presence of a young child. Self-employed parents earn less per week than paid employees, and self-employed husbands also earn less per hour, though self-employed wives' mean hourly income is higher, reflecting the wide, right-skewed self-employment earnings distribution. Their earnings standard deviations are substantially larger, confirming that the distributional spread reflects genuine income volatility rather than compositional differences alone. The within-person standard deviation of log weekly earnings is 1.67 for self-employed husbands versus 0.18 for paid employed husbands, and 1.81 versus 0.25 for wives. Households therefore face a flexibility-risk trade-off: Self-employment relaxes hours constraints but exposes them to substantially larger income inequalities across households.

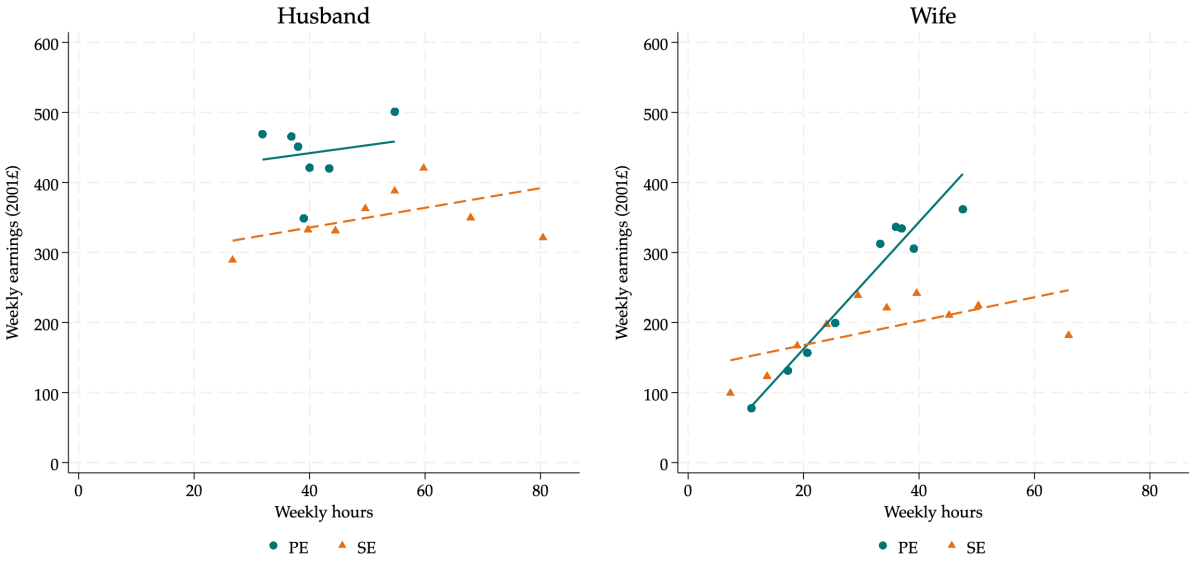
Table 1: Parental Income by Young Child and Work Arrangement

	Weekly Income		Hours Worked		Hourly Income	
Work Arrangement	NChild	YChild	NChild	YChild	NChild	YChild
Husband						
Paid Employed	423.6	479.4	39.5	39.9	10.9	12.2
	(230.0)	(325.1)	(6.6)	(7.0)	(6.1)	(8.5)
Self-Employed	339.6	387.4	48.0	49.2	8.1	9.0
	(391.4)	(390.9)	(15.2)	(14.7)	(13.9)	(14.3)
Wife						
Paid Employed/PT	153.8	148.6	20.7	17.8	7.6	8.2
	(125.6)	(107.1)	(6.6)	(6.6)	(12.5)	(5.8)
Paid Employed/FT	326.7	350.7	37.8	37.4	8.7	9.4
	(166.5)	(159.4)	(3.9)	(3.7)	(4.4)	(4.3)
Self-Employed	236.7	164.7	36.1	25.7	8.0	10.3
	(329.4)	(220.0)	(18.2)	(16.7)	(12.7)	(27.7)

Note: Means and standard deviations (in parentheses) of weekly income (£), weekly hours worked, and hourly wage (£) by parental work arrangement and young-child (age five or below) status. Weekly income for self-employed workers is gross monthly pay and profit converted to a weekly measure; self-employment earnings are bottom-coded at zero in the BHPS, so zero-earnings observations are included and pull the self-employed means down relative to positive-earners only. All amounts in 2001 £.
Source: British Household Panel Study.

The arrangements differ not only in the level of pay but in how it varies with hours (Figure 4). For wives, paid employment rewards long hours steeply while self-employment is flat. A wife who shortens her paid employment hours gives up a more-than-proportional share of earnings, the part-time penalty, while self-employment's flatter schedule carries less penalty. This relates to the long-hours premium that Goldin (2014) identifies as a source of the gender gap. It is present in paid employment and largely absent in self-employment. For husbands, whose hours cluster at full-time, the difference is instead in the level: self-employed men work more hours at lower pay. The fact that they accept lower pay to remain self-employed points to the non-pecuniary value of the arrangement.

In addition to parental time inputs, sending the child to formal childcare is another lever of



Note: Each marker is the mean of weekly earnings and weekly hours within one of ten hours-decile bins, computed separately by work arrangement and gender. PE denotes paid employment and SE self-employment.
Source: British Household Panel Study.

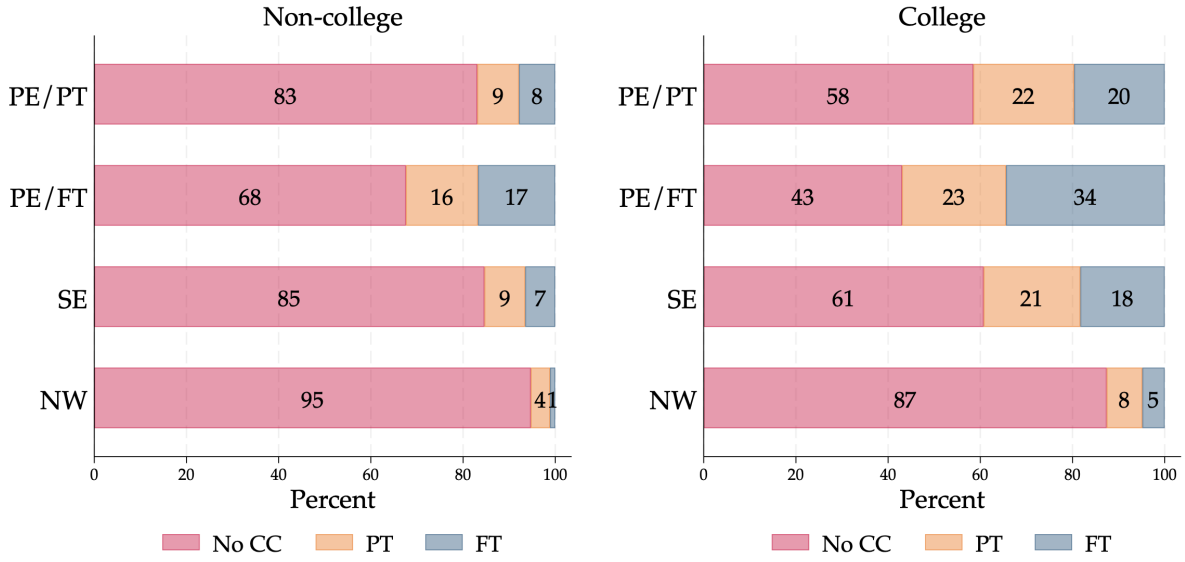
Figure 4: Hours-Earnings Schedule by Work Arrangement and Gender

investment. Figure 5 shows the share of households using no formal childcare, part-time formal childcare (up to 15 hours/week), and full-time formal childcare, by maternal work arrangement and education.[17] Three patterns stand out. First, self-employed mothers substitute parental time for formal childcare more, relative to paid employed mothers: Among non-college households in particular, 85% of self-employed mothers use no formal childcare, compared with 68% of full-time paid employed mothers. Second, the education gradient is sharp for paid employment but muted for self-employment. College full-time paid employed mothers use full-time formal childcare at a rate of 34%, more than double the 17% rate for non-college full-time paid employed mothers, reflecting both higher earnings and stronger labor market attachment. College self-employed mothers use full-time childcare at a rate of only 18%, barely above the non-college rate of 17%, consistent with the idea that self-employment provides scheduling flexibility that substitutes for formal care regardless of education. Third, non-working mothers overwhelmingly rely on parental care across both education groups.

2.4 Implications for Child Development

The time-allocation and childcare patterns above translate into measurable differences in early child development outcomes. In Suh (2026), I estimate a child production function via three-stage

[17] Formal childcare includes childcare centers, childminders, nannies, and breakfast and after-school clubs. I classify breakfast and after-school clubs as formal childcare following Del Bono et al. (2016), as they primarily serve children aged five. I treat formal childcare as homogeneous in quality within the regulated sector; see Appendix D for supporting evidence.



Note: Bars show the share of mothers by childcare type, work arrangement, and maternal education. No CC = no formal childcare; PT = part-time formal childcare (≤15 hours/week); FT = full-time formal childcare (>15 hours/week).
Source: Millennium Cohort Study.

Figure 5: Formal Childcare Arrangements by Maternal Work Arrangement and Education

least squares, treating parental work arrangement and time as jointly determined inputs, using sibling sex composition to instrument for parental time, age-earnings profiles to instrument for income and work arrangement, and lagged work arrangement to identify state dependence in work-arrangement choice. Conditional on parental time and earnings, having a working mother is associated with higher cognitive and non-cognitive skills relative to non-working mothers. The cognitive advantage is concentrated in full-time paid employment and the non-cognitive advantage extends to self-employment. Self-employed fathers also raise non-cognitive skills relative to paid-employed fathers.[18]

Translating these patterns into welfare and policy implications requires a structural model for three reasons. First, the sorting documented in Section 2.2 means parents select work arrangements based on preferences, human capital, and household resources that independently affect child outcomes. Separating the effect of the arrangement from the preferences that drive the choice requires jointly modeling selection and skill production. This is in a similar vein to Bernal (2008), who models maternal employment and childcare together and finds negative effects on child cognitive skills. Second, the input patterns in Section 2.3 show that parental time and formal

[18]The child production function parameters reported here differ from those in Suh (2026) for two reasons. First, Suh (2026) uses demographic instruments (sibling sex composition and age-earnings profiles). The current estimates use FORTAX-simulated instruments that exploit variation in the UK tax-benefit system, which is plausibly orthogonal to such preferences. Second, Suh (2026) corrects for the joint determination of inputs within the estimating system but does not model selection into work arrangements and childcare use; the life-cycle framework here addresses selection by modeling the work arrangement and hours choices that feed into child skill production.

childcare respond together to the same work-arrangement choice, so their quantities are jointly endogenous. Understanding how parental inputs substitute, and why that substitution differs, especially by maternal education, requires a structural model. The model places their productivities within the household's budget and time constraints. Third, the lower and more volatile self-employment earnings and nonlinear returns to hours imply that a policy changing incentives shifts the entire household budget set, as well as the time constraints. Evaluating childcare subsidies or flexibility mandates therefore requires tracing the household's joint re-optimization.

3 Household Life-Cycle Model

I build a household life-cycle model that connects parents' work arrangements to their child's development. A household consists of two adults, a wife ($i = w$) and a husband ($i = h$), and their biological child(ren).[19] Spouses jointly choose a work arrangement and, conditional on it, consumption, time allocations, savings, and formal childcare. The household faces several constraints, so its allocation responds as a whole when incentives change. Let t represent the woman's age, from 21 to 51: at 51, the household faces a terminal value. Each period covers 2 years, matching the biennial child development data.[20]

3.1 Household Choices and Preferences

The household enters each period with a state vector that consists of each spouse's college degree θ^i, the wife's human capital hc_t^w, household assets a_t, the number of children k_t, and the youngest child's age g_t. The number of children and child's age evolve through a weakly endogenous fertility process: At the end of each period the wife faces a birth probability p_t^w that depends on her education, her current number of children, and both spouses' current work arrangements.[21]

[19] I include cohabiting couples whose children are biological. Cohabiting couples constitute roughly 22% of UK couples (Office for National Statistics, 2022a) and a larger share of households with young children. The channels through which cohabitation might differ from marriage in this setting would be changes in family structure. Since I do not model divorce or re-partnering, I focus on families with only biological children. During the couple formation, UK tax-and-transfer treatment is equivalent for cohabitants and spouses, so the budget constraint applies identically to both. Throughout, I refer to the two adults as wife and husband in their labor-market roles, and as mother and father in their child-rearing roles.

[20] The choice of model period interval is driven by data availability. For the household life-cycle labor supply, I use the BHPS, which is available annually; however, the MCS for child's development skills is available every two years for the age range I focus on.

[21] This assumption does not require that fertility be unconditionally random. It requires only that birth timing be exogenous to unobserved work-arrangement preferences conditional on the state vector (age, education, parity, and work arrangement), which together determine p_t^w. A birth is realized as an end-of-period shock given the state, so within-period choices respond to the birth probability rather than to the realization. The assumption binds on unobserved heterogeneity, since p_t^w depends on the state and not on the wife's latent ability type. I assume birth timing is independent of that type conditional on the state, and were higher-ability types to delay fertility while also holding more productive maternal time, the maternal-time return for college mothers would be overstated. The event-study analysis in Figure 1 also supports the assumption. Work arrangements show flat pre-trends in the years before first birth, which is inconsistent with parents strategically timing births around planned work arrangement transitions. The forward-looking nature of the model also mitigates the concern because households solve the full life-cycle problem taking the fertility process as given. Hence, the optimal work-arrangement choices already incorporate the anticipated

Given this state, the household chooses savings a_{t+1}, work arrangement q_t, and each spouse's hours worked n_t^i and home production time τ_t^i. The work arrangement $q_t \in \mathcal{Q}$ summarizes the joint status of both spouses, indexed by indicators $d_{j,t}^i \in \{0,1\}$ for individual i in work arrangement j. The husband always works and chooses between paid employment ($j = \text{PE}$) and self-employment ($j = \text{SE}$);[22] the wife chooses among paid employment, self-employment, and not working ($j = \text{NW}$), yielding six mutually exclusive household arrangements ($|\mathcal{Q}| = 6$), from dual paid employment ($q_t = 1$) to a self-employed husband with a non-working wife ($q_t = 6$). When there is a young child aged 5 or below, the household additionally chooses a formal childcare arrangement, part-time or full-time: $cc_t \in \{0, \text{PT}, \text{FT}\}$, at a cost.[23] When the child is older (ages 6–15), childcare is no longer a choice and child-related expenditures are a fixed fraction of household income. I provide more details in Appendix D. The child investment phase ends at age 5, when the child enters school and parents stop making direct time investments.

I assume spouses are paternalistic: they derive utility directly from their child's skill outcomes rather than from the child's own lifetime utility. This warm-glow formulation (Andreoni, 1990; De Nardi, 2004) is standard in structural models of child investment (Del Boca, Flinn, and Wiswall, 2014; Daruich, 2018; Lee and Seshadri, 2019; Caucutt et al., 2026).[24] Flow utility can be separated into three parts: the outcomes the household directly values, its preferences over the inputs that produce the child's skills, and non-pecuniary terms tied to the work arrangement. The household directly values consumption C_t, each spouse's leisure l_t^i, and the child's latent cognitive and non-cognitive skills $\log \theta_{D,t}$, $D \in \{\text{cog}, \text{ncog}\}$. It holds direct preferences over the inputs that produce those skills, each spouse's home production time τ_t^i and formal childcare. These inputs enter utility directly, not only through the skills they produce, so a parent's chosen time and childcare reflect a direct taste or effort cost alongside the skill return. The remaining terms are non-pecuniary and tied to the work arrangement, the extra cost of a young child and a preference shock ξ_t.[25] I further allow interaction terms with the presence of a young child, capturing how formal childcare demands and work-arrangement disutility shift when there is a young child.

arrival of children.

[22] In the BHPS sample frame of married women aged 21–51, about 10% of couple-waves have a non-employed husband, and among them, very few have a self-employed wife, so the excluded cells contain almost none of the variation that identifies the work arrangement margin. The husband's employment margin is also the least responsive, as fathers' work arrangements do not respond to childbirth (Section 2; Kleven, Landais, and Søgaard, 2019). Fixing the husband's employment while modeling his arrangement and hours choices is richer than the common practice of treating the husband's earnings as an exogenous income process.

[23] Part-time care is 15 hours per week and full-time is 30 hours per week, each priced at the per-hour price of the chosen type. I focus on the main childcare type.

[24] I assume the child has no bargaining power as I focus on ages before school entry and parents make investment decisions.

[25] I adopt a unitary framework as a benchmark. Although bargaining power might vary over the life cycle (Lundberg, Pollak, and Wales, 1997; Guo and Xie, 2024), the paper's focus is on joint time-use and child investment decisions rather than intra-household resource distribution. Appendix M shows that the results are robust to re-solving the counterfactuals under education-specific Pareto weights.

$$
\begin{aligned}
\mathcal{U}_t = \; & \underbrace{\log C_t}_{\text{Consumption}} + \underbrace{\alpha_l^w \log l_t^w + \alpha_l^h \log l_t^h}_{\text{Couple's leisure}} + \underbrace{(\alpha_{cog} + \alpha_{cog}^{ed}\theta^w)\log\theta_{cog,t} + (\alpha_{ncog} + \alpha_{ncog}^{ed}\theta^w)\log\theta_{ncog,t}}_{\text{Child's skills by education}} \\
& + \underbrace{\left[\alpha_\tau^w + \alpha_{\tau,n}^w \tfrac{n_t^w}{\bar{n}} + \alpha_{\tau,k}^w \mathbb{1}[ychild_t = 1]\right]\cdot \log\tau_t^w + \left[\alpha_\tau^h + \alpha_{\tau,n}^h \tfrac{n_t^h}{\bar{n}} + \alpha_{\tau,k}^h \mathbb{1}[ychild_t = 1]\right]\cdot \log\tau_t^h}_{\text{Couple's home production}} \\
& + \underbrace{(\alpha_{cc,PT} + \alpha_{cc,PT}^{ed}\theta^w)\mathbb{1}[cc_t = PT] + (\alpha_{cc,FT} + \alpha_{cc,FT}^{ed}\theta^w)\mathbb{1}[cc_t = FT]}_{\text{formal childcare}} \\
& + \underbrace{\sum_{j\in\{PE,SE\}} \left[\alpha_{j,1}^w \log l_t^w + \alpha_{j,2}^w \mathbb{1}[ychild_t = 1] + (\alpha_{j,3,PT}^w + \alpha_{j,3,PT}^{w,ed}\theta^w)\mathbb{1}[cc_t = PT] + (\alpha_{j,3,FT}^w + \alpha_{j,3,FT}^{w,ed}\theta^w)\mathbb{1}[cc_t = FT]\right]\cdot d_{j,t}^w}_{\text{Wife's (dis)utility by work arrangement}} \\
& + \underbrace{[\alpha_{SE,1}^h \log l_t^h + \alpha_{SE,2}^h \mathbb{1}[ychild_t = 1]]\cdot d_{SE,t}^h}_{\text{Husband's (dis)utility for SE}} + \underbrace{\xi_t}_{\substack{\text{preference}\\ \text{shocks}}}
\end{aligned}
\tag{1}
$$

For identification, I normalize the reference categories to a paid-employed father and a non-working, non-college mother; the base parameters $\alpha_{cc,PT}$ and $\alpha_{cc,FT}$ then capture the (dis)utility to her of part-time and full-time formal childcare. The education interactions $\alpha_{cc,PT}^{ed}$ and $\alpha_{cc,FT}^{ed}$ allow college mothers to value formal childcare differently, and the work-arrangement interactions $\alpha_{j,3,\mathrm{PT}}^w$ and $\alpha_{j,3,\mathrm{FT}}^w$, $j \in \{\mathrm{PE}, \mathrm{SE}\}$ measure how paid and self-employed mothers differ from the non-working baseline in valuing each intensity, and the education interactions $\alpha_{j,3,\mathrm{PT}}^{w,ed}$ and $\alpha_{j,3,\mathrm{FT}}^{w,ed}$ let that difference vary by maternal education. The skill-valuation interactions α_{cog}^{ed} and α_{ncog}^{ed} let college mothers weight their child's cognitive and non-cognitive development differently, so that the education gradient in child investment is not attributed to the production technology alone.

Two features of the specification require motivation. First, the non-pecuniary terms are important for self-employment. Self-employed workers often earn less and work longer hours than paid employees yet remain self-employed, which income and leisure alone cannot explain. The parameters $\alpha_{SE,1}^i$ and $\alpha_{SE,2}^i$ capture the non-pecuniary value of each work arrangement. This may reflect the flexibility and autonomy that draw women and men into self-employment. Together with the hours flexibility in Section 3.3, these amenities explain why parents may accept lower pay to remain self-employed. Second, home production carries a net time cost. If the base weights α_τ^i are negative, parents do not value home production for its own sake, but invest time because it produces child skills they value and because a young child must be supervised. The weight also varies with own market hours through $\alpha_{\tau,n}^i$, scaled by hours relative to their maximum $\bar{n}$. A negative value means the effort cost of home production rises with the workday, so a parent working longer supplies less of it. This is a reduced form way to capture effort that is convex in total work (market plus home).

The model features two levels of heterogeneity. Observed heterogeneity enters through parental education, which shifts earnings profiles, childcare and child-skill valuation, and fertility. Unob-

served heterogeneity enters through the wife's latent ability type, a permanent low/high factor that shifts her initial experience stock and her participation propensity. There are four dynamic state variables: household assets a_t, the wife's human capital hc_t^w, and the child's latent cognitive and non-cognitive skills $\log \theta_{D,t}$, $D \in \{\text{cog,ncog}\}$, indexed by age $g_t \in \{1, 3, 5\}$.[26] Without loss of generality, I normalize the mean of initial log skills to zero at age 1 since pre-natal investment is unobserved and the level of initial skills is not identified.[27] At the terminal period, when the wife reaches age 51, the household no longer makes choices and faces the terminal value

$$\mathbb{E}V_{T+1} = \psi_{hc}\, hc_{T+1}^w, \tag{2}$$

where hc_{T+1}^w denotes the wife's human capital at the end of the life cycle.[28]

3.2 Household Constraints

The household faces three constraints. First, each spouse faces a time constraint across leisure, hours worked, and home production subject to time endowment $\bar{T}$. Home production τ_t^i includes the time a parent spends caring for the child. Second, when there is a young child, the household faces a supervision constraint, which requires that parental home production and formal childcare together meet a minimum care requirement $\bar{T}^{cc}$.[29]

$$\underbrace{l_t^i}_{\text{leisure}} + \underbrace{n_t^i}_{\text{labor}} + \underbrace{\tau_t^i}_{\text{home production}} = \bar{T} \tag{3}$$

$$\tau_t^w + \tau_t^h + \bar{\tau}(cc_t) \geq \bar{T}^{cc}, \qquad cc_t \in \{0, \text{PT}, \text{FT}\}, \quad \bar{\tau}(cc_t) \in \{0, 15, 30\} \tag{4}$$

Meeting this requirement embodies the time-money trade-off: Parental care costs hours worked and leisure, while formal childcare costs money. Third, the household faces the intertemporal budget constraint:

$$C_t + M_t + a_{t+1} = (1 + r) \cdot a_t + \mathcal{T}(Y_{j,t}^w, Y_{j,t}^h, \mathbf{x}_t) \tag{5}$$

Here, household disposable income $\mathcal{T}(Y_{j,t}^w, Y_{j,t}^h, \mathbf{x}_t)$ finances consumption C_t, child-related expenditures M_t, and next-period savings a_{t+1} at net interest rate r. For young children, $M_t = p_{cc} \cdot \bar{\tau}(cc_t)$, and for school-age children (ages 6–15), M_t is a fixed share of net income.[30] Assets are subject to the borrowing constraint $a_{t+1} \geq \underline{a}$. I approximate after-tax income using the FORTAX micro-simulation system (Shephard, 2009), which maps gross household earnings and family characteristics to net

[26] Ages correspond to nine months, age three, and age five in the MCS. I refer to nine months as age one throughout.

[27] I estimate the initial dispersion from wave-1 MCS measures, as described in Appendix G.6. The test measures are age standardized, and I normalize the reference measure to load one-for-one on the latent log-skill (Section 4.1); $\log \theta_{D,t}$ is therefore expressed on the scale of the standardized score, so a one-unit change corresponds to one standard deviation of the test-score distribution, the units in which I report all counterfactual skill effects.

[28] The terminal value rewards the wife's human capital but not terminal financial wealth, so the model imposes no bequest motive.

[29] Spousal time endowment is set to $\bar{T} = 112$ hours/week, and the care requirement is set to $\bar{T}^{cc} = 20$ hours/week. Results are robust to the choice of $\bar{T}$ and $\bar{T}^{cc}$.

[30] Appendix D describes the calibrated prices of formal childcare per age.

disposable income. Following Gayle, Hincapié, and Miller (2024), I parametrize the tax-transfer schedule as follows:

$$\mathcal{T}(Y^w_{j,t}, Y^h_{j,t}, \mathbf{x}_t) = \chi_1(\mathbf{x}_t) + \chi_2(\mathbf{x}_t) \cdot \left(Y^w_{j,t} + Y^h_{j,t}\right)^{\chi_3(\mathbf{x}_t)} \tag{6}$$

Here, $\chi_1(\mathbf{x}_t)$ are lump-sum taxes or transfers, $\chi_2(\mathbf{x}_t)$ scales net income in gross earnings, and $\chi_3(\mathbf{x}_t)$ captures the degree of tax progressivity.[31] Including $\chi_1(\mathbf{x}_t)$ is essential because self-employed parents may incur losses or report zero income. I allow these coefficients to vary by the child's age to account for benefits that vary by household composition, such as Child Benefit, Working Tax Credit, and Child Tax Credit.[32]

3.3 Couple's Income and Wife's Human Capital

Weekly earnings depend on an earnings rate and hours worked with a curvature $\eta_{i,j}$,

$$Y^i_{j,t} = y^i_{j,t} \cdot \left(n^i_t\right)^{\eta_{i,j}}, \qquad j \in \{PE, SE\}, \tag{7}$$

so average hourly earnings, $Y^i_{j,t}/n^i_t = y^i_{j,t}(n^i_t)^{\eta_{i,j}-1}$, vary with hours whenever $\eta_{i,j} \neq 1$. Flexibility enters through two channels: the menu of hours a worker can choose and this curvature. The hours menu is the primary channel, which is coarse in paid employment and wider in self-employment. The curvature is a reduced-form way of capturing how pay varies with hours: when $\eta_{i,j} < 1$, cutting hours forfeits less than a proportional share of earnings. The same nonlinear pricing underlies Goldin (2014)'s account of the gender pay gap, where convex hours-pay imposes the largest penalties on flexibility, and self-employment lies at the flexible, less-convex end. Because self-employment bundles this flexibility with a lower and more volatile wage process, the arrangement choice trades the value of flexible hours against the level and risk of pay. Appendix F documents the grid construction, within-person hours adjustment, and robustness to grid size.

Parental hourly earnings depend on their level of education and experience. Since the husband always works, his experience accumulates deterministically with age. His log hourly earnings follow a quadratic age profile (Van der Klaauw, 1996; Attanasio, Low, and Sánchez-Marcos, 2008):

$$\log y^h_{j,t} = \phi^h_{j,0} + \phi^h_{j,1}\theta^h + \phi^h_{j,2}t + \phi^h_{j,3}t^2/100 + \varepsilon^h_{j,t} \tag{8}$$

The wife's hourly earnings are a function of her education and accumulated work experience:

$$\log y^w_{j,t} = \phi^w_{j,0} + \phi^w_{j,1}\theta^w + \phi^w_{j,2}\log(1 + hc^w_t) + \phi^w_{j,3}\log(1 + hc^w_t)\,\theta^w + \varepsilon^w_{j,t} \tag{9}$$

[31] The UK taxes spouses individually, but over the sample period the individual schedule is a personal allowance plus a nearly flat basic rate; thus, the rate aggregates across spouses, and the work arrangement fixes the number of allowances the household uses, which χ_1 and χ_2 absorb. The progressivity that χ_3 captures comes from the family-assessed credits and benefits, which depend on joint income by statute. Conditional on the work arrangement, household net income is then approximately a function of household gross income, and I fit the coefficients to FORTAX net incomes computed under the actual individual-tax and family-assessment rules.

[32] More details are described in Appendix A.3.

where hc_t^w is the wife's human capital, a continuous stock of accumulated work experience, and $\phi_{j,2}^w$ and $\phi_{j,3}^w$ are the baseline and college-specific returns to experience. Allowing the return to differ by education captures the well-documented pattern that experience raises wages far more for educated women (Blundell et al., 2016). Income shocks are correlated across spouses to capture assortative matching on unobservable characteristics. Conditional on the work arrangement q_t, they are drawn from a bivariate normal distribution:

$$\begin{bmatrix} \varepsilon_{j,t}^w \\ \varepsilon_{j,t}^h \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0},\ \mathbf{\Sigma}_{j_t^w, j_t^h}\right), \qquad \mathbf{\Sigma}_{j^w, j^h} = \begin{pmatrix} (\sigma_{j^w}^w)^2 & \rho\, \sigma_{j^w}^w\, \sigma_{j^h}^h \\ \rho\, \sigma_{j^w}^w\, \sigma_{j^h}^h & (\sigma_{j^h}^h)^2 \end{pmatrix}$$

Here, ρ represents the correlation between unobserved income shocks across spouses.[33]

The wife's human capital hc_t^w is a continuous stock of accumulated work experience that enters her wage in Equation 9 and evolves with her labor supply. It accumulates with hours worked and depreciates during non-employment spells:

$$hc_{t+1}^w = hc_t^w \left(1 - \delta - \delta^{ed}\theta^w\right)^{2 \cdot \mathbb{1}[j=\text{NW}]} + 2\left(\frac{n_t^w}{\bar{n}}\right)^{\nu} \mathbb{1}[j \neq \text{NW}] \tag{10}$$

Full-time work at the reference hours $\bar{n}$ adds one unit of experience per year, while not working depreciates the stock at an annual rate $\delta + \delta^{ed}\theta^w$ that I allow to differ by education, since career interruptions scar more-educated women's wages more. The curvature ν governs how part-time hours translate into experience: with $\nu > 1$, part-time work accumulates disproportionately less than full-time, so that cutting hours slows wage growth, consistent with Blundell et al. (2016). Work arrangement shapes experience only through the feasible hours menu (Section 3.3): a paid-employed mother who reduces her hours must move to part-time or out of work, incurring an interruption and depreciation, whereas a self-employed mother can lower her hours while staying attached and continuing to accumulate, so flexibility also protects her own wage path.

Because women enter the sample at different ages, I treat the initial experience stock following Gayle, Lott, and Shephard (2019). Each wife belongs to one of two latent ability types $k \in \{\text{low}, \text{high}\}$ with education-dependent probabilities, following the finite-mixture approach of Heckman and Singer (1984). The type governs her experience at entry, which scales with her age at entry:

$$hc_1^w = (age_1 - a_0)\,\pi_k, \qquad \Pr(k = \text{high} \mid \theta^w) = \frac{\exp(\iota_0 + \iota_1\,\theta^w)}{1 + \exp(\iota_0 + \iota_1\,\theta^w)} \tag{11}$$

where a_0 is the labor-market entry age and π_k is the participation rate of type k, so the high type enters with more accumulated experience, and ι_0, ι_1 govern the type's prevalence by education. The human-capital parameters are $\{\iota_0, \iota_1, \boldsymbol{\pi}, \delta, \delta^{ed}, \nu\}$.

[33] I assume a common correlation parameter ρ across work arrangements, so the heterogeneity in dependence between parents' earnings is instead captured through the work-arrangement-specific variances of the income shocks.

3.4 Child Production Function

Child cognitive and non-cognitive skills, $D \in \{\text{cog}, \text{ncog}\}$, evolve dynamically over developmental stages g_t.[34] The evolution of each skill follows a log-linear production function[35] with heterogeneous elasticities by parental education:

$$\begin{aligned}\log \theta_{D,g_t+1} = {} & \underbrace{\log A_{D,g_t}}_{\text{TFP}} + \gamma_{D,1} \underbrace{\log \theta_{D,g_t}}_{\text{own skill}} + \gamma_{D,2} \underbrace{\log \theta_{D',g_t}}_{\text{cross skill}} + \underbrace{\gamma_{D,3} \log \tau^w_{g_t} + \gamma_{D,4} \log \tau^w_{g_t} \times \theta^w}_{\text{mother's time}} \\ & + \underbrace{\gamma_{D,5} \log \tau^h_{g_t} + \gamma_{D,6} \log \tau^h_{g_t} \times \theta^h}_{\text{father's time}} \\ & + \underbrace{\gamma_{D,7}\, cc^{PT}_{g_t} + \gamma_{D,8}\, cc^{FT}_{g_t} + \gamma_{D,9}\, cc^{PT}_{g_t} \times \theta^w + \gamma_{D,10}\, cc^{FT}_{g_t} \times \theta^w}_{\text{formal childcare}} + \gamma_{D,11} \underbrace{sib_{g_t}}_{\text{siblings}} + \nu_{D,g_t+1}\end{aligned} \tag{12}$$

Each skill depends on its own lagged skill and the cross-domain lagged skill, both parents' time, formal childcare, and the presence of siblings, plus a mean-zero production shock ν_{D,g_t+1}. Formal childcare enters as separate part-time and full-time indicators, and the college interactions allow the productivity of each childcare intensity to vary with maternal education. Because the elasticities $\gamma_{D,\cdot}$ are indexed by skill domain, maternal time, paternal time, and formal childcare can have different returns to cognitive versus non-cognitive skills. This domain-specific productivity is what makes the allocation of parental time across spouses matter for child outcomes.

3.5 Model Timeline and Solution

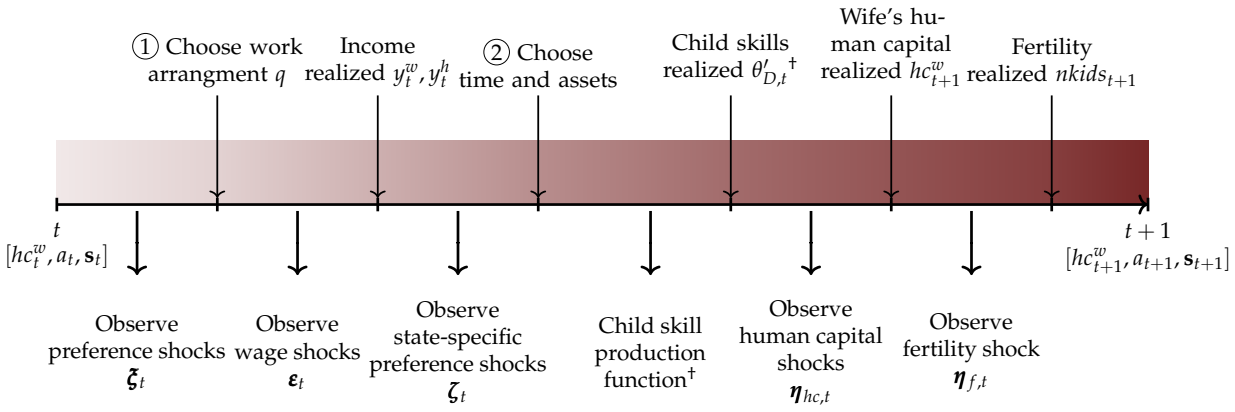
Figure 6 illustrates the intratemporal timeline, including the child skill realization step that occurs after Stage 2 allocations are finalized. The model is initialized with state vector $[\theta^w, \theta^h, hc^w_1, k_1, g_1, a_1]$. There are four dynamic states: mother's human capital, child cognitive and non-cognitive skills, and assets. Each period follows a two-stage intratemporal structure.

Stage 1 The household receives realizations of the preference shocks ξ_t and decides the optimal work arrangement choice. The ex-post value function for Stage 1 is:

$$\tilde{\mathcal{V}}^{s1}(\Omega^{s1}, \boldsymbol{\xi}) = \max_{q \in \mathcal{Q}} \left\{ \mathcal{V}^{s2}_q(\Omega^{s1}) + \xi_q \right\} \tag{13}$$

[34] The subscript g_t is used for notational consistency, even though the model is indexed by the mother's age t. The child production function becomes relevant only after a fertility shock occurs, such that $g_t \in \{1, 3, 5\}$ denotes the child's age in subsequent developmental periods.

[35] Because the production function is log-additive, its inputs have no technological interaction. When I describe formal childcare as a stronger or weaker substitute for maternal time, I mean that it replaces maternal time as an input in the household's investment problem. How well it does so is governed by the two inputs' relative marginal productivities, which differ by education, rather than by a technological elasticity of substitution.



†Applies only when a young child is present; childless couples skip the childcare choice in Stage 2 and this step.

Figure 6: Intratemporal Timeline

Because preference shocks follow a Type-I extreme value distribution, the conditional choice probability has a closed-form logit representation (Rust, 1987):

$$p(d_q = 1|\Omega^{s1}) = \frac{\exp(\mathcal{V}_q^{s2}(\Omega^{s1})/\sigma_\xi)}{\sum_{q\in\mathcal{Q}} \exp(\mathcal{V}_q^{s2}(\Omega^{s1})/\sigma_\xi)} \tag{14}$$

The ex-ante value function is written as:

$$\mathbb{E}_{\boldsymbol{\xi}}[\tilde{\mathcal{V}}^{s1}(\Omega^{s1}, \boldsymbol{\xi})] = \sigma_\xi \log\Big(\sum_{q\in\mathcal{Q}} \exp\Big(\frac{\mathcal{V}_q^{s2}(\Omega^{s1})}{\sigma_\xi}\Big)\Big) + \sigma_\xi \gamma_E, \tag{15}$$

where σ_ξ is the scale of the preference shocks, common across all alternatives, and γ_E is the Euler–Mascheroni constant.

Stage 2 Conditional on the work arrangement, the household draws wage shocks $\boldsymbol{\varepsilon}$ and state-specific preference shocks $\boldsymbol{\zeta}$, then maximizes flow utility over combination of discretized choices $r \in \mathcal{R}$. With a young child, the choice set adds formal childcare, $r = (n^w, n^h, \tau^w, \tau^h, a', cc)$, and the child skill production function enters the problem:

$$
\begin{aligned}
\tilde{V}_q^{s2}(\Omega^{s2}, \boldsymbol{\zeta}) &= \max_{r \in \mathcal{R}} \Big\{ \hat{V}_{q,r}^{s2}(\Omega^{s2}) + \zeta_{q,r} \Big\} = \max_{r \in \mathcal{R}} \Big\{ \mathcal{U}(\cdot) + \zeta_{q,r} + \beta \mathbb{E}[V^{s1}(\Omega^{s1,\prime}) | \Omega^{s2}, r] \Big\} \\
\text{s.t} \quad & (\text{TC}^w) \quad l^w + \tau^w + n^w = \bar{T} \\
& (\text{TC}^h) \quad l^h + \tau^h + n^h = \bar{T} \\
& (\text{Sup}) \quad \tau^w + \tau^h + \bar{\tau}(cc) \geq \bar{T}^{cc}, \quad cc \in \{0, PT, FT\} \\
& (\text{BC}) \quad C + M + a' = (1 + r) \cdot a + \mathcal{T}(Y^w, Y^h, \mathbf{x}) \\
& (\text{I}) \quad Y_j^i = y_j^i \cdot (n^i)^{\eta_{i,j}}, \quad j \in \{PE, SE\} \\
& (\text{IC}^h) \quad \log y_j^h = \phi_{j,0}^h + \phi_{j,1}^h \theta^h + \phi_{j,2}^h \, t + \phi_{j,3}^h \, t^2 + \varepsilon_j^h \\
& (\text{IC}^w) \quad \log y_j^w = \phi_{j,0}^w + \phi_{j,1}^w \theta^w + \phi_{j,2}^w \log(1 + hc^w) + \phi_{j,3}^w \log(1 + hc^w) \, \theta^w + \varepsilon_j^w \\
& (\text{HC}^w) \quad hc^{w\prime} \text{ evolves according to Equation 10} \\
& (\text{CPF}^D) \quad \log \theta_D' \quad \text{evolves according to Equation 12}
\end{aligned}
\tag{16}
$$

The conditional choice probability takes the same logit form:

$$
p(d^r = 1 | \Omega^{s2}, q) = \frac{\exp(\hat{V}_{q,r}^{s2}(\Omega^{s2}) / \sigma_\zeta)}{\sum_{r' \in \mathcal{R}} \exp(\hat{V}_{q,r'}^{s2}(\Omega^{s2}) / \sigma_\zeta)} \tag{17}
$$

The ex-ante value function is:

$$
\mathcal{V}_q^{s2}(\Omega^{s2}) \equiv \mathbb{E}_{\boldsymbol{\zeta}}[\tilde{V}_q^{s2}(\Omega^{s2}, \boldsymbol{\zeta})] = \sigma_\zeta \log \Big(\sum_{r \in \mathcal{R}} \exp \Big(\frac{\hat{V}_{q,r}^{s2}(\Omega^{s2})}{\sigma_\zeta} \Big) \Big) + \sigma_\zeta \gamma_E \tag{18}
$$

Here, σ_ζ is the scale of the state-specific preference shocks, common across alternatives.

Wife's human capital updates deterministically at the end of each period, after Stage 2, through the law of motion in Equation 10. The household's continuation value therefore integrates only over the fertility shock. Child skills evolve in parallel but drop out of the choice problem by log-additivity:

$$
\begin{aligned}
\mathbb{E}\left[V^{s1}(\Omega^{s1,\prime}) \,\middle|\, \Omega^{s2}, \, r \right] &= p_t^w \, V^{s1}\big(a', \, hc^{w\prime}, \, \Omega^{s1,\prime} \,\big|\, \text{yes child}\big) \\
&\quad + \big(1 - p_t^w\big) \, V^{s1}\big(a', \, hc^{w\prime}, \, \Omega^{s1,\prime} \,\big|\, \text{no child}\big)
\end{aligned}
$$

where $hc^{w\prime} = hc^{w\prime}\big(hc^w, n^w(r)\big)$ is the human-capital stock implied by the law of motion.

To solve the model, I use backward induction from the terminal period T, using the terminal value in Equation 2. For tractability, I restrict backward recursion to reachable states, given each household's initial data observation.[36] In each period, I first solve Stage 2, computing conditional choice probabilities and expected value functions over the discrete grids of choices

[36] For example, a wife observed at age 35 with two children cannot have a third birth in the already-completed periods; the reachable state space conditions on observed initial conditions.

$(n^w, n^h, \tau^w, \tau^h, a')$ given realized income shocks, then integrate over income shocks to obtain the Stage 1 expected value function over work arrangements. The Stage 1 value at period t serves as the continuation value entering period $t-1$. Because of the log-additivity in utility, the value functions are additively separable in the skill states (Appendix E.1), so optimal parental investment decisions are independent of the current skill level in each period. Hence, I solve the model over two dynamic states $(hc_t^w,\, a_t)$ instead of four, making the model more tractable. I provide more computational details in Appendix G.

4 Identification and Estimation Strategy

This section describes the identification and estimation strategy for the structural parameters, those governing the child skill production functions, household preferences, parental income processes, and the wife's human capital accumulation.

4.1 Identifying the Child Production Function

The child production function estimation faces two econometric challenges: measurement error in observed skill measures and endogeneity of parental inputs. First, cognitive and non-cognitive skills are latent and observed only through assessment scores, which are subject to measurement error. Following Agostinelli and Wiswall (2025), I specify the measurement system for each skill domain $D \in \{cog, ncog\}$ at child age g_t as:

$$\log \widetilde{\theta}^m_{D,g_t} = \mu^m_{\theta_D,g_t} + \alpha^m_{\theta_D,g_t} \log \theta_{D,g_t} + \epsilon^m_{\theta_D,g_t} \tag{19}$$

where $\log \widetilde{\theta}^m_{D,g_t}$ is observed measure m, $\log \theta_{D,g_t}$ is the latent skill, and ϵ^m is measurement error, uncorrelated across measures. I normalize one reference measure per domain to fix scale and location ($\mu^1_{\theta_D,g_t} = 0$, $\alpha^1_{\theta_D,g_t} = 1$). The reference measure therefore equals the latent skill plus noise. Entering it as the lagged skill attenuates the persistence coefficients and biases the coefficients on the inputs correlated with skill. I therefore instrument the reference measure of each lagged skill with a vector Z_D that is correlated with $\log \theta_{D,g_t}$ but orthogonal to the measurement error in the reference measure:[37]

$$\operatorname{cov}\left(Z_D,\, \log \theta_{D,g_t}\right) \neq 0, \qquad \operatorname{cov}\left(Z_D,\, \epsilon^1_{\theta_D,g_t}\right) = 0 \tag{20}$$

[37] The cognitive skill is measured by the gross-motor scale of the Denver Developmental Screening Test at nine months, the British Ability Scales Naming Vocabulary score at ages three and five, and the British Ability Scales Word Reading score at age seven. The non-cognitive skill is measured by the mood scale of the Carey Infant Temperament Scale at nine months and the Strengths and Difficulties Questionnaire total difficulties score at ages three, five, and seven. The instruments are second measures of the other domain from the preceding sweep. The lagged cognitive skill is instrumented with the adaptability scale of the Carey Infant Temperament Scale at nine months, for the nine-month and age-three skills, and with the emotional self-regulation score at age three, for the age-five skill. The lagged non-cognitive skill is instrumented with the fine-motor scale of the Denver test at nine months, for the nine-month skill, with its communicative-gestures scale at nine months, for the age-three skill, and with the Bracken School Readiness score at age three, for the age-five skill. The nine-month skills have no preceding sweep, so their instruments come from the nine-month sweep. All scores are standardized within wave using survey weights.

For Z_D I use lagged cross-domain instruments. Skills are persistent and correlated across domains, so the instrument moves with the latent skill. Its measurement error comes from a different assessment of a different domain from a different wave, so it is uncorrelated with the error in the reference measure. In Appendix E, I show the first-stage and that the instrument is orthogonal to the production shock.

The second concern is input endogeneity. Parental inputs (time and formal childcare) are subject to household constraints. Households who invest differently may differ in unobserved determinants of child skills, which the model does not rule out. In particular, the channel of formal childcare goes through the supervision constraint, which ties its take-up to labor supply, and it responds to the price of care and in turn couples it to parental time. Therefore, I estimate the child production function instrumenting (τ^w, τ^h, cc) with FORTAX-simulated instruments spanning distinct UK policy channels via joint GMM. I provide more details on the exogeneity of parental inputs and estimation of child skills in Appendix E.3.

4.2 Identifying Preferences, Income, and Human Capital

All parameters are jointly identified by the full set of moments, but distinct sources of variation discipline distinct parameters. Appendix H develops the formal identification, and Table 2 provides the moment-parameter mapping. Here, I describe the general identification strategy.

Preferences The preference weights are identified from how households choose work arrangements, hours worked, home production, and formal childcare as their budget and child status vary. Hours worked within an arrangement identify the leisure weight of a working spouse, because the income process and the tax schedule fix the net-of-tax return to an additional hour. For the wife, this weight is the sum of her base leisure weight and the arrangement-leisure interaction. Her participation rate with and without a young child separates the work-arrangement specific intercept. Home production without a young child identifies the base home-production weights, and its shift with the arrival of a young child identifies the young-child interactions. The workload interactions are identified separately from the levels by the within-person slope of home production in own market hours. The child production function estimates separate the child-skill weights α_{cog} and α_{ncog} from the young-child interactions: the skill return to each parent's time and to formal childcare is known and differs by parent, skill domain, and education, so both parents' home production and childcare take-up with a young child pin down the two weights. Their education interactions are identified from the education gradient in these choices. Shifts in work arrangement shares after childbirth identify the work-arrangement amenities, and formal childcare take-up by intensity, maternal work arrangement, and education identifies the childcare valuations.

Income and Human Capital Mean log hourly income by work arrangement and education identifies the earnings levels and the education returns. The husband's hourly income by age identifies his age profile, and the within-arrangement dispersion of hourly income identifies the

earnings volatility. The correlation of the spouses' hourly incomes identifies the shock correlation ρ. The curvature $\eta_{i,j}$ is identified from the mean and dispersion of log hours within each arrangement together with hourly income. The curvature sets the return to a marginal hour relative to the average hour, so, given the leisure weights, it governs how much mass households place on long relative to short hours.

The wife's income dynamics pin down her human-capital process. The wage growth of continuously employed wives, by age and education, identifies the returns to experience $\phi^w_{j,2}$ and $\phi^w_{j,3}$. The depreciation rate δ, and its education interaction δ^{ed}, are identified from the within-person wage change when a woman returns from a non-employment spell, measured against her own last wage before the spell and against the wage growth of continuously employed women, separately by education. The gap between full-time and part-time wage growth identifies the accumulation curvature ν, since part-time hours build less experience. For the initial experience distribution, mean entry earnings by education and their slope in age at entry pin down the type participation rates π_k, and the dispersion of entry earnings and the auto-covariance of wages separate the type by education from the transitory shocks.

4.3 Estimation Procedure

Estimation relies on a fully specified parametric model and proceeds in two stages. First, I preset a subset of parameters: the discount factor $\beta = 0.98^2$ and the gross return on assets $(1+r)^2$ with annual risk-free interest rate $r = 0.0469$, each compounded over the two-year model period.[38] In the first stage, I estimate directly from the data the probability of child arrival and the parameters of the tax-and-transfer system based on family composition and parental labor market characteristics (Equation 6). The child skill production function is estimated via joint GMM, correcting for skill measurement error and endogeneity of parental inputs.

In the second stage, I recover the remaining structural parameters using Method of Simulated Moments (McFadden, 1989). The estimation targets empirical moments capturing key patterns in work arrangements, earnings, and time allocations across households and the presence of a young child. Let $\mathcal{M}$ be the vector of sample moments. For each of S simulated paths of household choices over the wife's age, I compute the corresponding simulated moments and match them to their data counterparts. The simulated moment vector $\widetilde{\mathcal{M}}_S(\Theta)$ is determined by the vector of structural parameters Θ and the draws used in the data generating process. The SMM estimator minimizes the weighted distance between simulated and actual moments:

$$\widehat{\Theta}_{S,W} = \underset{\Theta}{\operatorname{argmin}} \Big(\mathcal{M} - \widetilde{\mathcal{M}}_S(\Theta)\Big)' W \Big(\mathcal{M} - \widetilde{\mathcal{M}}_S(\Theta)\Big) \tag{21}$$

Here, W is a symmetric, positive-definite weighting matrix.[39]

[38] For the UK, Blundell, Pistaferri, and Saporta-Eksten (2016) assume a discount factor of 0.98 and Bolt et al. (2023) and Jordà et al. (2019) set the interest rate to $r = 0.0469$.

[39] I use the diagonal of the inverse of the variance of the empirical moments. Since the moments are computed

Table 2: Moments and Parameter Identification

Moments	Parameters
Work-Arrangement Choices	
Couple's and household's work-arrangement shares by child bin	α_l^w, $\alpha_{j,1}^w$, $\alpha_{j,2}^w$, $\alpha_{SE,1}^h$, $\alpha_{SE,2}^h$, σ_ξ, ψ_{hc}
Labor supply and time allocation	
Couple's hours worked by work arrangement and child bin	α_l^h, $\alpha_l^w + \alpha_{j,1}^w$, $\alpha_l^h + \alpha_{SE,1}^h$, σ_ζ
Couple's home production by work arrangement and child bin	α_τ^i, $\alpha_{\tau,k}^i$, $\alpha_{\tau,n}^i$, α_{cog}, α_{ncog}
Within-person slope of home production on hours worked	
Income Processes	
Couple's log hourly income by work arrangement and education (mean, std)	$\phi_{j,0}^i$, $\phi_{j,1}^i$, σ_j^i
Husband's log hourly income by work arrangement and age	$\phi_{j,2}^h$, $\phi_{j,3}^h$
Couple's log hours by work arrangement (mean, std)	$\eta_{w,j}$, $\eta_{h,j}$
Correlation between husband's and wife's income	ρ
Human Capital Dynamics	
Wife's initial log hourly income by education (mean, std)	ι_0, ι_1, $\boldsymbol{\pi}$
Entry-age slope of wife's initial log income by education	$\boldsymbol{\pi}$
Autocovariance of wife's log income	ι_0, ι_1, σ_j^w
Wife's log income change after non-employment vs employment spells, by education	δ, δ^{ed}
Wife's log income growth, continuously employed, by age and education	$\phi_{j,2}^w$, $\phi_{j,3}^w$
Std. of wife's log income growth	σ_j^w
Wife's income growth gap, full-time vs lower hours	ν
Childcare Choices	
Part-time and full-time formal childcare usage by wife's work arrangement and education	$\alpha_{cc,PT}$, $\alpha_{cc,FT}$, $\alpha_{cc,PT}^{ed}$, $\alpha_{cc,FT}^{ed}$, $\alpha_{j,3,PT}^w$, $\alpha_{j,3,FT}^w$, $\alpha_{j,3,PT}^{w,ed}$, $\alpha_{j,3,FT}^{w,ed}$, α_{cog}^{ed}, α_{ncog}^{ed}

Note: The BHPS and MCS are used to compute data moments. The BHPS, with its longitudinal panel structure, provides rich life-cycle information on household work arrangements, income, and time investments. The MCS is used for estimating the child skill production functions and the moments on formal childcare usage. Each row lists the parameters a moment group chiefly disciplines; all parameters are estimated jointly. PE denotes paid employment; SE denotes self-employment. Parameters follow the notation of Section 3; $i \in \{w, h\}$ for wife and husband, and $j \in \{PE, SE\}$ indexes the work arrangement, and bold symbols denote vectors.

5 Estimation Results

5.1 Child Skill Production Function Estimates

The GMM estimates in Table 3 show that child cognitive and non-cognitive skills are persistent, so early gaps persist and transmit across domains. The productivity of maternal time is strongest for cognitive skills and for college-educated mothers. The evidence is consistent with the idea that maternal time is a quantitatively important early input (Del Bono et al., 2016; Del Boca, Flinn, and Wiswall, 2014), where educational activities with parents are the most productive use of a child's time (Fiorini and Keane, 2014). Educated mothers also match their time to the child's developmental stage (Kalil, Ryan, and Corey, 2012). The mother-father productivity ranking matches Del Boca, Flinn, and Wiswall (2014).

Table 3: Child Production Function Estimates

	Cognitive$_{t+1}$		Non-Cognitive$_{t+1}$	
Variable	Coef	SE	Coef	SE
TFP intercept	-1.136	0.565	-0.846	0.501
Cognitive	0.748	0.089	0.149	0.082
Non-Cognitive	0.154	0.073	0.721	0.069
log(Mother's time)	0.317	0.127	0.209	0.123
log(Mother's time) $\times$ College	0.111	0.021	0.103	0.021
log(Father's time)	0.059	0.296	-0.028	0.262
log(Father's time) $\times$ College	0.045	0.020	0.058	0.019
PT Childcare	1.117	0.445	1.624	0.409
FT Childcare	0.675	0.286	0.496	0.268
PT Childcare $\times$ Mother's college	-1.120	0.438	-1.657	0.400
FT Childcare $\times$ Mother's college	-0.517	0.265	-0.283	0.251
Sibling	-0.168	0.043	0.072	0.040
Observations		6,726		

Note: Joint GMM estimates of the child production function. The system stacks four first-stage moment conditions for the parental inputs with the second-stage child production function moment condition, correcting measurement error in lagged skills via cross-domain skill instruments and endogeneity of τ^w, τ^h, cc via a control function built from FORTAX-simulated instruments from distinct UK policy channels; see Appendix E.2 for instrument construction, moment conditions, and diagnostics. Cognitive skills are measured using the Denver Developmental Screening Test at nine months and the British Ability Scales verbal scores at ages three and five. Non-cognitive skills are measured using the Carey Temperament Assessment Tool at nine months and the Total Difficulties Score from the Strengths and Difficulties Questionnaire at ages three to seven. All skill measures are standardized by age. Standard errors are analytical two-step sandwich standard errors clustered at the child-household level.

The formal childcare estimates reproduce the central finding of the quasi-experimental liter-

using two data sources, I calculate the variance-covariance matrix for each dataset independently. In forming the final weighting matrix, I assume the datasets are statistically independent, which means the off-diagonal blocks representing covariances between the two datasets are zero. Additional computational details are in Appendix G.

ature:[40] The returns concentrate among less advantaged children (Cornelissen et al., 2018; Felfe and Lalive, 2018; Havnes and Mogstad, 2015; Kottelenberg and Lehrer, 2017), while children of advantaged families gain little or lose (Baker, Gruber, and Milligan, 2008; Fort, Ichino, and Zanella, 2020). In the UK, the part-time childcare entitlement shows small, fading average effects (Blanden et al., 2016). Part-time care yields the highest returns for children of non-college mothers, consistent with German evidence that full-day care lowered children's socio-emotional well-being relative to half-day care (Felfe and Zierow, 2018). The sibling coefficients match resource dilution for cognitive skills (Blake, 1981) and the social-skill benefit of siblings for non-cognitive skills (Downey and Condron, 2004).

5.2 Preferences and Household Division of Labor

The utility estimates in Table 4 show that both spouses value leisure. It is higher for the wife than the husband because given each spouse's net-of-tax return, her shorter market hours and lower participation identify the higher value she places on time out of paid work. The marginal utility of leisure also depends on the work arrangement. For the wife, it is lower under self-employment than paid employment, so self-employed hours carry less disutility and she works at a lower average pay. For the husband, the interaction has the opposite sign, so self-employment carries no such amenity. His self-employment reflects autonomy rather than flexibility (Taylor, 1996), so it does not pull his hours down. This captures self-employment's non-pecuniary benefits (Hamilton, 2000; Hurst and Pugsley, 2011) and workers' willingness to pay for schedule control (Mas and Pallais, 2020; Lachowska et al., 2026). Together with the hours-earnings non-linearity discussed below, these preferences produce the observed gender asymmetry in self-employment hours.

A young child raises the time cost of market work, and most sharply for the wife. Her disutility of paid employment with a young child is the largest and negative compared to self-employment and not working. The workload interactions reveal a second gender asymmetry, in how home production responds to market hours. The wife's is negative, so her home production is crowded out as her market hours rise, especially at full-time work. The husband's is small and positive, matching his flat home-production response in the data.

Both parents value child's cognitive and non-cognitive skills, more so for non-cognitive skills. Caring for a young child forgoes leisure and carries a direct disutility, but parents put in home production time because the child-skill return they value outweighs both costs. The education interaction is positive but small and insignificant, so the valuation does not vary by maternal education. This matters because the education gap in child development is then not a difference in how much parents value their children, but works through the productivity of their inputs and the resources they have.

[40] The child production function treats formal childcare as homogeneous in quality. Both Working Tax Credit and the age-3 free entitlement operate within the regulated sector, so if families sorting into higher-quality settings also invested more on unobservables, the childcare coefficients would partly reflect quality-driven selection. In fact, observed setting quality does not differ by maternal education in the MCS childcare setting study (Table D.12).

Table 4: Preference Parameter Estimates

Description	Coef	SE
Wife's leisure	0.541	0.053
Husband's leisure	0.356	0.094
Wife's home production	-0.053	0.016
Wife's home production × own hours	-0.100	0.016
Wife's home production × young child	-0.889	0.132
Husband's home production	-0.170	0.012
Husband's home production × own hours	0.080	0.021
Husband's home production × young child	-0.161	0.031
Child's cognitive skills	0.104	0.062
Child's cognitive skills × college	0.054	0.162
Child's non-cognitive skills	0.421	0.107
Child's non-cognitive skills × college	0.020	0.155
Childcare		
PT childcare	-6.920	1.179
FT childcare	-2.422	0.306
PT childcare × college	6.770	1.155
FT childcare × college	1.470	0.242
Work-Arrangement Interactions		
Wife PE × leisure	-0.026	0.007
Wife PE × young child	-0.668	0.064
Wife PE × PT childcare	0.087	0.105
Wife PE × FT childcare	0.054	0.136
Wife PE × PT childcare × college	-0.012	0.121
Wife PE × FT childcare × college	0.109	0.157
Wife SE × leisure	-0.122	0.011
Wife SE × young child	-0.065	0.051
Wife SE × PT childcare	0.193	0.238
Wife SE × FT childcare	0.060	0.062
Wife SE × PT childcare × college	-0.315	0.317
Wife SE × FT childcare × college	-0.037	1.321
Husband SE × leisure	0.073	0.009
Husband SE × young child	0.457	0.064
Wife's human capital terminal value	0.708	0.121
Variance of preference shock: Stage 1	0.665	0.037
Variance of preference shock: Stage 2	0.260	0.012

Note: The own-hours rows report the workload interactions $\alpha^i_{\tau,n}$. The reference category for childcare base parameters corresponds to non-working, non-college mothers. College interactions capture differential valuation by maternal education. Work arrangement interactions are deviations from the non-working baseline for the wife and paid employment for the husband. The childcare×arrangement×college terms let the education gradient in childcare valuation differ across work arrangements. Standard errors follow Appendix G.5. PE denotes paid employment; SE denotes self-employment.

Part-time childcare carries a large base disutility for non-college mothers that the college interaction offsets. This pattern matches the production function estimates. College-educated mothers face near-zero net skill returns from part-time care, so less disutility implies lower take-up. For non-college families the two margins pull apart. Formal part-time care raises the child's skill, but it carries a large utility cost beyond its price, so take-up stays low even where the child would gain. The cost is non-pecuniary, plausibly a preference for household care. The wife's human capital enters utility positively, capturing the dynamic returns to labor-market attachment beyond current earnings.

Table 5: Comparative Advantage in Child Production and the Child-Skill Cost of an Equal Division of Child-Rearing

	Wife non-college		Wife college	
	Husband NC	Husband C	Husband NC	Husband C
Panel A: Comparative advantage in child production				
Mother's coefficient, cognitive a^w_{cog}	0.32		0.43	
Father's coefficient, cognitive a^h_{cog}	0.06	0.10	0.06	0.10
Mother's coefficient, non-cognitive a^w_{ncog}	0.21		0.31	
Father's coefficient, non-cognitive a^h_{ncog}	-0.03	0.03	-0.03	0.03
Productivity ratio $a^w_{\text{cog}}/a^h_{\text{cog}}$	5.41	3.07	7.31	4.14
Wage ratio w^w/w^h	0.65	0.44	1.09	0.76
Comparative-advantage index	8.38	6.94	6.72	5.42
Mother's forgone earnings (£/yr per hour)	413	398	708	697
Father's forgone earnings (£/yr per hour)	1408	2220	1320	2122
Marginal comparative-advantage ratio	4.57	8.42	2.05	3.30
Panel B: Cost of an equal division of the child's time (behavior fixed)				
Δ cognitive skill (SD)	-0.35	-0.10	-0.59	-0.35
Δ non-cognitive skill (SD)	-0.40	-0.14	-0.63	-0.37
CEV, child channel (% lifetime cons.)	-6.46	-2.15	-9.35	-4.36
in £/yr	-1936	-827	-3580	-1995

Note: Columns are the four wife × husband education cells, over periods with a young child. Panel A reports each parent's marginal coefficient on log home time in the two production functions. Forgone earnings per childcare hour is each parent's marginal earnings $\eta_j y\, n^{\eta_j - 1}$ at the baseline optimum, annualized. The marginal comparative-advantage ratio is each parent's child-skill return per pound of forgone earnings, mother relative to father. Panel B splits total parental time equally between the two parents, holding that total, arrangements, hours, childcare, and earnings at baseline, and recomputes child skills. CEV is consumption-equivalent variation as a percent of lifetime consumption. The £ figure applies each household's CEV to its gross annual income.

The estimates imply a gender division of labor, where the mother specializes in home production. I quantify what drives it, and what an equal division of child-rearing would cost in consumption-equivalent terms (Table 5). Panel A shows that the mother specializes not because she earns less, but because she is the lower-cost producer of the child's skills. Her time is the child's most productive input, three to seven times the father's across the education cells, and her advantage is largest when she is college educated and he is not. What matters is not who earns more but whose forgone earnings per unit of child skill are lower, which the comparative-advantage index $(a^w/a^h)/(w^w/w^h)$ captures. The index exceeds five in every cell, which implies that producing a

given gain in the child's skill through the father forgoes more than five times the earnings it would through the mother, so the mother becomes the margin of allocating more home production time. It is highest when both spouses are non-college and lowest when both are college, because the father's child-skill productivity rises faster with education than the mother's, so a college father substitutes better, while a college wife's higher relative wage raises the opportunity cost of her time. This holds even where the wife out-earns her husband, as in college-wife, non-college-husband couples, where the wage ratio is 1.09 and the index is 6.7. Her time is scarce on both margins, so the household decides to keep her partly out of full-time work rather than shift home production to the father. One weekly hour of the father's child time forgoes between £1,300 and £2,200 a year in earnings, against £400 to £700 for the mother.

Equalizing the division of the child's time is costly. Holding every household choice at the baseline and splitting the child's time equally between the two parents, child skills fall in every cell, because the father is a poor substitute throughout. A college wife with a non-college husband loses about 0.6 SD of cognitive skill, which the household values at about 9% of lifetime consumption, roughly £3,600 a year. The loss is smallest, though still sizeable, for a non-college wife with a college husband, at about 0.1 SD and 2% of lifetime consumption.

This relates to the debate on gender specialization in home production and the gender wage gap. The couple-inequality literature traces the within-couple division of labor to greedy jobs that penalize the parent who steps back (Goldin, 2021). I study a different consequence of the same division of labor, its effect on the child. The mother provides the majority of the child's time because she is the lower-cost producer of child skills. Because the household is unitary, I quantify the cost of an equal division at the household level.[41]

5.3 Income and Human Capital Estimates

The income estimates in Table 6 confirm the earnings patterns in Section 2. Paid employment pays a higher baseline and rises with the wife's human capital, while self-employment pays a lower baseline but offers larger returns to college for women, and earnings dispersion is substantially higher in self-employment for both spouses (Giupponi and Xu, 2020). The hours-earnings schedules of the two arrangements differ sharply for women. Paid employed wives face a mildly convex schedule, so hourly earnings rise with hours and part-time jobs carry a pay penalty (Goldin, 2014).

Self-employed wives face no comparable premium for long hours, so a wife who shortens her week forfeits little additional pay. This is the flexibility self-employment offers because she has more control over hours without the part-time penalty that paid employment imposes.[42] This feature

[41] A model with intra-household bargaining would also reveal how this cost is divided between the spouses. In Appendix M, I re-solve the counterfactual under education-specific Pareto weights and find the by-education pattern unchanged, so the household-level cost does not hinge on the unitary assumption.

[42] In the BHPS, the pooled hours-earnings elasticity for women is 1.15 in paid employment and 0.57 in self-employment, so paid employment rewards long hours while self-employment offers no comparable premium. The paid-employment premium is a between-job feature, the greedy-jobs premium of Goldin (2014): within person the elasticity falls to 0.86, so the convexity reflects the sorting of workers across jobs that reward hours differently rather than a within-job return to marginal hours. The structural η summarizes the hours-wage menu a worker chooses from.

Table 6: Couple's Log Hourly Income Process Parameters

Description	Coef	SE
Wife PE		
Constant	1.326	0.026
College	0.107	0.013
Experience	0.224	0.012
Experience × college	0.162	0.017
Standard deviation	0.309	0.026
Curvature	1.085	0.028
Wife SE		
Constant	1.347	0.376
College	0.415	0.801
Experience	0.041	0.142
Experience × college	0.125	0.316
Standard deviation	0.775	0.115
Curvature	0.525	0.174
Husband PE		
Constant	2.168	0.096
College	0.385	0.027
Wife's age	-0.007	0.004
Wife's age^2/100	0.018	0.004
Standard deviation	0.435	0.032
Curvature	1.266	0.019
Husband SE		
Constant	2.071	0.161
College	0.212	0.062
Wife's age	0.002	0.006
Wife's age^2/100	-0.016	0.009
Standard deviation	0.739	0.052
Curvature	0.736	0.056
Correlation between husband's and wife's income shock	0.325	0.235

Note: Wife wage equations depend on education and accumulated experience; husband wage equations follow a deterministic age profile. Standard errors follow Appendix G.5. PE denotes paid employment; SE denotes self-employment.

explains why self-employed wives choose shorter, more flexible hours and why paid employment pulls hours upward. For men, the reduced-form schedules are flatter and less cleanly separated, since male hours vary little and the cross-sectional slope reflects job composition rather than a within-person return; the model recovers the male hours-earnings curvature from the estimated parameters. The premium is also gendered. Paid employment's schedule is more convex for men, and fathers work long hours in both arrangements and do not adjust them around childbirth (Kleven, Landais, and Søgaard, 2019). The long-hours premium accrues largely to husbands while the part-time penalty binds on wives, who are the ones who reduce hours. Between spouses, the positive correlation in earnings shocks reflects their shared exposure to common shocks.

Table 7: Wife's Human Capital Parameters

Description	Coef	SE
Experience accumulation		
Depreciation when not working: Non-college	0.040	0.022
Depreciation when not working: College	0.052	0.026
Curvature of part-time accumulation	1.086	0.130
Terminal value of experience	0.708	0.121
Permanent ability type		
P(high ability type): Non-college	0.498	
P(high ability type): College	0.604	
Initial experience rate: low type	0.453	0.338
Initial experience rate: high type	0.650	0.230

Note: Experience depreciates when the wife is not working and accumulates with market hours (part-time accumulation is governed by the curvature). Each wife is one of two latent ability types with education-dependent probability. Standard errors follow Appendix G.5.

Last, the estimates in Table 7 show that the wife's human capital depreciates when she is not working and barely at all under full-time employment. An employment interruption lowers her experience stock by about 4% a year for non-college women and 5% for college women, so career breaks scar the more educated more, slightly lower but in line with Blundell et al. (2016). Labor-market attachment is therefore the key mechanism for preserving her earning capacity. This connects to the motherhood penalty: An employment gap around childbirth depresses her human capital, with persistent effects.

5.4 Model Fit

The model fits the work-arrangement, hours, home-production, and income-distribution moments well across households and child-presence states. It also reproduces untargeted moments, including education-specific work arrangement shares and labor supply elasticities, and, approximately, the hours and wages of households with older children. The first-stage tax-and-transfer and fertility processes fit the data well (see Appendix I). Appendix J documents the comparisons.

5.5 Model Mechanisms

Three mechanisms govern how the work arrangement affects child development: households' preferences, the productivity of their inputs, and the constraints they face. The arrangement's three attributes enter through the first and third. *Mechanism 1 (preferences)*: Spouses value leisure, home production, and their child's cognitive and non-cognitive skills, and a young child raises the cost of market work. They also value flexibility – the schedule control an arrangement provides – so the wife sorts into self-employment despite its lower pay. *Mechanism 2 (productivity)*: Parental inputs differ in child skill productivity. Mothers have a comparative advantage in both cognitive and non-cognitive skills, and the father's contribution to non-cognitive skill is negligible. Formal childcare sits between college and non-college maternal time in productivity, so replacing maternal time with formal care trades down for a college-educated mother's child but up for a non-college mother's. The same entry into work therefore carries opposite child consequences by education. *Mechanism 3 (constraints)*: The supervision constraint couples both spouses' time but mostly affects the mother's, requiring parental time and formal care to jointly cover a young child. The budget constraint is where the arrangement's pay binds, through self-employment's lower earnings and higher volatility. Table K.27 shows the mapping of parameters.

I probe each mechanism with a comparative static, altering the component that governs it and re-solving while holding the rest at their estimated values. *Mechanism 1*: The valuation of child skills drives parental time. Removing child skills from utility leads parents to cut home production, and skills fall. A household that values only consumption and child skills raises them substantially, moving both parents, especially the mother, into self-employment for its flexibility. The value of leisure and the disutility of home production are what hold child investment below its potential. *Mechanism 2*: The father's time is the least productive input on average, though his main-effect productivity is imprecisely estimated and is significant only through the education interaction of time and education. *Mechanism 3*: The supervision constraint determines who supplies that time. Removing it lets mothers cut home production and formal childcare and move into paid employment, again differing by education. The arrangement's flexibility and pay then pull maternal labor supply in opposite directions. Granting paid employment an expanded hours menu draws mothers into work at intermediate hours and lowers the cognitive skills of children of college-educated mothers. Raising paid-employment dispersion to the self-employment level pushes maternal non-employment up. Stripping self-employment of its income dispersion instead draws husbands into it.

6 Flexibility and Childcare Subsidies as Family Policy

With the model, I examine two policies at the center of the family-policy debate: expanding access to flexible work and subsidizing formal childcare. Both are promoted as ways to raise maternal employment and ease the work-family trade-off, but their consequences for children depend on how they reshape the parental time and formal care a child receives. I ask what flexibility is worth

on its own and what the role of childcare is. Then, I propose a way to design and target the two policies by maternal education. Last, I describe what their skill effects mean for children's lifetime earnings.

6.1 What Is the Value of Flexibility?

I first show that the model reproduces Bernal (2008)'s finding that maternal full-time work with childcare lowers child cognitive skills. I then evaluate the value of flexibility and its trade-off. Flexibility is the margin I isolate: It is a price the model can grant without self-employment's lower, riskier pay. I do so by granting paid employed workers an expanded hours menu, holding their pay, hours-earnings curvature, and risk at paid-employment values. The counterfactual therefore separates the hours-menu component of flexibility – that is, control over the quantity of hours. It anchors to the UK's *Right to Request Flexible Working*, which is a statutory right to request changes regarding when and how long one works. The policy is meant to help parents combine work and family. It was introduced by the Employment Act 2002 (effective April 2003) for parents of young children, extended to all employees in 2014, and made a day-one right from April 2024 under the Employment Relations (Flexible Working) Act 2023 and accompanying regulations. Employers may refuse to offer the benefit only on statutory business grounds (House of Commons Library, SN01086).[43] Flexibility reaches the child through changes in hours; I model the policy as extending the paid-employment hours menu to shorter, intermediate, and longer hours.[44]

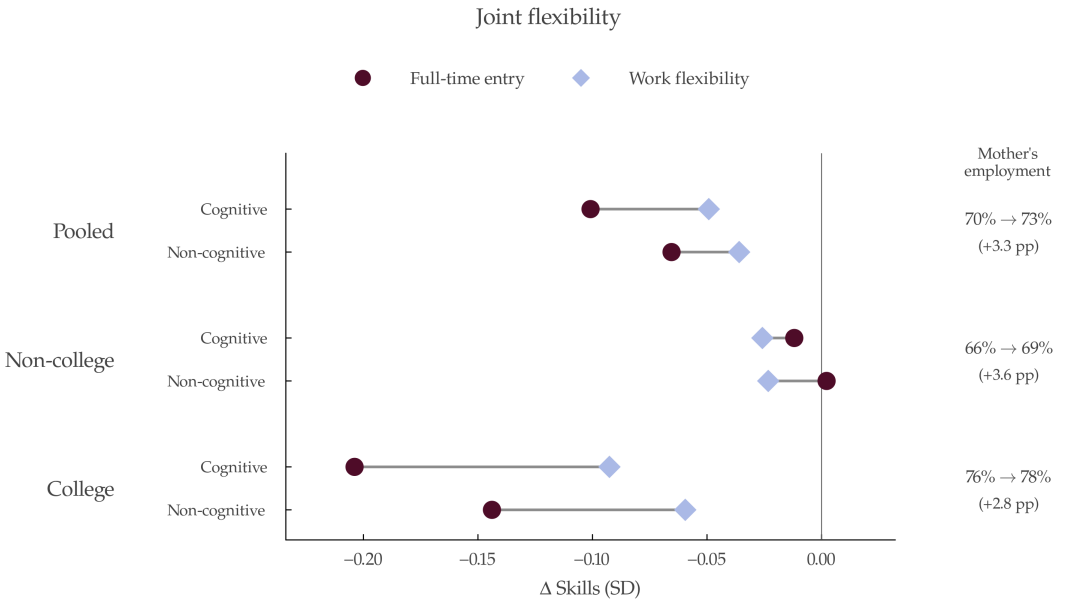
Figure 7 shows that child cognitive skills fall on average when a mother enters full-time employment and uses childcare, more so for children of college-educated mothers. Full-time entry lowers the cognitive and non-cognitive skills of a college-educated mother's child by 0.20 and 0.14 SD but leaves those of a non-college mother's child essentially unchanged, so the pooled effect masks the heterogeneity across maternal education.

Flexibility, however, draws a different group into work: mothers who enter below full-time. Table K.29 shows that self-employment allows mothers to preserve their time more than full-time paid employment. Because these mothers enter at lower hours, their children bear a smaller cost, which shrinks as entry hours fall. The difference is compositional: It reflects who enters and at

[43]The Right to Request Flexible Working overlaps my sample period. About a fifth of paid employed mothers of young children report flexitime after 2003, a limited form of flexibility that shifts when they work around fixed core hours but leaves the number of hours fixed. The model prices control over the quantity of hours, which governs time with the child and which self-employment's near-continuous menu provides but flexitime does not. A difference-in-differences regression around the 2003 reform finds no effect on flexitime take-up or maternal labor supply, so it does not confound the baseline. Appendix F.5 presents the estimates, and external evidence on the 2014 expansion finds that the response emerges as reduced hours among women, not men (Xue et al., 2025).

[44]The choice of expanded hours is explained in Appendix F.4. The counterfactual abstracts from demand-side frictions by assuming workers can freely choose among work arrangements conditional on the available hours menu. This free-choice assumption biases the estimated response in two offsetting ways. On one side, if some workers are self-employed because they are shut out of paid employment rather than because they prefer it, the model reads their choice as attachment to self-employment; it then overstates that attachment and understates how many would move into paid employment once flexibility is available, biasing the response down. On the other side, the frictionless setup ignores the employer costs, refusals, and demand-side frictions that would limit access to flexible hours in practice, so it overstates take-up and biases the response up. Thus, the net sign is ambiguous.

what hours, not a lower cost for any given full-time mother. A full-time mother does not cut her hours when offered the menu. Flexibility's margin is who works, not the inputs of any given child, and it raises maternal employment by about 4 percentage points. Averaged over all households, the pooled effect on children is statistically indistinguishable from zero, though a small cost to children of college-educated mothers remains.



Note: This figure plots the effect on a child's cognitive and non-cognitive skill at age seven of the mother entering paid work during the child's ages one to five, relative to remaining out of work, by maternal education. Circles are full-time entry, the treatment-on-treated effect for the mothers who enter full-time in the baseline; diamonds are work flexibility, the treatment-on-treated effect for the mothers the wider menu draws into work, at the intermediate hours they choose. The right column reports the rise in the maternal employment rate.

Figure 7: Effects of Flexibility on Child Skills

Behind the split is a quantity-quality trade-off between the mother's market hours and her child's skills. It binds only for college mothers. Because a college mother's time is her child's most productive input and formal care is an inferior substitute, raising her market work hours draws down input quality that the market cannot replace. For non-college mothers it does not bind, since formal care substitutes well for the time they withdraw, so their entry raises household income at no quality cost. The flexible menu relaxes this trade-off for the mothers it draws into work: Instead of facing the part-time/full-time choice, they add market hours while their home production stays relatively inelastic.

Child skills barely move because mothers protect their time with children, absorbing the added work out of leisure. Under joint flexibility, mothers' work arrangement shifts 4.7 percentage points into paid employment and 3.4 points out of non-work. Among those who work, a mother raises her market hours by about 1.2 while cutting home production by only 0.2, with the father's time unchanged. The small residual loss concentrates on the mothers who instead use flexibility to lengthen their hours. They are the lower-wage earners within each education group who work longer to raise income, and among them the loss falls on children of college-educated mothers

(Figure K.20). On average the effect on children is null: The pooled cognitive change is statistically indistinguishable from zero. Even for children of college-educated mothers, the cost is small in absolute terms – 0.022 SD averaged over college households, about a ninth of the 0.20 SD such a child loses from a mother's full-time entry – because flexibility raises employment without pushing mothers to take full-time hours.[45]

Flexibility draws parents into paid employment, and employment responds through whichever spouse has flexibility. Directing it to the husband moves him from self-employment into paid employment, trading self-employment's flexible hours for comparable hours at lower income volatility while her hours barely change. The patterns align with Lachowska et al. (2026), which finds workers, especially men, value longer hours. Directing it to the wife raises her participation the most, by 6.8 percentage points. Granting it to both draws both into work (Figure K.19). Given a wider menu, parents work somewhat more and vary their hours more across periods. Table K.28 shows who moves and how. Most of the reallocation is full-time mothers moving into intermediate hours. About two-thirds to four-fifths of baseline full-timers cut their market hours by nine to twelve a week, gaining home time and using less full-time formal childcare, so they protect their child's time directly rather than through care. Not working mothers who enter do so at low hours, where the effect is bigger for college mothers. Since they are more often full-time at baseline, flexibility reaches their children most. It also reflects gendered sorting on hours: Women concentrate in lower-hours positions (Gallen, Lesner, and Vejlin, 2019) and report more satisfaction than men with fewer hours (Kahn and Lang, 1995).

6.2 What Is the Role of Childcare?

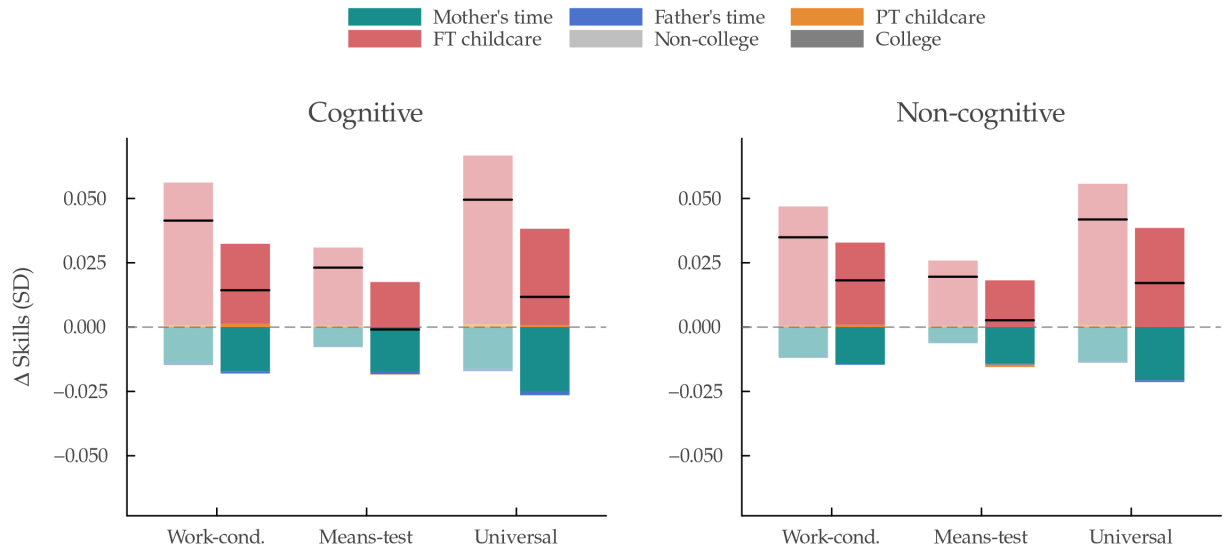
Flexibility affects the time allocation and labor supply, while childcare affects the child more directly through the formal care it finances. Its effect again depends on whose time the care replaces. It substitutes well for a non-college mother's time but poorly for a college-educated mother's, so the same subsidy raises the skills of a non-college mother's child but does little for a college-educated mother's. The design question is therefore not whether to subsidize childcare but whom to subsidize, which the eligibility rule decides.

I evaluate three rules that mirror the UK's programs. (i) *Work-conditional* childcare is free for working families and withdrawn once household income exceeds £100,000, mirroring the UK's tax-free and 30-hours-free childcare. (ii) *Means-tested* childcare is free for households in the bottom third of the net income distribution, mirroring the childcare element of the Working Tax Credit and Universal Credit and the free entitlement for disadvantaged children. (iii) *Universal* childcare is free for all families, mirroring the universal free early-education entitlement.[46] Figure 8 decomposes the change in cognitive and non-cognitive skills at age seven, relative to the baseline, into the four

[45] A cluster bootstrap of the production function confirms this. Under joint flexibility the pooled cognitive loss of 0.015 SD is not distinguishable from zero (95% confidence interval $[-0.041, +0.024]$). Neither the college loss (0.022 SD, $[-0.052, +0.015]$) nor the non-college change ($[-0.035, +0.029]$) is statistically distinguishable from zero.

[46] I focus on free formal childcare, which is in current debate in the UK. Other subsidies, such as lowering the price, yield similar results.

child production function input channels by a Shapley decomposition, and Table K.30 reports the underlying input changes and numerical detail.[47]



Note: Stacked-bar Shapley decomposition of the change in cognitive (left) and non-cognitive (right) skill at age seven, relative to the baseline, into the four child production function input channels, split by maternal education. Within each policy the left bar (lighter) is non-college and the right bar (solid) is college; the black tick marks the net change. Free formal childcare is divided by its income rule: work-conditional (free for working families with household income below £100,000), means-tested (free for the bottom third of net income), or universal (free for all). See Table K.30 for numerical detail.

Figure 8: Skill Effect Decomposition of Free Formal Childcare by Maternal Education

Childcare subsidies raise the skills of children of non-college mothers. Free care relaxes the household budget constraint. When eligibility is conditional on work, it draws mothers modestly into paid work and lowers their home production, while part-time childcare barely responds. The subsidy operates through full-time care. Universal provision raises the cognitive skill of a non-college mother's child by 0.05 SD and non-cognitive by 0.04 SD, but raises those of a college-educated mother's child by only about 0.01 to 0.02 SD, so the skill gains concentrate among non-college children and narrow the gap in child skills across maternal education. Universal provision therefore spends on college children for almost no skill return. The father's home production does not respond, because his productivity in child skills is low and formal care supplies the supervision the mother's added work requires. Take-up of free full-time care is low and concentrated among mothers who already work: it shifts the kind of care a working mother uses rather than drawing non-working mothers into the labor force. Childcare is therefore a child-skills instrument, not an employment lever, unlike flexibility.

The childcare literature confirms this education gradient. UK evidence shows the gains for

[47] For each subset S of the four inputs, I re-simulate the household's skill trajectory with inputs in S at counterfactual values and the rest at baseline, and input k's Shapley contribution is the weighted average of its marginal effect across subsets that exclude k.

disadvantaged children, most sharply in Sure Start, whose GCSE benefits were six times larger for children eligible for free school meals (Carneiro, Cattan, and Ridpath, 2024). Gains from the part-time entitlement are smaller and fade (Blanden et al., 2016). The losses for advantaged children appear under full-day provision, as in Norway, Quebec, and Italy (Havnes and Mogstad, 2015; Baker, Gruber, and Milligan, 2008; Fort, Ichino, and Zanella, 2020). The same universal provision carries a large maternal payoff, which is where the trade-off is critical. In Quebec, the program raised mothers' employment for two decades after the children left care, and their long-run earnings by twice the participation effect (Baker, Gruber, and Milligan, 2026). In my model the effect on advantaged children is close to zero rather than negative, so universal childcare buys maternal earnings while delivering its child-skill gains almost entirely to non-college children. Self-employment is the other side of this trade-off because mothers substitute their own flexible time for formal childcare rather than purchasing it.

6.3 Family-Friendly Flexibility and Targeted Childcare

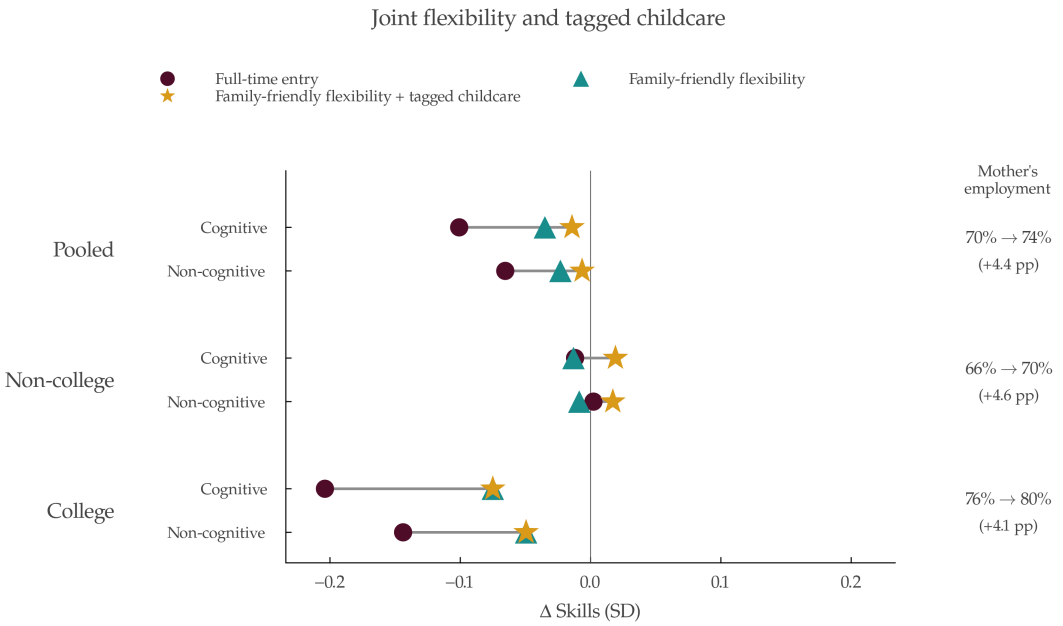
Evaluated without targeting, neither instrument reaches every child efficiently. Flexibility is close to child-neutral, with a small cost to children of college mothers, while universal childcare raises non-college children's skills but does little for college children. I now design each instrument to boost child development. These policies have important implications for intergenerational returns. Because they differ by maternal education, the design also determines whether a policy narrows or widens the skill gap children would face.

Specifically, I combine two instruments: (i) *Family-friendly Flexibility* offers a reduced-hours menu, closer to the shorter-hours policies now debated in the UK, such as the four-day week, than to the full work-flexibility menu. (ii) *Education-tagged childcare* provides free care to non-college households only. Conditioning free childcare on maternal education is a form of tagging (Akerlof, 1978). The transfer, tied to a fixed observable trait that correlates with its return, is more efficient than income testing (Nichols and Zeckhauser, 1982; Mankiw and Weinzierl, 2010). Education is the better tag here because it is time invariant, so a mother cannot game it by entering or exiting work as she could an income test. It also captures both household income and which children gain from formal care.[48]

Figure 9 builds on Figure 7 and shows that family-friendly flexibility, a reduced-hours menu, avoids the college quantity-quality trade-off while drawing more mothers into work.[49] It raises maternal employment by about 4-5 percentage points, from 69 to 74%, drawing mothers into the intermediate hours they would have preferred (Booth and van Ours, 2013), and by keeping those

[48]The estimation sample is limited to two-parent households, so single mothers, whom childcare subsidies most directly target, are out of sample. Because the tag conditions on maternal education, it would still reach them, as single mothers are disproportionately non-college-educated. They also resemble the low-income households in the model, who are more budget-constrained and use little formal care (Blundell and Shephard, 2012), so free provision would raise their take-up. The estimated childcare gain is therefore a lower bound on their children's gain since it is the low-care households whose children gain most from a shift into formal care.

[49]Effects are similar when flexibility is granted either only to the wife or the husband, but more so for wives (Figure K.21).



Note: This figure plots the effect on a child's cognitive and non-cognitive skill at age seven of the mother entering paid work during the child's ages one to five, relative to remaining out of work, by maternal education. Circles are full-time entry, triangles family-friendly flexibility entry, and stars the package of family-friendly flexibility with tagged childcare. The right column reports the rise in the maternal employment rate.

Figure 9: Effects of Family-Friendly Flexibility and Tagged Childcare on Child Skills

hours short it shrinks the full-time entry cost to college children's cognitive skill from about 0.20 to 0.07 SD. On its own, though, family-friendly flexibility does little for children of non-college mothers. It draws mothers into reduced hours and away from full-time formal care, so they gain less than the children of full-time entrants. Bundling it with tagged childcare restores that productive input and turns the non-college children's skill change positive. The bundled policy makes both education groups better off: family-friendly flexibility protects the maternal time of children of college-educated mothers, and tagged childcare supplies the formal care of children of non-college mothers.

Flexibility raises maternal employment, narrowing the gap in participation, but the mothers it draws in work shorter hours, so it does not close the within-couple earnings gap conditional on working. This matches evidence that flexible work and family policy raise women's attachment but do little for gender convergence in pay (Cook et al., 2021; Kleven et al., 2024; Olivetti and Petrongolo, 2017; Blau et al., 2026).

6.4 Willingness to Pay for Flexibility and Childcare Policies

What is the household's willingness to pay for the policies? Table 8 decomposes annual household welfare through consumption equivalent variation (Appendix L) into a *parental* component, covering consumption, leisure, work-arrangement amenities, and childcare costs, and a *child-skill* component, the consumption equivalent variation to the household for child skill returns. This

allows me to see how much of each policy's value accrues to parents and how much to the child, and how that split varies by maternal education.

The policy costs differ by instrument. The childcare subsidy has a direct fiscal cost because the government provides the subsidized expenditure on formal childcare.[50] In contrast, flexibility has no direct fiscal cost in the model since employer obligations are not modeled. I treat zero as an upper bound on net welfare rather than a realistic estimate.[51] To provide realistic figures, I calibrate a cost for flexibility from the literature. Workers value flexibility, requiring a 12% higher wage to forgo ideal hours (Lachowska et al., 2026) and forgoing about 20% of wages to avoid employer-discretionary scheduling (Mas and Pallais, 2017). In a competitive labor market, employers must therefore bear some cost to provide it, or firms would offer full flexibility and cut wages (Flabbi and Moro, 2012). To bound net welfare, I scale wages by the 12% productivity loss that Emanuel and Harrington (2024) estimate for flexible work. I use this employer-side estimate rather than the worker-side valuations above. Because the model is a partial equilibrium, the relevant object is the social cost of flexibility, so wages equal the marginal product of labor and that cost is the employer's productivity loss. Two caveats make the welfare figures conservative. The Emanuel and Harrington (2024) estimate is measured for remote work rather than for the hours flexibility, so I treat the 12% as an upper bound on the cost rather than a precise match. And the worker-side valuations are revealed-preference lower bounds on the social value of flexibility. They capture what parents forgo for schedule control but not the child-development return that flexibility also delivers, nor any value firms derive from offering it. The net welfare estimates are therefore conservative on both sides.

These welfare values average over all households with young children. This contrasts with the entry effects in Figure 9, which condition on mothers newly drawn into work. The split shows who captures each policy's welfare effects and how much they would pay for it. For college households, most parents already work. Family-friendly flexibility therefore raises welfare only slightly through child skills, as incumbent mothers use reduced hours to keep their home production time nearly intact. For non-college households, flexibility is a parental gain, raising parental welfare by about £4,200 a year with essentially no gain from child skills. Tagged childcare is its mirror image. Parents' welfare falls by about £400, as they work more and bear the childcare disutility, offset by child-skill gains of about £500, at a fiscal cost near £400. The combined package delivers both. For college households, welfare runs entirely through flexibility, with a large parental gain of about £5,400.

[50] This direct cost overstates the net fiscal burden, since higher maternal employment returns part of it. Baker, Gruber, and Milligan (2026) estimate that the long-run rise in mothers' tax payments and fall in benefit receipt recovers 75 to 117% of the upfront cost of Quebec's universal childcare program.

[51] The model prices flexibility from the self-employment income process, which is imperfectly measured: Self-employment income is bottom-coded at zero in the data, its estimated dispersion is large, and the below-one wife hours-earnings curvature shapes the hours response. I therefore probe sensitivity to the self-employment process rather than bound bottom-coding directly; I re-solve for both spouses under lower dispersion (standard deviations of 2.0 and 1.5) and flatter curvature (0.85 and linear). The self-employment share rises across these cases, but the flexibility results hold. The maternal employment gain stays positive throughout, about 1.8 to 2.5 percentage points for college mothers and 2.9 to 3.5 for non-college mothers. Mothers still enter at intermediate hours and the value of flexibility stays positive, within about 15% of its baseline.

Table 8: Policy Welfare by Maternal Education

Policy	Welfare (£/hh/yr)			Cost (£)	Δ Parental (£)	Δ Child Skills (£)
	Total	Parental	Child Skills			
Non-college mother						
Family-friendly flexibility	+4204	+4232	−28	[−75, 0]	[4157, 4232]	[−103, −28]
Tagged childcare	+112	−431	+543	−379	−810	+164
Combined	+4334	+3755	+579	[−455, −383]	[3300, 3372]	[124, 196]
College mother						
Family-friendly flexibility	+5445	+5399	+46	[−121, 0]	[5278, 5399]	[−75, 46]

Note: Population-average willingness to pay in consumption-equivalent variation over all young-child households, by the wife's education. Parental and Child Skills sum to Total. Family-friendly flexibility grants a reduced-hours menu to both spouses; tagged childcare provides free formal childcare to non-college households only; combined applies both. College households are excluded from tagged childcare, so for them tagged childcare has no effect and the combined package coincides with family-friendly flexibility, the single college row shown. Cost is the employer productivity loss at 12% of affected wages for flexibility (range [−12%, 0]) following Emanuel and Harrington (2024), or the government childcare expenditure per household-year with a young child for that education group. Δ Parental = Parental − Cost; Δ Child Skills = Child Skills − Cost.

Flexibility is worth more to college households, while non-college households gain most of their welfare through child skills.

Table 9: The Value of Policy by Maternal Education

Policy	Total	Channel (% of lifetime consumption)			
	(% cons.)	Consumption	Wife's time	Husband's time	Child skills
Non-college mother					
Family-friendly flexibility	+12.21	+15.42	−2.60	−0.53	−0.08
Tagged childcare	+0.35	+0.68	−2.09	+0.08	+1.69
Combined	+12.62	+16.11	−4.86	−0.44	+1.81
College mother					
Family-friendly flexibility	+12.29	+14.49	−0.99	−1.33	+0.12

Note: The value of each policy in consumption-equivalent variation, the percent of lifetime consumption a household would give up to have it, by the wife's education. The channel columns decompose the total by the utility margin it flows through: consumption (a household public good), the wife's and husband's time (leisure and home production), and child skills; the two time channels are spouse-specific, consumption and child are shared. Because the household is unitary the split is by margin, not by spouse. Means over households with at least one young-child period. College households are excluded from tagged childcare, so the combined package coincides with work flexibility for them, the single college row shown.

Table 9 shows the value of family-friendly flexibility and childcare, in percent of lifetime consumption, decomposed by the utility channel and maternal education. Because household utility is unitary, I separate parental from child welfare but cannot divide the parental gain between the two spouses. Consumption is a household public good, so only leisure and home production are spouse-specific. Both education groups value flexibility, but for different reasons. Non-college households value it almost entirely through consumption, financed from the income from entering work at intermediate hours. Little comes from the child-skill or time channels. College households value it mainly through consumption, as the reduced-hours menu lets the mother work and earn

income, with a small positive child-skill channel from the maternal time the menu preserves. When the policies are combined, non-college families receive a child-skill gain from childcare on top of the income from work. College families receive an income gain from flexibility.

6.5 Returns to Child Skills and Lifetime Earnings

The welfare values above measure what parents would pay for their child's skills, but parents internalize only part of what those skills are worth. A gain that is modest to the household can be large for the child who carries it into adulthood. To capture that return, I project each policy's skill change onto the child's expected lifetime earnings, whose payoff accrues to the next generation. This exercise is an earnings projection, distinct from the willingness-to-pay values.

I value a standard deviation of skill through its effect on lifetime earnings, following the approach of Chetty et al. (2011a) and the estimates of Mullins (2025), who finds that a one standard deviation increase raises lifetime earnings by 32% for cognitive skills and 18% for non-cognitive skills.[52] I apply this to the UK context, using a UK expected-lifetime-earnings present value of about £590,600.[53] A standard deviation of cognitive skill is then worth roughly £189,000 and of non-cognitive skill about £106,300. UK evidence confirms that both translate into adult earnings (Carneiro, Crawford, and Goodman, 2007; Hanushek et al., 2015), so these figures are an upper benchmark. The combined total sums the cognitive and non-cognitive projections and thus treats their returns as additive. It does not net out any component the two returns share, so it is an additive upper bound rather than a joint estimate.

Table 10 translates each policy's population skill change into the present value of a child's lifetime earnings. I further split the results by maternal education and by the cognitive and non-cognitive channels. For each policy, education group, and skill domain, I take the mean change in the child's skill at age seven induced by the policy. Since it is already in standard-deviation units, I multiply it by the corresponding per-standard-deviation value (£189,000 for cognitive and £106,300 for non-cognitive skill). The total shows the sum of the two channels, and the final column shows the total lifetime earnings gain as a share of the £590,600 present value. For children of non-college mothers, tagged childcare is by far the most potent lever, raising expected lifetime earnings by up to £13,900 (2.3%). The earnings gain from family-friendly flexibility is near zero on its own, so the combined package raises expected lifetime earnings by £14,700 (2.5%), almost all of it from childcare. For children of college-educated mothers, family-friendly flexibility yields a small positive gain of about £2,200 (0.4%). Flexibility's gains run through the cognitive channel, via

[52] Chetty et al. (2011a) report that a one-percentile-rank gain in kindergarten test scores is associated with a $132 rise in age-27 earnings in the raw data, and $94 after controlling for parental characteristics. Their Project STAR randomization identifies classroom-quality effects, with a class one standard deviation higher in quality raising age-27 earnings by about 3%. Scaling the raw gradient by the roughly 34 percentile ranks in a standard deviation gives a cross-sectional cognitive return of about $4,500 per SD annually. Mullins (2025) estimates both the cognitive and non-cognitive return from PSID-CDS earnings in young adulthood.

[53] Median full-time earnings of £35,000 over 45 working years, discounted at 3.5%, the UK Treasury Green Book's social discount rate (HM Treasury, 2022), and scaled by the 75% UK employment rate (Office for National Statistics, 2024).

Table 10: Monetary Returns to Policy-Induced Child Skill Changes by Maternal Education

	Δ Skills (SD)		Lifetime earnings (£)			
Policy	Cog	Ncog	Cog	Ncog	Total	% of PV
Non-college mother						
Family-friendly flexibility	+0.00	−0.00	+100	+0	+0	+0.0
Tagged childcare	+0.05	+0.04	+9,400	+4,500	+13,900	+2.3
Combined	+0.05	+0.04	+9,900	+4,700	+14,700	+2.5
College mother						
Family-friendly flexibility	+0.01	+0.01	+1,500	+700	+2,200	+0.4

Note: Present value of the change in a child's expected lifetime earnings by the wife's education, from the population-average skill change over all young-child households, decomposed into cognitive and non-cognitive channels. The cognitive (non-cognitive) channel applies a 32% (18%) earnings return per standard deviation (Chetty et al., 2011a; Mullins, 2025) to a UK expected-lifetime-earnings present value of £590,600, about £189,000 (£106,300) per SD; the final column expresses the total as a share of that present value. Skill changes are mean changes in log skills at age 7 within each education group. College households are excluded from tagged childcare, so the combined package coincides with work flexibility for them, the single college row shown.

the mother's time, while childcare's run through both.

7 Conclusion

As flexible work and free childcare expand, how a mother's employment reaches her child has become a central question for family and labor policy. Existing work traces the effects of flexibility and childcare on women's labor supply, earnings, and welfare. This paper is the first to quantify the child-development consequences of workplace flexibility and to weigh it against childcare within a single framework. Using the margin between paid employment and self-employment, I decompose a work arrangement into three attributes: flexibility, earnings level, and earnings volatility. A structural life-cycle model then traces how that choice reaches the child and evaluates work flexibility and childcare policies.

Beneath the policy results lies the household's division of labor. The mother specializes in the child's time because her comparative advantage is in producing child skills, not in earnings, so her work arrangement, not the father's, is the margin that adjusts. Granting flexibility to paid employed workers without the pay penalty has small effects on children on average, which mask sharp heterogeneity by maternal education. Full-time employment of a college-educated mother is associated with child skill loss, which flexibility alleviates. For a non-college mother full-time employment has little effect on the child, and flexibility leaves child skills unchanged. The difference reflects a quantity-quality trade-off that binds only for college mothers. Because her time is her child's most productive input and the market does not offer a good substitute for it, adding market hours detracts from child quality. Flexibility eases the trade-off by letting her enter the labor market while keeping her home-production time nearly intact. For its part, formal childcare substitutes well for the time a non-college mother withdraws, so she can enter work at no

cost to her child. Which policy reaches a child thus depends on which input is scarce.

In welfare terms, flexibility delivers larger parental gains than childcare because parents value flexibility. These results speak directly to policy debate. The Employment Relations (Flexible Working) Act 2023 made the right to request flexible working a day-one right for all UK employees. My findings imply that expanding effective access to flexible hours raises maternal employment and household welfare substantially. To reach children, the instruments must be targeted by education: Family-friendly flexibility protects the time of children of college-educated mothers, whose mothers choose to enter below full-time. Childcare tagged by maternal education delivers the non-college gain while concentrating public funds on the children who gain, since college children gain little from universal care.

Two natural extensions would further sharpen the policy analysis. The first is including single mothers, the group childcare subsidies most directly target. For a single mother the time-and-money trade-off is most acute, so the ranking of flexibility against childcare may provide different policy implications. The second endogenizes the supply of flexibility. I treat flexible hours as freely available at a calibrated employer cost. Letting firms choose how much to offer, and at what wage, would determine who bears that cost and turn the partial-equilibrium welfare estimates into general-equilibrium ones.

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Appendix A Data and Variable Construction

Appendix A.1 British Household Panel Survey

Household Identifier: BHPS individuals are identified by an individual specific identifier across all waves. Individuals are also identified by the person number (*pno*) in household (*hidp*). Each wave, individuals have a unique combination of *pno* and *hidp*; however, household identifiers cannot be linked across waves. Since household structures can change over time (e.g. couples get divorced, the child ($age < 16$) at wave t becomes independent and forms his or her own household at wave $t + 1$), BHPS does not include a longitudinal household identifier.[54]

Table A.1: BHPS Sample Selection Criteria

Criteria	Obs	Ind	HH
Original Sample	329,993	43,206	
Keep adults (age > 15)	257,502	35,570	
Keep singles or couples	195,043	24,885	
Drop couples after divorce	183,048	24,846	14,075
Keep married couples, stable family structure	174,961	24,846	14,075
Keep wife age 21–51	91,366	14,923	7,586
Drop if missing education	86,666	13,545	7,463
Drop if missing labor market characteristics	75,517	13,069	7,305
Keep working husbands	57,980	10,809	5,717
Keep households with natural children (max 3)	55,101	10,325	5,461
Drop early or late pregnancies	54,048	10,085	5,330
Keep both couples observed	49,958	9,619	4,871

Since BHPS does not provide a longitudinal household identifier, I construct one from the cross-sectional person identifiers. I restrict to reference persons and their partners, extract the woman's individual identifier (*pidp_women*) for each household-wave, and assign a sequential *hh_id* using *pidp_women* as the grouping key. This identifier is unique across couples and stable across waves as long as the woman remains in the sample. I keep only households whose children are all biological, and drop a household entirely if it ever becomes a blended or step-family.

Marital status: Marital status takes value 1 for married and cohabiting couples and 0 otherwise. I retain each household until the first observed dissolution event and drop all subsequent waves, so each retained observation comes from the stable two-parent spell before any separation. Among households that dissolve, the child is on average 6 years old at the time of separation.

Employment status: Individuals are considered working if they did paid work last week and have work. For wives, I distinguish four work arrangements: paid employed part-time (PE/PT, $\leq$ 30 hours per week), paid employed full-time (PE/FT, $>$ 30 hours per week), self-employed, and not working. The 30-hour threshold follows Manning and Petrongolo (2008). For husbands, I distinguish paid employed and self-employed; the sample conditions on husbands being always

[54] More details are described in the Understanding Society: website.

working.

Self-employment incorporation status: BHPS asks self-employed individuals whether they hire employees and the nature of their employment. I define *incorporated* as those who own a business or practice and have employees and *unincorporated* as those who own a business or practice but do not have employees or who work for others, both, or other form of self-employment. This employee-based split identifies employer versus own-account self-employment, which proxies for legal incorporation.

Educational attainment: The highest educational qualification is reported for each individual. I construct a binary indicator for college degree, which takes value 1 if the individual holds a degree or higher. Those with A-levels, GCSEs, other qualifications, or no formal education are assigned a value of 0. I construct a categorical variable *educ_level*: low for GCSE or lower, medium for A-level, and high for degree or other higher degree.

Income: BHPS asks the usual/non-usual payments and the usual/non-usual weeks of payments. I first replace non-usual payments with usual payments and compute weekly payment by dividing the gross-pay by the number of weeks received. For hourly payment, I use the number of hours worked. Self-employed individuals do not report the number of weeks so I use information on their monthly self-employed gross pay and profit. For all working individuals, I compute the weekly gross pay. The timing of occupation and income variables is described in Table A.2. I deflate earnings and wages with the CPI (2001=100).

Assets: BHPS asks households questions on home ownership and home values every year. If adequate, they are asked about whether or not they have a mortgage and the corresponding amount. Asset information is collected every 5 years (wave 5, 10, 15), where individuals are asked in greater detail on financial issues. Assets are categorized into savings, investments, and debt. Savings include both regular and non-regular savings. Investment includes money from national savings certificates, premium bonds, unit trusts, personal equity plans, shares in the UK or foreign countries, NS/BS insurance bonds, and other investments or securities. Debt includes owing money from hire purchase, personal loan, credit cards, mail order purchase, DSS social fund loan, individual loans, and other. Further details on the survey instruments are described in Table A.3. I measure savings, investments, and debt as point-in-time stocks at the wealth waves (5, 10, and 15), and construct net financial assets $=$ savings $+$ investment $-$ debt, the model's asset state. Net home value is the home value (for homeowners) minus the outstanding mortgage balance. Home ownership in the sample is 80% for couples with children and 83% for couples without, consistent with Crossley and O'Dea (2010) (Table 3.7). The share owning outright is low, as households with dependent children are less likely to have paid off a mortgage. For missing home ownership between waves, I impute to the closest year.

Table A.2: Timing of Occupation and Income

Variable	Question	Timing
Work Arrangement		
Did paid work last week	"*Did you do any paid work last week that is in the seven days ending last Sunday either as an employee or self employed?*"	*t*
Employee or self-employed: current job	"*Are you an employee or self-employed?*"	*t*
No. of hours normally worked per week	"*Thinking about your (main) job, how many hours, excluding overtime and meal breaks, are you expected to work in a normal week?*"	*t*
Occupation (SOC): current main job	"*What was your (main) job last week? Please tell me the exact job title and describe fully the sort of work you do.*"	*t*
Income		
Gross pay at last payment	"*The last time you were paid, what was your gross pay that is including any overtime, bonuses, commission, tips or tax refund, but before any deductions for tax, national insurance or pension contributions, union dues and so on?*"	
Pay period (weeks): last gross pay	"*How long a period did that (pay) cover?*"	
Was last gross pay the usual amount	"*Your gross pay last time was (amount at E23a). Is this the amount you usually receive (before any statutory sick pay or statutory maternity pay)?*"	
Usual pay	"*How much are you usually paid?*"	
Usual pay: pay period (weeks)	"*What period does that (pay) cover?*"	
Total gross earnings in last 12 months	"*In all, before tax and other deductions, how much have you earned in the past 12 months?*"	
Self-Employment		
S/emp: hours normally worked per week	"*How many hours in total do you usually work a week in your job?*"	*t*
Monthly self employed profit	"*Computes a monthly self-employed profit variable for self-employed respondents who draw up profit and loss accounts. Inapplicable for those who are not self-employed, and the self employed who do not draw up accounts. (cf. AJSPAYG). Includes imputed data.*"	
Monthly self employed gross pay	"*Computes a monthly self-employed gross pay variable if a self-employed respondent does not draw up profit and loss accounts. Inapplicable for those who are not self employed, and the self employed who draw up accounts. (cf. AJSPROF). Includes imputed data.*"	

Table A.3: Asset Variables

Variable	**Question**
Amount in account(s) – regular saving	"*And about how much do you currently have in your savings account(s)? Please do not include money you have in share purchase schemes or Personal Equity Plans (PEP) schemes. ... In the last 12 months.*"
Amount in account(s) – non-reg saving	"*About how much do you currently hold in this/these account(s)? ... In the last 12 months.*"
Amount held in investments	"*Thinking of all your investments, about how much do you have invested in total? ... In the last 12 months.*"
	"*Which ones? National Savings Certificate*"
	"*Which ones? Premium Bonds*"
	"*Which ones? Unit trusts*"
	"*Which ones? Personal Equity Plans*"
	"*Which ones? Shares (UK or foreign*"
	"*Which ones? National Savings/Building Society/Insurance Bonds*"
	"*Which ones? Other investments, government or company securities*"
Currently owe money	"*I would like to ask you now about any other financial commitments you may have apart from mortgages and housing related loans. Do you currently owe any money on the things listed on this card (43)? ... In the last 12 months.*"
House owned or rented	"*Does your household own or rent this accommodation or does it come rent-free?*"
Amount borrowed at purchase: mortgagee	"*How much did you borrow originally when you bought the property, that is excluding any later additions to the mortgage?*"
Amount still owed on mortgage	"*About how much do you still owe on the mortgage(s) or loan(s) on this property?*"
Last total monthly mortgage payment	"*How much was your total last monthly instalment on the morgage(s) or loan(s)?*"
Value of property: home owners	"*About how much would you expect to get for your home if you sold it today?*"

Table A.4: Summary Statistics by Young Child: BHPS

		No Young Child	Young Child
Wife	Age	39.13	32.76
		(8.218)	(5.065)
	Annual pay	11044.7	7176.8
		(9199.1)	(8486.2)
	College degree	0.253	0.308
		(0.435)	(0.462)
	Work arrangement		
	Paid Employed/PT	31.81%	42.04%
	Hours worked	19.58	17.81
		(6.750)	(6.573)
	Home production	19.37	18.90
		(9.930)	(10.724)
	Paid Employed/FT	48.30%	19.34%
	Hours worked	37.61	37.42
		(3.850)	(3.698)
	Home production	11.49	12.63
		(7.113)	(7.556)
	Self-Employed	6.13%	4.99%
	Hours worked	34.03	25.69
		(17.797)	(16.676)
	Home production	16.25	18.57
		(10.320)	(13.013)
	Not Working	13.77%	33.64%
	Home production	23.98	25.66
		(13.375)	(13.789)
Husband	Age	41.49	35.01
		(8.858)	(5.601)
	Annual pay	22360.6	23469.8
		(13637.2)	(13722.6)
	College degree	0.259	0.303
		(0.438)	(0.460)
	Work arrangement		
	Paid Employed	81.74%	83.28%
	Hours worked	39.61	39.85
		(6.887)	(6.966)
	Home production	5.11	5.08
		(4.560)	(4.733)
	Self-Employed	18.26%	16.72%
	Hours worked	49.37	49.15
		(14.755)	(14.724)
	Home production	4.39	5.07
		(4.603)	(6.456)
Household	Net assets	11140.1	8094.2
		(43961.0)	(59131.0)
	Net home value	111678.8	110326.6
		(112006.9)	(109169.1)
	Annual household pay	33405.3	30646.6
		(17138.1)	(16528.7)
Observations		17,468	7,511

Note: Sample: married couples, BHPS waves 1–18, wife aged 21–51, husband always working. No young child = no child aged zero to five present; young child = at least one child aged zero to five. Work arrangement categories sum to 100%. Home production hours from BHPS housework questions. Net assets = savings plus investment holdings minus debt. Net home value = home market value minus mortgage. All monetary values in 2001 £. Standard deviations in parentheses.

Appendix A.2 Millennium Cohort Study

For the main analysis, I use MCS Sweeps 1–4 (9 months to age 7). In Sweep 1, 18,552 children were interviewed across England, Wales, Scotland, and Northern Ireland; the sample is nationally representative. In Sweep 2, an additional 1,389 families were added for whom addresses were reached too late for Sweep 1; I do not include them. I restrict to singleton births with both parents present, and keep children observed in at least two consecutive sweeps so that skill transitions are identified. Table A.5 describes the sample selection steps, yielding 5,219 children and 28,802 parent-wave observations.

Table A.5: MCS Sample Selection Criteria

	All		Wave 1		Wave 2		Wave 3		Wave 4	
	Obs	Child	Obs	Child	Obs	Child	Obs	Child	Obs	Child
Original sample (Sweeps 1–4)	116,179	19,201	34,044	18,388	28,844	15,579	28,014	15,236	25,277	13,847
Singleton birth only	104,816	18,951	31,080	18,147	25,654	15,381	25,363	15,041	22,719	13,681
Drop if added after 2001	101,658	18,147	31,080	18,147	24,445	14,595	24,314	14,369	21,819	13,097
Both natural parents in household	86,972	15,684	27,848	14,964	21,280	11,705	20,270	10,852	17,574	9,499
Mother's age at birth 21–45	81,449	14,377	25,740	13,802	19,904	10,940	19,137	10,236	16,668	8,997
Drop if missing child development outcomes	76,773	14,125	24,550	13,117	17,921	9,770	18,496	9,858	15,806	8,476
Drop if missing child characteristics	75,931	13,936	24,281	12,966	17,749	9,673	18,294	9,746	15,607	8,367
Drop if missing parent labor characteristics	61,481	13,207	19,238	10,491	11,869	6,623	16,338	8,799	14,036	7,603
Drop if missing parent demographics	60,817	13,081	19,221	10,481	11,682	6,528	15,933	8,592	13,981	7,574
Drop if missing childcare	60,199	13,071	19,220	10,480	11,587	6,474	15,427	8,357	13,965	7,558
Drop if father not working	54,382	12,299	17,145	9,441	10,414	5,887	14,054	7,660	12,769	6,954
Drop households with half sibling	45,455	10,287	14,701	8,039	8,836	4,930	11,592	6,279	10,326	5,576
Keep if 2 consecutive waves observed	32,599	6,373	6,654	3,814	8,004	4,526	9,735	5,347	8,206	4,511
Keep both parents	28,802	5,219	5,680	2,840	6,956	3,478	8,776	4,388	7,390	3,695

Incorporated vs Unincorporated Status: The MCS does not include an explicit variable for incorporation status. I use whether the self-employed individual has employees as a proxy: *incorporated* for those with employees, *unincorporated* (sole trader) for those working alone or with partners but without employees. As in the BHPS, this identifies employer versus own-account status rather than legal incorporation directly.

Income: MCS includes gross and net income, the derived OECD equivalised income, and predicted family income. Hawkes, Plewis, and Verropoulou (2008) and Anyaegbu (2010) document missing income in the MCS and show within-household and within-individual correlation in non-response. To address this issue, MCS imputed missing and continuous income using interval regression with observable household characteristics such as housing tenure, parents combined labor market status and parents' combined highest educational attainment. They also derived both McClements scale (McClements, 1977) and OECD equivalence scale Hagenaars, De Vos, and Zaidi (1994) for family income (Centre for Longitudinal Studies, 2020).[55] Instead of using derived

[55]The OECD equivalised income scale is widely used in benefit analysis. According to the OECD scale, the first

variables, I calculate weekly gross pay based on the total gross pay amount and the number of periods during which it was received in the last financial year. For self-employed individuals, I only observe the net (take-home) income in the last financial year. Hourly wage is calculated by dividing weekly income by the average number of hours worked. Observations with hourly wages below the 2001 minimum wage of £3 for paid employment (Department for Business, Energy & Industrial Strategy, 2019), below £1 for self-employment, and above £150 were excluded from the analysis. All incomes are deflated to 2001 pounds.

Child Development Outcomes: Cognitive skills are measured using age-appropriate tests (Frankenburg and Dodds, 1967; Elliott, Smith, and McCulloch, 1997). At 9 months, I use the Denver Development Screening Test. At ages 3 and 5, children take the BAS Naming Vocabulary Test, in which they name objects depicted in 36 colored pictures shown by a trained interviewer. At age 7, they take the BAS Word Reading Test, in which they read aloud a series of 90 words arranged in increasing difficulty. All responses are recorded via Computer Assisted Personal Interviewing (CAPI), and I use the age-adjusted ability scores.

For non-cognitive skills, I use the Carey Infant Temperament Scale (Carey and McDevitt, 1978) at 9 months and the Total Difficulties Score from the Strengths and Difficulties Questionnaire (SDQ) at ages 3 to 7. The Carey scale measures infant temperament (the child's characteristic behavioral style and patterns of response to stimuli) and is designed for infants aged 4 to 8 months. The SDQ is a parent-completed behavioral screening questionnaire widely used in research and clinical settings (Goodman, 1997, 2001). It consists of 25 items across five components: (i) emotional symptoms, (ii) conduct problems, (iii) hyperactivity/inattention, (iv) peer relationship problems, and (v) prosocial behavior. Each item is rated on a three-point scale (not true, somewhat true, certainly true). The first four components sum to the Total Difficulties Score, where a higher score indicates more behavioral problems. Following the literature (Del Bono and Ermisch, 2009; Nicoletti and Tonei, 2020), I reverse-code the score so that a higher value reflects better socio-emotional skills.

Appendix A.3 Tax and Transfer System: Fortax

For the tax and transfer system, the FORTAX library must be first installed, which is accessible from the online library written by Shephard (2009): Github Page. From the BHPS, I first get the data on both mother and father's gross weekly income, hours worked, self-employment status, and child's age. I compute the household disposable income given the parent labor market and family characteristics.

During the sample period (2001–2006), the UK government introduced several tax reforms to support working families and reduce income inequality. In 2003, the government replaced the Working Families' Tax Credit (WFTC) with two targeted programs: the Working Tax Credit (WTC) and Child Tax Credit (CTC). While WTC provided supplemental income for low-earning

adult accounts for 0.67 weight, the spouse 0.33, dependent child between 14 and 18 years old 0.33, and the child aged under 14 years old 0.20. The baseline is set to 1 for couples without children. This measure is relatively complete in the data; however, it captures annual income at the household level rather than the individual level, which is critical in my analysis. I use this scale to cross-validate the individually computed annual income.

Table A.6: Summary Statistics by Child Age: MCS

	9 Months	Age 3	Age 5	Age 7
Child Outcomes				
Cognitive‡	9.14	78.31	112.75	115.03
	(1.56)	(15.36)	(15.07)	(26.33)
Non-Cognitive‡	57.36	31.52	33.88	33.75
	(5.92)	(4.55)	(4.16)	(4.58)
Child Characteristics				
Female	0.49	0.49	0.49	0.49
	(0.50)	(0.50)	(0.50)	(0.50)
Parity	0.48	0.23	0.13	0.09
	(0.50)	(0.42)	(0.34)	(0.29)
Birth weight (g)	3465.80	3455.03	3442.58	3439.11
	(545.38)	(551.98)	(559.04)	(551.45)
Low birth weight (<2.5kg)	0.04	0.04	0.05	0.05
	(0.20)	(0.20)	(0.21)	(0.21)
Number of siblings	0.63	0.93	1.10	1.19
	(0.67)	(0.63)	(0.60)	(0.58)
Mother's Characteristics				
Work arrangement				
Paid Employed/PT	34.7%	34.0%	46.0%	49.1%
Paid Employed/FT	12.8%	10.1%	12.6%	15.8%
Self-Employed	4.2%	6.3%	6.7%	7.5%
Not Working	48.4%	49.6%	34.6%	27.6%
Age	31.39	33.53	35.62	37.53
	(4.36)	(4.38)	(4.45)	(4.48)
Annual gross income	7590.59	7276.52	9918.98	12132.53
	(11467.02)	(12529.34)	(15083.64)	(17789.45)
College degree	0.30	0.28	0.28	0.27
	(0.46)	(0.45)	(0.45)	(0.45)
Father's Characteristics				
Work arrangement				
Paid Employed	83.7%	82.0%	81.7%	81.8%
Self-Employed	16.3%	18.0%	18.3%	18.2%
Age	33.78	35.96	37.97	39.82
	(5.17)	(5.22)	(5.16)	(5.04)
Annual gross income	32919.07	36941.51	39098.49	46853.22
	(27178.95)	(35806.84)	(40069.74)	(79865.50)
College degree	0.30	0.28	0.29	0.29
	(0.46)	(0.45)	(0.45)	(0.45)
Household Characteristics				
Equivalized weekly family income†	430.36	441.04	441.33	452.25
	(203.25)	(216.42)	(198.80)	(193.00)
N	2,840	3,478	4,388	3,695

Note: Sample: Millennium Cohort Study singleton births, both parents present, no change in family structure, mother aged 21–45. ‡ Raw test scores: Denver Development Screening Test (nine months), British Ability Scales (ages three to seven) for cognitive; Carey Infant Temperament Scale (nine months), Strength and Difficulties Questionnaire Total Difficulties (ages three to seven) for non-cognitive. † OECD equivalized income scale. All monetary values in 2001 £. Standard deviations in parentheses.

working adults, CTC offered financial support to families with children regardless of employment status. These credits aimed to alleviate child poverty and encourage workforce participation among low-income households. Given these reforms, I calculate net household income for the relevant period and adjust it to £2001 for consistency.

Appendix A.4 UK Time Use Survey

I use the UK Time Use Survey (2000–2001) (Office for National Statistics, 2003) to validate the parental time measures. The BHPS remains the primary source for home production hours because the TUS covers self-employed parents in only a single wave, providing insufficient variation by work arrangement. I use time-diary measures rather than the MCS frequency-based proxies (e.g., times per week reading to the child) because frequency conflates intensity with duration and carries higher social desirability bias (Hofferth, 2009).

The TUS covers 11,664 individuals aged 8 or over. I restrict to married households with the wife aged 21–45 and the youngest child under 9, using the youngest child's age as a proxy for the child of interest (Bolt et al., 2023). I construct two composite measures: (i) time with the child, covering physical care and supervision, teaching, reading/playing/talking, and travel escorting to or from education; and (ii) housework time, covering food preparation, housekeeping, and gardening. The resulting sample is described in Table A.7.

Table A.7: UKTUS Sample Selection Criteria

	Obs
Original sample	11,664
Households with women aged 21–45	6,123
Married couples	2,608
Youngest child under 9 years old	1,378
Completed both weekday and weekend diaries, husband working	2,080

Appendix A.5 Additional Datasets

I supplement empirical evidence from the Labour Force Survey (LFS; Office for National Statistics, 2022b), the Family Resources Survey (FRS; Department for Work and Pensions, 2005), and O*Net (O*NET Resource Center, 2019) on labor market characteristics by work arrangement. LFS is the largest household survey with measures related to the labor market and income and I use the five-quarter longitudinal version to explore the individual-level variation in working hours. Unfortunately, the Labour Force Survey does not have information on self-employment income, which is crucial in my analysis; hence, I use the annual Family Resources Survey (FRS) for more details on earnings. The sample is from 1997 to 2021/2, consisting of household heads and their spouses of ages 16-64, after full-time education and not missing labor market characteristics (e.g. employment status and hours worked). To cover all UK, I look from 1997 onwards.

To get more details by occupation codes, I construct a work flexibility measure following Goldin

(2014) and Bang (2022) and classify into Time pressure, Contact with others, Establishing and maintaining interpersonal relationships, Structured vs. unstructured work, and Freedom to make decisions. Using the O*Net dataset, I first normalize the values to mean 0 and variance 1 and get the average per 3 digit, 2 digit, and 1 digit groups. Since the UK dataset is based on UK SOC Codes which are different than the O*Net occupation codes and there were several changes in the SOC codes after the 2000s, 2010 and 2020, I describe in detail how the O*Net is mapped to the UK datasets. First, I convert UK SOC2000 to ISCO-08 following Office for National Statistics (2022c).[56] Then, I crosswalk between the International Standard Classification of Occupations (ISCO-08) and the 2010 Standard Occupational Classification (SOC).[57] Lastly, I match ONET SOC codes from 2010 to 2019 (O*NET Resource Center, 2019). The sequential mapping is intended to reduce errors in converting the codes.[58]

For the BHPS, which codes occupation in ISCO-88, I map occupations to the O*Net scores through the US SOC 2000 and 2010 classifications and form a single flexibility index that rises with autonomy and unstructured work and falls with time pressure and required interpersonal contact. Self-employed workers sort into more flexible occupations than paid employees for both genders (Table A.8). The index is 0.18 standard deviations higher for self-employed husbands than paid employed husbands, and 0.5 to 0.6 standard deviations higher for self-employed wives than part- or full-time paid employed wives. Greater freedom to make decisions and less structured work drive the gap; self-employed mothers also report more interpersonal contact, consistent with their concentration in personal services and childminding, but the autonomy dimensions dominate. This gives flexibility an observable occupational counterpart rather than treating self-employment as flexible by assumption.

[56] Authors note that "These conversions were developed for this research project, and do not constitute official guidance on classification conversion. They should be used with caution."

[57] In the report, BLS acknowledges that there are some weak connections listed in Table 1.

[58] Note that some data are not collected. e.g. Occupation 55-1011 "Air Crew Officers" is a military occupation. O*NET does not collect data on military occupations. The military services are the best resource for information on this occupation.

Table A.8: O*Net occupational flexibility by work arrangement and gender

	Husband		Wife		
	PE	SE	PE/PT	PE/FT	SE
Flexibility index	-0.44	-0.26	-0.86	-0.92	-0.35
Freedom to make decisions	-0.15	-0.00	-0.43	-0.22	0.04
Unstructured work	-0.09	-0.00	-0.27	-0.01	0.15
Time pressure	0.23	0.20	-0.11	0.11	-0.01
Contact with others	0.06	0.05	0.41	0.42	0.33
Maintaining relationships	-0.09	0.01	-0.14	0.16	0.21
Observations	35,486	8,645	20,477	22,392	3,775

Note: Mean O*Net work-context scores are computed by work arrangement and gender. Occupation is measured in ISCO-88 three-digit groups mapped to O*Net through the US SOC classifications. Each component is normalized to mean zero and unit variance across occupations. The flexibility index is freedom to make decisions plus unstructured work minus time pressure, contact with others, and maintaining interpersonal relationships, so higher values indicate more scheduling discretion.

Appendix B Additional Motivational Facts

This appendix documents additional empirical patterns that support the findings in Section 2. Figure B.1 shows how labor market outcomes evolve around first birth separately by gender and work arrangement. Figure B.8 documents self-employment by incorporation and sectoral status. Figure B.9 documents household asset dynamics around first birth.

Appendix B.1 Event-Study Specification

The event-study plots the within-person specification around the birth of the first child, following the event-study design of Kleven, Landais, and Søgaard (2019). For each outcome Y_{it}, I estimate for each gender,

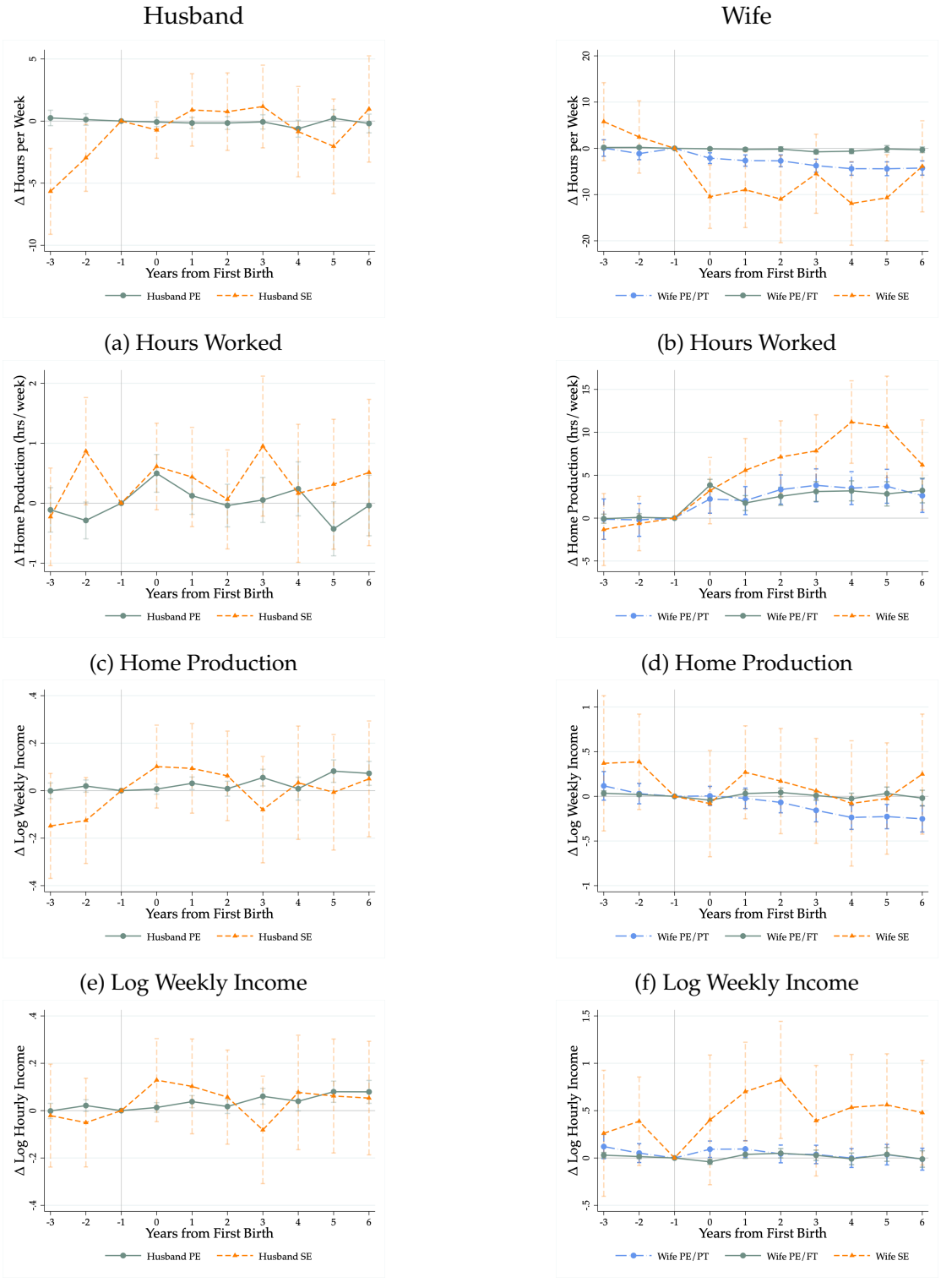
$$Y_{it} = \sum_{e \neq -1} \beta_e \mathbb{1}[e_{it} = e] + \alpha_i + \gamma_{\text{age}(it)} + \delta_t + \varepsilon_{it}, \tag{B.1}$$

where α_i is an individual fixed effect and γ_{age} and δ_t are full sets of age and calendar-year dummies. I cluster standard errors by individual and plot 90% confidence intervals. The event-time coefficients β_e measure outcomes after the first birth against the counterfactual of that same parent not having had the child.

Appendix B.2 Labor Market Outcomes

Figure B.1 reports event-study estimates for hours worked, home production, log weekly income, and log hourly wage around first birth, separately for husbands and wives within each work arrangement. Wives reduce market hours and increase home production at birth regardless of work arrangement, but the adjustment is sharpest for self-employed mothers. For paid-employed wives, the decline in weekly income tracks the hours reduction. On the other hand, for self-employed wives, weekly income falls less sharply because hourly wages do not decline. In some specifications, they rise modestly, consistent with positive selection remaining in the labor market after birth. Husbands show limited adjustment across all outcomes and work arrangements. These patterns confirm the gender asymmetry documented in Section 2. Figure B.2 complements the event-study evidence with the cross-sectional distribution of weekly earnings by work arrangement.

Self-employed workers also adjust hours more within individuals over time. Figure B.3 shows the distribution of within-person standard deviations in weekly hours, computed from individuals observed in at least two waves in the same work arrangement. Self-employed husbands and wives show substantially wider within-person hour variation than their paid-employed counterparts. The dispersion visible in Figure 2 reflects genuine scheduling flexibility rather than permanent sorting into different jobs. Figure B.4 documents the same pattern using the five-quarter Labour Force Survey.



(a) Hours Worked (b) Hours Worked

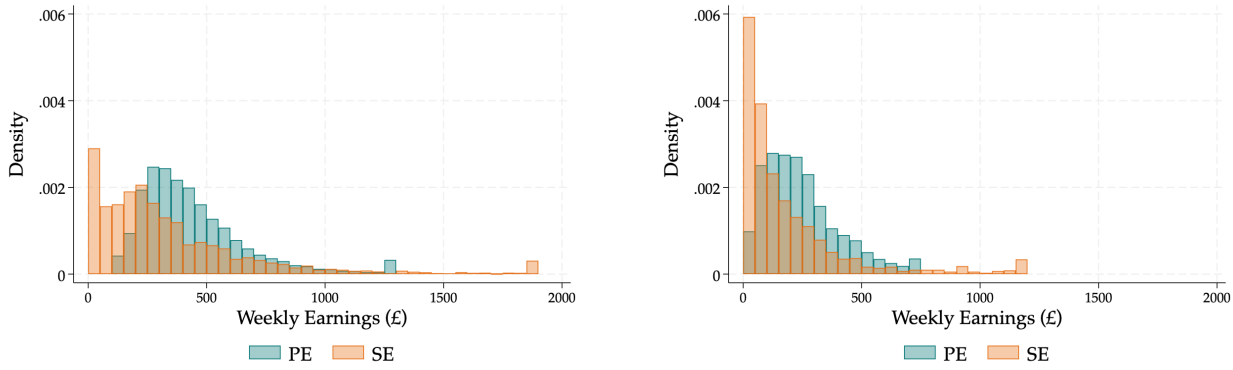
(c) Home Production (d) Home Production

(e) Log Weekly Income (f) Log Weekly Income

(g) Log Hourly Income (h) Log Hourly Income

Note: Event-study estimates by work arrangement around first birth. Hours panels show absolute changes in hours per week relative to $t = -1$; income panels show log-point changes relative to $t = -1$. Wife panels: PE/PT, PE/FT, SE; husband panels: PE and SE. Controls include age, year, education, and marital status; standard errors clustered at the individual level. Confidence intervals at 90%.
Source: British Household Panel Study.

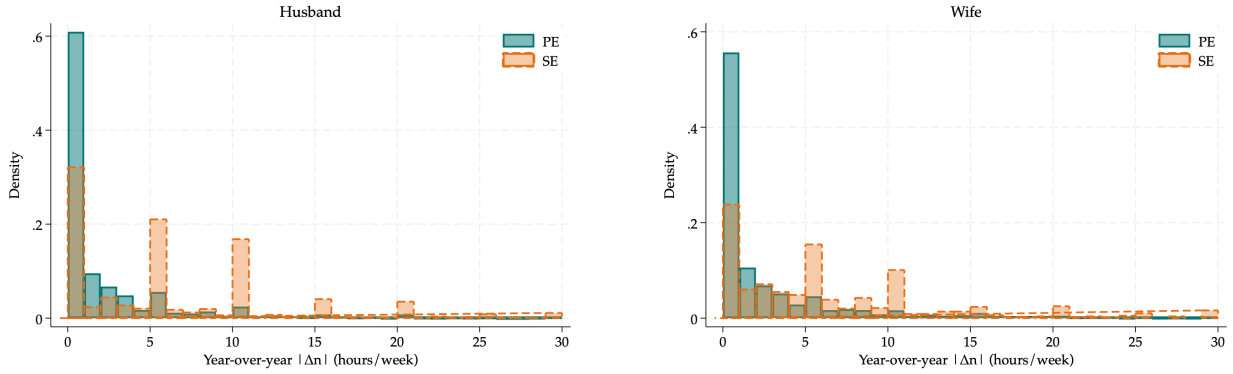
Figure B.1: Event Study: Labor Market Outcomes by Gender and Work Arrangement



(a) Husband (b) Wife

Note: Bars report the share of person-year observations in £50 bins. Self-employment earnings are bottom-coded at zero in the BHPS, so the self-employment spike at the zero bin collects observations with zero or negative earnings recorded as zero; all other bins report positive earnings.
Source: British Household Panel Study.

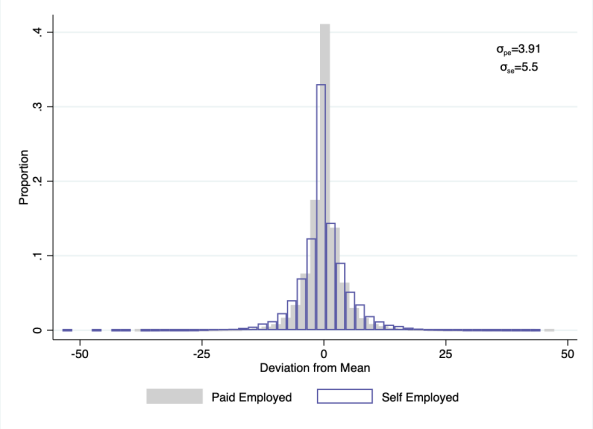
Figure B.2: Distribution of Weekly Earnings by Work Arrangement



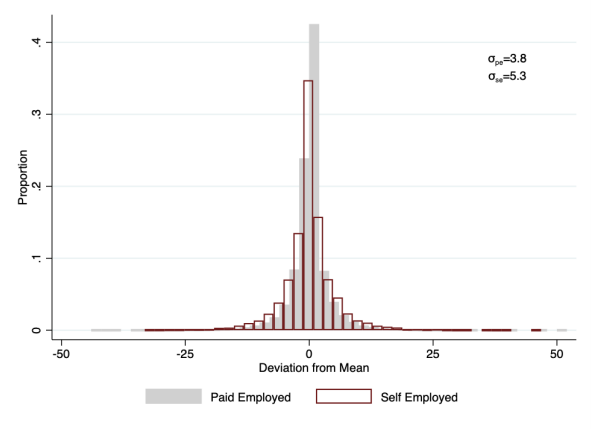
(a) Husband (b) Wife

Note: Distribution of within-person standard deviations in weekly hours worked, for individuals observed in at least two waves in the same work arrangement.
Source: British Household Panel Study.

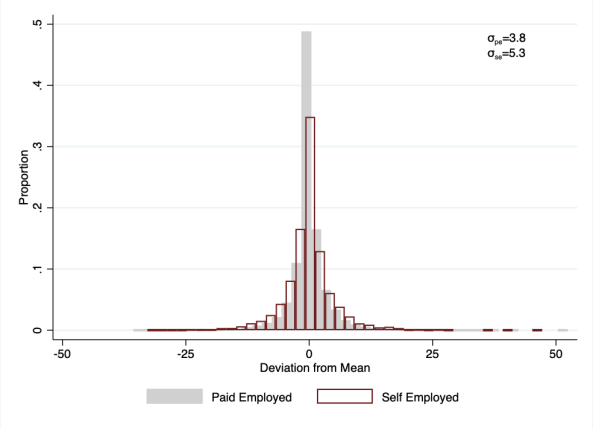
Figure B.3: Within-Person Hours Variation by Work Arrangement



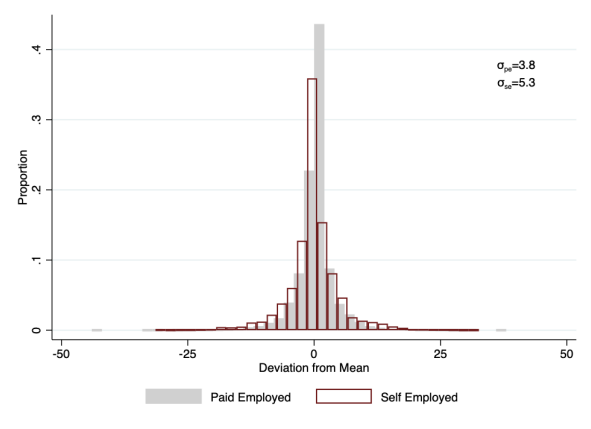
(a) Husband



(b) Wife



(c) Husband in Paid Employment



(d) Husband in Self-Employment

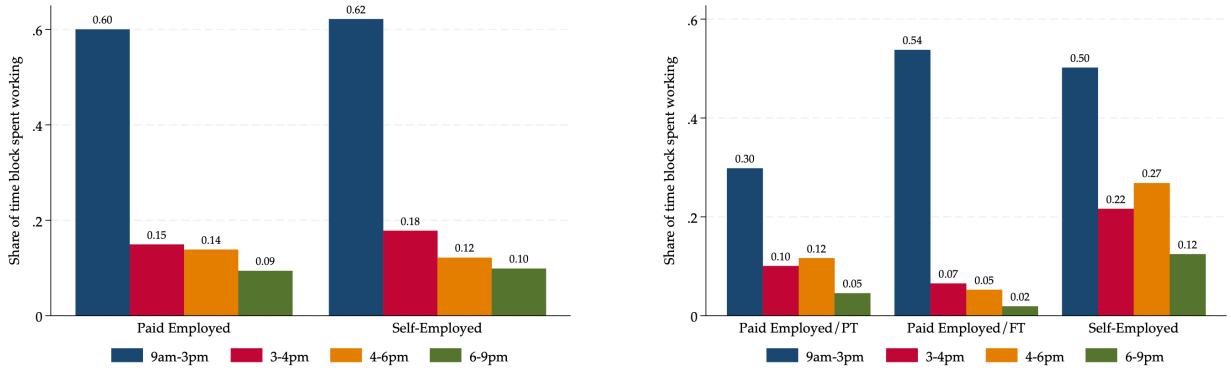
Note: Each panel shows deviations from a respondent's mean hours across five consecutive quarters, for married individuals with a youngest child aged 5. Controls include the husband's work arrangement.
Source: Five-Quarter Longitudinal Labour Force Survey.

Figure B.4: Within-Individual Variation in Hours Worked by Work Arrangement

Appendix B.3 Self-Employment: Work Flexibility

Time-use diaries from the UK Time Use Survey show how self-employed parents use scheduling flexibility. Self-employed mothers keep a morning work rate similar to paid employees but carry more work into the afternoon and evening: 22% work during the 3–4pm childcare window versus 7% of paid-employed mothers, and 27% during 4–6pm versus 5%. This shows up as split shifts, work in two or more non-contiguous blocks: 33% of self-employed mothers split their day versus 7% of paid employees, stepping away during the afternoon childcare window and resuming later. Fathers show the same pattern. These diaries indicate that the flexibility of self-employment is used to interleave market work with childcare, not simply to reduce total hours.

Self-employed parents are also more likely to interleave childcare and housework within their market work spans. Among those with a work span of at least three hours, self-employed mothers record childcare as a secondary activity during work at nearly twice the rate of paid-employed mothers. Self-employed fathers show a similar pattern relative to paid-employed fathers. Figure B.7 shows that work-from-home rates are higher for both self-employed wives (44% vs. 12% for paid-employed wives, rising to 56% when a young child is present) and self-employed husbands. This relates to the physical proximity that makes schedule interruptions feasible. Together, self-employment flexibility operates through the timing and structure of the workday, not only through total hours.



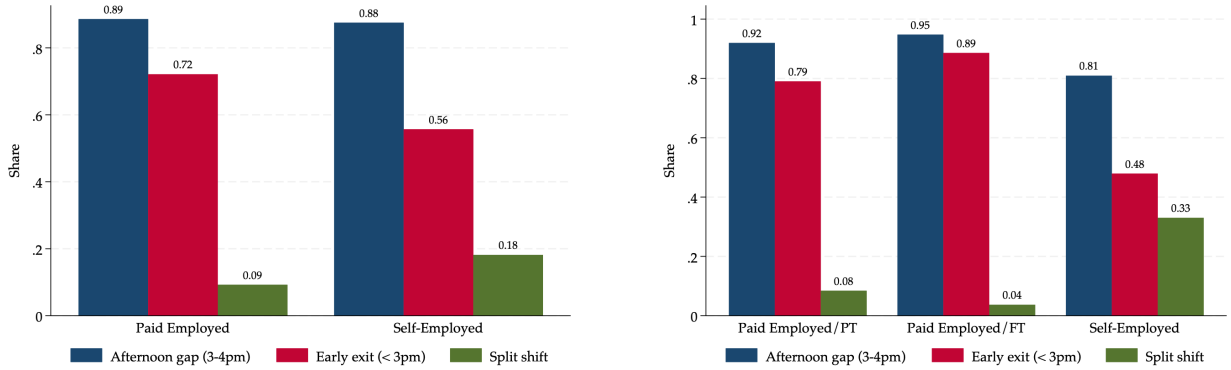
(a) Fathers (b) Mothers

Note: Share of each time block spent working by work arrangement.
Source: UK Time Use Survey (2000–2001).

Figure B.5: Work Rate by Time Block

Appendix B.4 Self-Employment: Incorporation and Sectoral Sorting

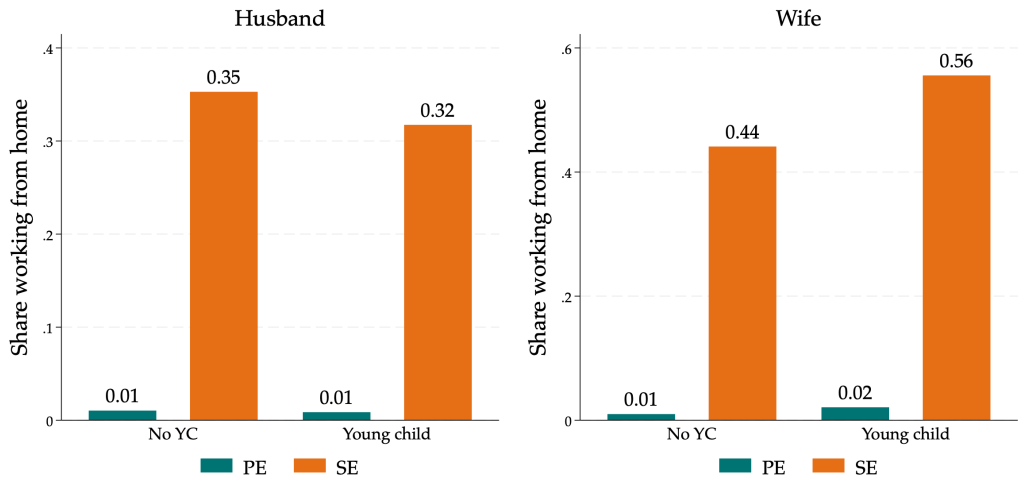
The self-employment rise documented in Section 2 is driven almost entirely by unincorporated self-employment among women. Figure B.8 shows event-study estimates for incorporated and unincorporated self-employment separately, conditional on remaining employed. The unincorporated rate rises sharply for wives after childbirth, while the incorporated rate remains flat. For husbands,



(a) Fathers (b) Mothers

Note: Afternoon gap = share with a work break during the 3–4pm block; early exit = share finishing work before 3pm; split shift = share working in two or more non-contiguous time blocks.
Source: UK Time Use Survey (2000–2001).

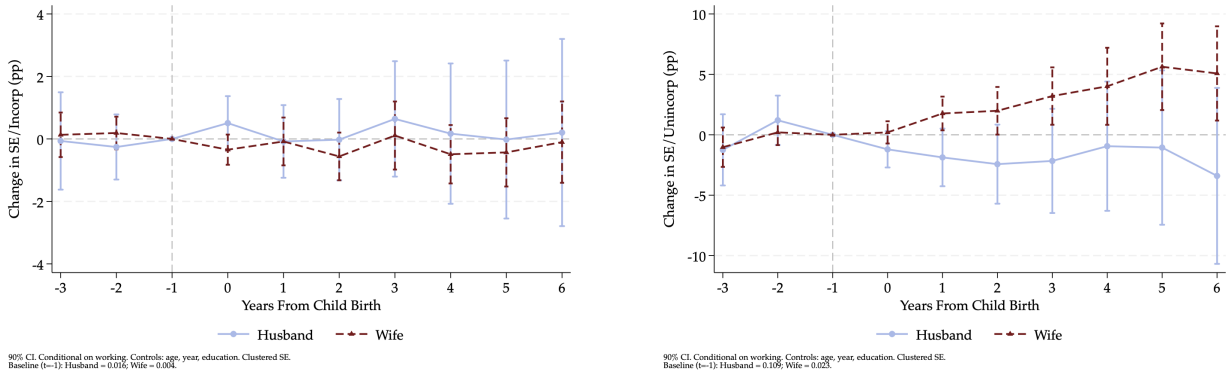
Figure B.6: Schedule Interruptions by Work Arrangement



Note: Share of workers usually working from home, by work arrangement, separately for husbands and wives and by the presence of a young child (under five).
Source: British Household Panel Study.

Figure B.7: Working from Home by Work Arrangement

neither form of self-employment shows a substantial post-birth adjustment. Mothers are more likely to sort into sole-trader arrangements rather than into formal businesses with employees.



(a) Incorporated Self-Employment (b) Unincorporated Self-Employment

Note: Both panels are conditional on being employed. The y-axis shows percentage point changes relative to the year before first birth ($t = -1$). Controls include age, year, education, and marital status; standard errors clustered at the individual level. Confidence intervals at 90%.
Source: British Household Panel Study.

Figure B.8: Event Study: Self-Employment by Incorporation Status (Conditional on Working)

Paid employed and self-employed workers sort into systematically different occupations and industries (Tables B.9 and B.10). Both spouses avoid large-employer, fixed-schedule sectors when entering self-employment. Manufacturing, Public Administration, and Financial Intermediation are heavily PE-concentrated for husbands and wives. Self-employed husbands concentrate in Crafts and Construction, with little response to presence of a young child. Self-employed wives concentrate at the top of the occupational ladder and in personal services, while the rate is low for clerical and technical roles, which tend to have more rigid schedules. The presence of a young child shifts self-employed wives further toward services and Health and Social Work. Within Health and Social Work, self-employed mothers are almost entirely in childminding and daycare rather than residential or clinical care, occupations with high scheduling autonomy even though they involve substantial interpersonal contact. This sorting is what the O*Net flexibility index in Table A.8 alludes to: self-employment concentrates in higher-discretion occupations for both parents.

Firm-level data reinforce these patterns. Paid employees work for large organizations, with a plurality at establishments of fifty or more employees, while self-employed workers are overwhelmingly in establishments of fewer than ten people. Within self-employment, roughly one-third of husbands and one-fifth to one-third of wives are employers with paid employees. The remainder are own-account workers. The share of self-employed wives with employees falls sharply with a young child.

Table B.9: Occupation composition by work arrangement and gender (ISCO-88)

	Husband				Wife			
	PE		SE		PE		SE	
	Nchild	Ychild	Nchild	Ychild	Nchild	Ychild	Nchild	Ychild
Legislators, senior officials and managers	0.21	0.21	0.27	0.23	0.11	0.08	0.29	0.19
	(0.41)	(0.41)	(0.44)	(0.42)	(0.31)	(0.27)	(0.45)	(0.39)
Professionals	0.16	0.15	0.13	0.11	0.13	0.15	0.18	0.22
	(0.36)	(0.36)	(0.34)	(0.32)	(0.34)	(0.36)	(0.38)	(0.42)
Technicians and associate professionals	0.12	0.13	0.10	0.10	0.17	0.20	0.11	0.08
	(0.32)	(0.33)	(0.30)	(0.30)	(0.37)	(0.40)	(0.32)	(0.28)
Clerks	0.08	0.08	0.00	0.00	0.28	0.28	0.07	0.06
	(0.27)	(0.27)	(0.06)	(0.06)	(0.45)	(0.45)	(0.25)	(0.23)
Service and sales workers	0.06	0.07	0.01	0.01	0.20	0.20	0.22	0.30
	(0.24)	(0.25)	(0.11)	(0.10)	(0.40)	(0.40)	(0.41)	(0.46)
Skilled agricultural and fishery workers	0.01	0.01	0.05	0.06	0.00	0.00	0.02	0.02
	(0.10)	(0.09)	(0.21)	(0.23)	(0.05)	(0.02)	(0.15)	(0.12)
Craft and related trades workers	0.19	0.17	0.31	0.34	0.01	0.01	0.03	0.01
	(0.39)	(0.38)	(0.46)	(0.47)	(0.12)	(0.12)	(0.16)	(0.09)
Plant and machine operators and assemblers	0.13	0.13	0.07	0.08	0.04	0.02	0.01	0.03
	(0.33)	(0.34)	(0.25)	(0.27)	(0.19)	(0.14)	(0.09)	(0.16)
Elementary occupations	0.05	0.05	0.06	0.07	0.06	0.05	0.08	0.10
	(0.23)	(0.22)	(0.24)	(0.25)	(0.24)	(0.23)	(0.27)	(0.30)
Armed forces	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Observations	13635	5743	2929	1199	13276	4275	979	380

Note: Share of person-waves in each ISCO-88 one-digit major group by work arrangement and gender, with standard deviations in parentheses. Nchild and Ychild denote the absence and presence of a young child (age five or below).

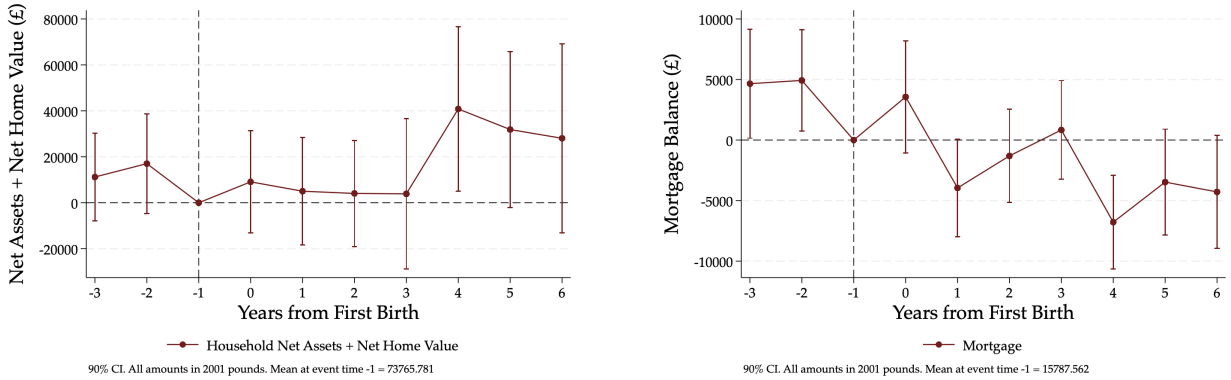
Table B.10: Industry composition by work arrangement and gender (SIC-92)

	Husband				Wife			
	PE		SE		PE		SE	
	Nchild	Ychild	Nchild	Ychild	Nchild	Ychild	Nchild	Ychild
A Agriculture, hunting, forestry	0.01	0.01	0.03	0.05	0.00	0.00	0.02	0.05
	(0.08)	(0.09)	(0.18)	(0.21)	(0.06)	(0.04)	(0.12)	(0.21)
B Fishing	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	(0.01)	(0.00)	(0.04)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
C Mining and quarrying	0.01	0.01	0.07	0.05	0.00	0.00	0.04	0.01
	(0.12)	(0.11)	(0.25)	(0.22)	(0.05)	(0.06)	(0.19)	(0.09)
D Manufacturing	0.32	0.30	0.11	0.12	0.13	0.09	0.04	0.04
	(0.46)	(0.46)	(0.31)	(0.32)	(0.34)	(0.29)	(0.19)	(0.20)
E Electricity, gas and water	0.02	0.01	0.00	0.00	0.01	0.00	0.00	0.00
	(0.15)	(0.10)	(0.03)	(0.03)	(0.09)	(0.06)	(0.00)	(0.00)
F Construction	0.05	0.07	0.20	0.27	0.01	0.01	0.02	0.02
	(0.22)	(0.25)	(0.40)	(0.44)	(0.10)	(0.11)	(0.13)	(0.15)
G Wholesale and retail	0.08	0.09	0.12	0.10	0.10	0.11	0.09	0.11
	(0.27)	(0.29)	(0.33)	(0.30)	(0.30)	(0.31)	(0.28)	(0.31)
H Hotels and restaurants	0.02	0.02	0.04	0.02	0.05	0.04	0.07	0.02
	(0.13)	(0.14)	(0.20)	(0.15)	(0.21)	(0.20)	(0.25)	(0.15)
I Transport, storage, communication	0.09	0.10	0.08	0.08	0.04	0.04	0.02	0.02
	(0.29)	(0.29)	(0.27)	(0.27)	(0.19)	(0.20)	(0.13)	(0.15)
J Financial intermediation	0.05	0.07	0.03	0.02	0.08	0.08	0.03	0.02
	(0.22)	(0.25)	(0.16)	(0.13)	(0.27)	(0.27)	(0.17)	(0.15)
K Real estate, business activities	0.11	0.11	0.19	0.16	0.10	0.09	0.17	0.14
	(0.32)	(0.31)	(0.39)	(0.36)	(0.30)	(0.29)	(0.38)	(0.35)
L Public administration	0.11	0.12	0.00	0.00	0.10	0.11	0.00	0.01
	(0.31)	(0.32)	(0.04)	(0.07)	(0.30)	(0.32)	(0.07)	(0.07)
M Education	0.04	0.03	0.04	0.03	0.11	0.10	0.08	0.09
	(0.19)	(0.17)	(0.18)	(0.18)	(0.31)	(0.30)	(0.27)	(0.28)
N Health and social work	0.04	0.04	0.03	0.02	0.22	0.25	0.15	0.25
	(0.20)	(0.19)	(0.17)	(0.15)	(0.42)	(0.44)	(0.35)	(0.43)
O Other community services	0.03	0.04	0.05	0.06	0.05	0.05	0.25	0.19
	(0.18)	(0.18)	(0.21)	(0.24)	(0.21)	(0.22)	(0.43)	(0.40)
P Private households as employers	0.00	0.00	0.01	0.00	0.01	0.00	0.04	0.03
	(0.02)	(0.00)	(0.09)	(0.07)	(0.07)	(0.04)	(0.19)	(0.17)
Q Extra-territorial	0.01	0.01	0.00	0.01	0.00	0.00	0.00	0.00
	(0.08)	(0.09)	(0.06)	(0.09)	(0.05)	(0.06)	(0.06)	(0.05)
Observations	13140	5727	2466	1095	12200	4222	917	371

Note: Share of person-waves in each SIC-92 section by work arrangement and gender, with standard deviations in parentheses. Nchild and Ychild denote the absence and presence of a young child (age five or below).

Appendix B.5 Household Assets

Households accumulate housing wealth in anticipation of childbirth and begin repayment afterward. Figure B.9 shows that net household assets, the sum of net assets and net home value, rise in the years after birth. The right panel reveals the mechanism. Mortgage balances peak pre-birth, indicating that households take on larger mortgages before childbirth, likely to secure larger or more stable housing. Net home value then rises as households begin repayment. This makes home equity the primary form of wealth accumulation around childbirth.



(a) Net Assets + Net Home Value (b) Mortgage

Note: Household net assets are defined as savings and investments, net of debt. Net home value is the difference between home value and mortgage balance. All amounts are in 2001 £. The sample includes married and cohabiting couples where the husband is always employed. Controls include year fixed effects, both spouses' age and education, household work arrangement, and the couple's annual income. Standard errors clustered at the household level. Confidence intervals at 90%.
Source: British Household Panel Study.

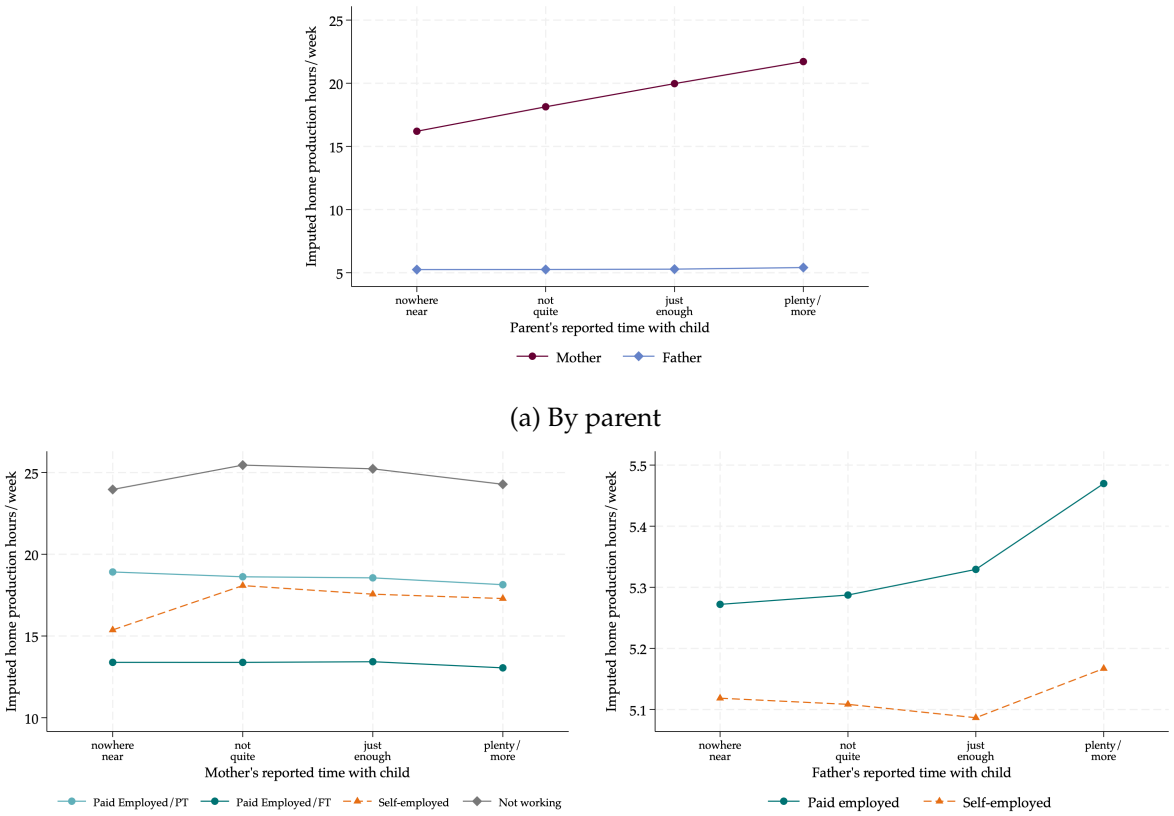
Figure B.9: Event Study: Household Assets

Appendix C Time Use Validation

The child production function uses maternal and paternal time inputs imputed from BHPS home-production hours. They are matched to each MCS household by work arrangement, education, number of children, and the age of the youngest child. The MCS does not record parental time use directly, so I validate these imputed inputs against a question the survey asks at every wave: whether the parent feels they spend enough time with the child. I harmonize the response across sweeps into a common four-point scale, ordered from "nowhere near enough" to "plenty". I then ask whether the households the imputation assigns more home-production time are the households that independently report more time with the child.

Figure C.10 Panel (a) plots mean imputed home-production time against the reported-time scale for each parent. Imputed maternal time rises monotonically with reported time, from 16.2 hours per week among mothers who report "nowhere near enough" to 21.7 hours among those who report "plenty". Panel (b) decomposes the maternal gradient by work arrangement. Non-working mothers

are assigned the most home-production time, at roughly 25 hours, and report the most time with their children; full-time employed mothers are assigned the least, at roughly 13 hours. Part-time and self-employed mothers fall in between. Within a given arrangement the relationship is flat, so the imputation encodes maternal time at the level of the work arrangement. This is the margin along which the model varies parental time. The validation suggests that the two datasets have the same ordering. The BHPS records that non-working women do the most home production, and MCS mothers independently report the most time with their children when not working. Fathers supply a low and arrangement-invariant amount of home production.



(a) By parent

(b) Mothers, by work arrangement

(c) Fathers, by work arrangement

Note: Mean imputed home-production time (BHPS cell means, hours per week) by the parent's reported adequacy of time with the child (MCS, harmonized four-point scale from "nowhere near enough" to "plenty"). Panel (a) plots mothers and fathers; panel (b) plots parents by work arrangement.

Figure C.10: Imputed home-production time and parents' reported time with the child

Since MCS questions are based on subjective adequacy measures, I next validate the imputed hours against the UK Time Use Survey. I construct weekly housework hours (cooking, cleaning, and home maintenance) and, separately, weekly primary childcare hours (physical care, teaching, playing, and escorting). I average both to the cells used in the imputation and compare them with the BHPS home-production grid. Figure C.11 shows that BHPS home production reproduces diary housework. For mothers the two coincide in level and across arrangements, averaging 18.3

hours per week in the grid against 18.8 in the diaries and tracing the same profile from full-time employment up to non-work. For fathers the ordering is the same, with a small diary-over-survey gap at paid employment that is second-order given how little home production fathers do.

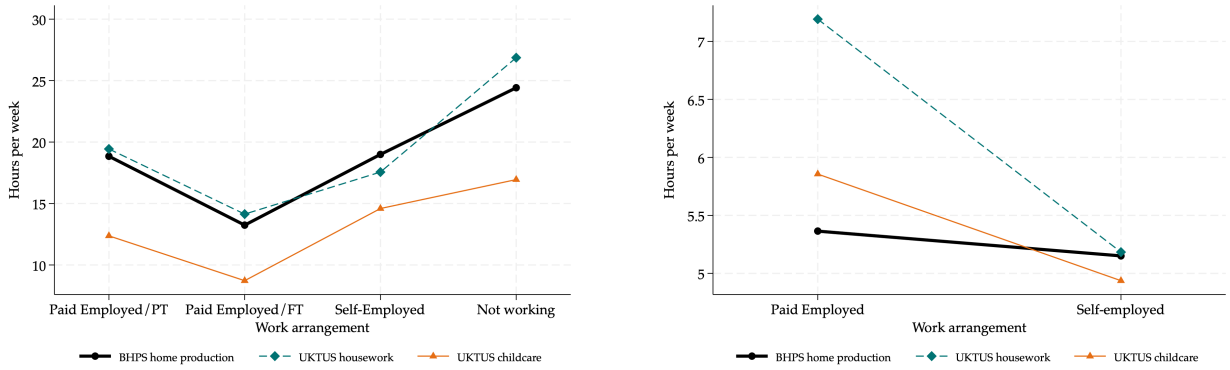
Home production tracks diary housework, with a cell-level correlation of 0.85 for mothers and 0.67 for fathers. It still co-moves with childcare across arrangements (correlation 0.57), so its variation indexes the broader gradient in time spent at home. Hence, it is a good proxy for the parent's time with the child because it tracks diary housework and moves with the parents' own reported time-with-child gradient. It enters skill formation as one input.

The maternal-education gradient in time use is concentrated in child-development activities rather than housework. Table C.11 reports diary time for mothers of young children by college status. College mothers spend 2.4 more hours per week with the child (17.2 against 14.8) and 1.2 more hours reading, playing, and talking (5.4 against 4.2), the most cognitively stimulating category. The differences across groups are statistically significant. Housework shows no significant education gradient and is, if anything, lower for college mothers (19.2 against 21.2). Combining housework with child time would attenuate the estimated skill productivity of educated mothers' time.

Table C.11: Maternal time use by education: child-development time versus housework

	Non-college	College	Difference
Total child time	14.84	17.21	2.37*
			(1.21)
Reading, playing, talking	4.18	5.40	1.22*
			(0.70)
Physical care and supervision	8.34	9.99	1.66**
			(0.78)
Teaching	0.40	0.35	-0.06
			(0.17)
Escorting and travel	1.88	1.40	-0.48
			(0.39)
Housework	21.23	19.23	-2.00
			(1.35)

Note: UK Time Use Survey (2000–2001) diary means, hours per week, for mothers whose youngest child is aged four or below, by whether the mother holds a college degree. The difference column reports the college minus non-college gap. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.



(a) Mothers (b) Fathers

Note: Mean home-production hours per week by work arrangement. Black line: the BHPS home-production grid (the imputed input). Teal dashed line: UKTUS diary housework. Orange line: UKTUS primary childcare. UKTUS means are diary-weighted and averaged to the imputation cells.

Figure C.11: BHPS home production and UKTUS diary time by work arrangement

Appendix D Childcare Price and Quality

I do not distinguish different types of formal childcare and assume homogeneous quality (nursery, childminder, nanny, etc.).[59] I allow childcare costs in the model to vary by the age of the youngest child, reflecting the substantial differences in the type and intensity of formal care used across child ages.

I verify this assumption directly in the MCS childcare setting observation study, which rated the center-based settings that cohort children attended around age four, observed in 2005, on the Early Childhood Environment Rating Scale-Revised (ECERS-R) and its curricular Extension (ECERS-E), the Caregiver Interaction Scale (CIS), staff qualifications, and child-adult ratios. Linking each setting to maternal education, Table D.12 shows that observed quality does not vary by education within the estimation sample. Process quality, curriculum quality, staff qualifications, and the child-adult ratio are statistically indistinguishable between children of college and non-college mothers, and the only significant difference, in caregiver positive interaction, slightly favors non-college children. The same pattern holds in the full childcare-study sample. The education gradient in the childcare coefficient therefore reflects how well formal care substitutes for maternal time, not differences in the quality of care used. These ratings cover center-based settings for about one in ten formal-childcare users in the estimation sample, so they are used as supporting evidence on the

[59] Three features of the UK regulated sector support this assumption. First, since the Care Standards Act 2000, all formal childcare providers must register with Ofsted and meet minimum National Standards; settings that fail inspection are required to improve or face closure, creating a quality floor that limits the lower tail of quality within the regulated sector. Second, the Working Tax Credit childcare element covered up to 70% of formal childcare costs during the MCS period, compressing price differentials across income groups within the regulated sector and limiting the ability of higher-income families to systematically sort into more expensive, higher-quality settings. Third, Gambaro, Stewart, and Waldfogel (2014) document that in the UK's regulated sector disadvantaged children are more likely than advantaged children to attend settings with qualified staff, so quality does not systematically favor advantaged children within the sector, though settings serving disadvantaged areas do receive somewhat lower inspection ratings.

regulated sector.

Table D.12: Observed Childcare Quality by Maternal Education

	College	Non-college	Difference
ECERS-R (process, 1–7)	4.68 (0.95)	4.75 (0.88)	-0.07
ECERS-E (curriculum, 1–7)	3.33 (0.87)	3.42 (0.77)	-0.09
CIS positive interaction (1–4)	3.39 (0.44)	3.53 (0.41)	-0.14
Mean staff qualification (1–4)	2.35 (0.61)	2.43 (0.62)	-0.08
Children per paid adult	4.95 (1.72)	5.15 (2.24)	-0.20
Settings (children)	121	174	

Note: Mean (standard deviation) of each quality measure for children of college and non-college mothers in the estimation sample, from the MCS childcare setting observation study (center-based settings attended by cohort children around age four, observed in 2005). College denotes a maternal degree. Process and curriculum quality are the ECERS-R and ECERS-E overall scores; CIS is the Caregiver Interaction Scale positive-relationships subscale; staff qualification is the mean childcare-qualification level of valid staff. None of the differences in process quality, curriculum quality, staff qualification, or the child-adult ratio is statistically significant; the positive-interaction difference is small and favors non-college children.

Ages 1 and 3: Nursery care. For children under age 2, the average cost of a full-time nursery place in England was £159 per week for 50 hours, implying an hourly rate of £3.18 (Daycare Trust, 2008, Table 1). For children aged 2 and over, the corresponding figure was £149 per week, or £2.98 per hour. In the model, households choose between part-time (15 hours per week) and full-time (30 hours per week) formal childcare. At the full-time intensity, these rates imply annual childcare costs of approximately £4,961 for a child aged 1 and £4,649 for a child aged 3, with part-time costing half.[60]

Age 5: After-school care. Children aged 5 attend primary school and require only wraparound care. The Daycare Trust (Daycare Trust, 2008) reports an average cost of £43 per week for 15 hours of out-of-school club provision in England, implying an hourly rate of £2.87. At 15 hours per week, the annual cost is approximately £2,236, roughly half the cost of full-time nursery care.[61]

Age 7 and above: Childcare as a fraction of income. For families whose youngest child is aged 7 or older, childcare expenditure is modeled as a fixed fraction of household income. School-age children require substantially less formal care. The Childcare and Early Years Survey of Parents

[60] I use the gross market price rather than adjusting for the free early education entitlement introduced for 3-year-olds in 2004 (12.5 hours per week, later extended to 15 hours), as the entitlement was phased in partway through the sample period and the quality of provision under the free entitlement may differ from that of privately purchased care. These figures are consistent with the average hourly childcare cost of £2.60 (s.e. 0.04) estimated from BHPS expenditure data by Blundell et al. (2016), who pool across child ages.

[61] This is consistent with the median out-of-pocket payment of £2.50 per hour for on-site after-school clubs reported in the Childcare and Early Years Survey of Parents 2009 (Smith et al., 2010).

2009 reports a median of 6.5 hours per week in childcare for school-age children, compared with 20 hours for pre-school children, and a median weekly payment of £15 for school-age-only families versus £50 for pre-school families (Smith et al., 2010, pp. 65, 74). Relative to median household income, this implies a childcare expenditure share of approximately 4–5%. The figure is consistent with evidence from the Families and Children Study 2005 that the overall average expenditure share was 11%, with lower shares for higher-income households (Hoxhallari, Conolly, and Lyon, 2007). I set the childcare income share for families with older children to 5%. Table D.13 summarizes the calibrated childcare cost parameters.

Table D.13: Calibrated Childcare Cost Parameters

Age of youngest child	Care type	Hourly rate	Hours/week	Annual cost
1 (under 2)	Nursery (full-time)	£3.18	30	£4,961
3 (aged 2–4)	Nursery (full-time)	£2.98	30	£4,649
5	After-school club	£2.87	15	£2,236
7+	5% of net household income			

Note: Source: Daycare Trust (2008) for ages one, three, and five; Smith et al. (2010) and Hoxhallari et al. (2007) for age seven or more. Rates are deflated to £2001.

Appendix E Log-Additivity, Exogeneity, and Estimation of the Child Production Function

Appendix E.1 Log-Additivity of Child Skills

I prove that the household conditional value functions are log-additive in child's skills (Mullins, 2025; Garcia-Vazquez, 2023). This reduces the computational burden of solving the model. The optimal policy functions do not depend on child skill levels, which removes child skills from the dynamic state space. However, this does not eliminate forward-looking investment motives. Parents still choose time and formal childcare to shift next-period skills through the CPF, anticipating the continuation value gains from higher future skills.

Proposition 1. For child's age $g_t = g_1, \ldots, g_T$ and $D \in \{cog, ncog\}$, the Stage-2 conditional value function admits the decomposition

$$\widetilde{\mathcal{V}}^{s2}_{q,r}(\Omega^{s2}) = \bar{\mathcal{V}}^{s2}_{q,r}(\Omega^{s2}_{-\theta}) + \sum_D \Gamma_{D,g_t} \log \theta_{D,g_t}$$

where the coefficients follow the backward recursion

$$\Gamma_{D,g_t} = \alpha_D + \beta\big(\gamma_{D,1}\,\Gamma_{D,g_t+1} + \gamma_{D',2}\,\Gamma_{D',g_t+1}\big), \qquad \Gamma_{D,g_T+1} = \alpha_D \sum_{s=0}^{J-1} \beta^s,$$

$J = J(a^w_t)$ is the number of remaining model periods from the child's age-7 point, at the wife's age a^w_t, until her terminal age. The coefficient therefore depends on the wife's age, not on the child's age alone: because a given child age occurs at different wife ages under stochastic fertility, Γ_{D,g_t} is more precisely $\Gamma_D(g_t, a^w_t)$ and is solved at each wife age. $\Omega^{s2}_{-\theta}$ is the state space with both child skill levels removed. The terminal condition reflects that skills are fixed after the investment window but continue to enter flow utility in every remaining period. The coefficient values the youngest child's skill to the wife's terminal age. Because the skill weights differ by maternal education, α_D stands throughout for $\alpha_D + \alpha^{ed}_D \theta^w$, so the coefficients also vary with θ^w.

Proof. The proof proceeds by backward recursion. At the last child investment period g_T (child age 5), parents choose Stage-2 actions $r_{g_T} = (n^w_{g_T}, n^h_{g_T}, \tau^w_{g_T}, \tau^h_{g_T}, cc_{g_T}, a_{g_T+1})$ to solve

$$\begin{aligned}
\widetilde{\mathcal{V}}^{s2}_{q,g_T}(\Omega^{s2}_{g_T}) &= \max_{r_{g_T} \in \mathcal{R}} \Big\{ \mathcal{U}_{g_T}(C_{g_T}, l^w_{g_T}, l^h_{g_T}, cc_{g_T}, q) + \zeta_{q,r_{g_T}} + \beta \mathbb{E}\big[V^{s1}(\Omega^{s1,\prime}_{g_T+1})\big] \Big\} \\
&= \max_{r_{g_T} \in \mathcal{R}} \Big\{ \mathcal{U}_{g_T}(\cdot) + \sum_D \alpha_D \log \theta_{D,g_T} \\
&\qquad + \beta \Big(\sum_D \Gamma_{D,g_T+1}\, \mathbb{E} \log \theta_{D,g_T+1} + \mathbb{E}\bar{V}_{g_T+1}(\Omega^{s1,\prime}_{-\theta,g_T+1}) \Big) + \zeta_{q,r_{g_T}} \Big\} \\
\text{s.t.} \quad & (\mathrm{TC}^w), (\mathrm{TC}^h), (\mathrm{Sup}), (\mathrm{BC}), (\mathrm{I}), (\mathrm{IC}^w), (\mathrm{IC}^h), (\mathrm{HC}^w), (\mathrm{CPF}^D)
\end{aligned} \tag{E.2}$$

where the second line uses the fact that $\sum_D \alpha_D \log \theta_{D,g_T}$ enters the current flow utility through the child skills term in $\mathcal{U}_{g_T}$. After the investment window closes, the skill vector is fixed at θ_{D,g_T+1} and adds $\alpha_D \log \theta_{D,g_T+1}$ to flow utility in each of the J remaining periods; collecting this discounted stream gives the coefficient $\Gamma_{D,g_T+1} = \alpha_D \sum_{s=0}^{J-1} \beta^s$, and $\bar{V}_{g_T+1}$ collects the remaining skill-free terms.

Substituting the child production function into $\log \theta_{D,g_T+1}$ yields

$$\begin{aligned}
\widetilde{\mathcal{V}}^{s2}_{q,g_T}(\Omega^{s2}_{g_T}) = \max_{r_{g_T}} \mathbb{E}_\nu \Big[& \mathcal{U}_{g_T}(\cdot) + \sum_D \alpha_D \log \theta_{D,g_T} \\
& + \beta \sum_D \Gamma_{D,g_T+1} \Big(\log A_{D,g_T} + \gamma_{D,1} \log \theta_{D,g_T} + \gamma_{D,2} \log \theta_{D',g_T} \\
& + \gamma_{D,3} \log \tau^w_{g_T} + \gamma_{D,4} \log \tau^w_{g_T} \cdot \theta^w + \gamma_{D,5} \log \tau^h_{g_T} + \gamma_{D,6} \log \tau^h_{g_T} \cdot \theta^h \\
& + \gamma_{D,7} cc^{PT}_{g_T} + \gamma_{D,8} cc^{FT}_{g_T} + \gamma_{D,9} cc^{PT}_{g_T} \cdot \theta^w + \gamma_{D,10} cc^{FT}_{g_T} \cdot \theta^w + \gamma_{D,11} sib_{g_T} \Big) \\
& + \beta \mathbb{E} \bar{V}_{g_T+1}(\cdot) + \zeta_{q,r_{g_T}} \Big] \\
\text{s.t.} \quad & (\text{TC}^w), (\text{TC}^h), (\text{Sup}), (\text{BC}), (\text{I}), (\text{IC}^w), (\text{IC}^h), (\text{HC}^w)
\end{aligned}$$

where the expectation over the production shock ν_{D,g_T+1} drops out because $\mathbb{E}[\nu_{D,g_T+1}] = 0$. Collecting the terms in $\log \theta_{D,g_T}$ from the current flow utility, the own-domain persistence ($\gamma_{D,1}$), and the cross-domain persistence ($\gamma_{D',2}$, after relabeling domains in the cross term), I can rewrite the conditional value function at g_T as

$$\widetilde{\mathcal{V}}^{s2}_{q,g_T}(\Omega^{s2}_{g_T}) = \bar{\mathcal{V}}^{s2}_{q,g_T}(\Omega^{s2}_{-\theta,g_T}) + \sum_D \Gamma_{D,g_T} \log \theta_{D,g_T}$$

with $\Gamma_{D,g_T} = \alpha_D + \beta(\gamma_{D,1} \Gamma_{D,g_T+1} + \gamma_{D',2} \Gamma_{D',g_T+1})$, the recursion evaluated at the terminal condition, and

$$\begin{aligned}
\bar{\mathcal{V}}^{s2}_{q,g_T}(\Omega^{s2}_{-\theta,g_T}) = \max_{r_{g_T}} \Big[& \mathcal{U}_{g_T}(\cdot) + \beta \sum_D \Gamma_{D,g_T+1} \Big(\log A_{D,g_T} + \gamma_{D,3} \log \tau^w_{g_T} + \gamma_{D,4} \log \tau^w_{g_T} \cdot \theta^w \\
& + \gamma_{D,5} \log \tau^h_{g_T} + \gamma_{D,6} \log \tau^h_{g_T} \cdot \theta^h + \gamma_{D,7} cc^{PT}_{g_T} + \gamma_{D,8} cc^{FT}_{g_T} \\
& + \gamma_{D,9} cc^{PT}_{g_T} \cdot \theta^w + \gamma_{D,10} cc^{FT}_{g_T} \cdot \theta^w + \gamma_{D,11} sib_{g_T} \Big) + \beta \mathbb{E} \bar{V}_{g_T+1}(\cdot) + \zeta_{q,r_{g_T}} \Big] \\
\text{s.t.} \quad & (\text{TC}^w), (\text{TC}^h), (\text{Sup}), (\text{BC}), (\text{I}), (\text{IC}^w), (\text{IC}^h), (\text{HC}^w)
\end{aligned}$$

which proves the proposition for the terminal child investment period g_T.

Now suppose the result holds at $g_t + 1$, so that $\widetilde{\mathcal{V}}^{s2}_{q,g_t+1} = \bar{\mathcal{V}}^{s2}_{q,g_t+1} + \sum_D \Gamma_{D,g_t+1} \log \theta_{D,g_t+1}$. The decomposition extends to the Stage-1 ex-ante value $V^{s1}_{g_t+1}$ because the skill term is common across hours-savings choices r and work arrangements q, so it passes through the maximization and the expectations over $\boldsymbol{\zeta}$ and $\boldsymbol{\xi}$ as an additive constant. For the inductive step, I show the same decomposition holds at g_t. Applying the induction hypothesis to the continuation value, the conditional value function at g_t is

$$\begin{aligned}
\widetilde{\mathcal{V}}^{s2}_{q,g_t}(\Omega^{s2}_{g_t}) = \max_{r_{g_t} \in \mathcal{R}} \Big\{ & \mathcal{U}_{g_t}(\cdot) + \sum_D \alpha_D \log \theta_{D,g_t} \\
& + \beta \Big(\sum_D \Gamma_{D,g_t+1} \, \mathbb{E} \log \theta_{D,g_t+1} + \mathbb{E} \bar{V}_{g_t+1}(\Omega^{s1,\prime}_{-\theta,g_t+1}) \Big) + \zeta_{q,r_{g_t}} \Big\}
\end{aligned}$$

where $\bar{V}_{g_t+1}$ collects the remaining skill-free terms. Substituting the CPF for $\log \theta_{D,g_t+1}$ and again collecting the terms in $\log \theta_{D,g_t}$ from the flow utility, the own-domain persistence, and the cross-domain persistence, I obtain

$$\widetilde{\mathcal{V}}^{s2}_{q,g_t}(\Omega^{s2}_{g_t}) = \bar{\mathcal{V}}^{s2}_{q,g_t}(\Omega^{s2}_{-\theta,g_t}) + \sum_D \Gamma_{D,g_t} \log \theta_{D,g_t}$$

with $\Gamma_{D,g_t} = \alpha_D + \beta(\gamma_{D,1}\,\Gamma_{D,g_t+1} + \gamma_{D',2}\,\Gamma_{D',g_t+1})$, completing the induction. Note that the CPF input terms enter $\bar{\mathcal{V}}^{s2}_{q,g_t}$ with weight Γ_{D,g_t+1} rather than α_D, since parents internalize that skills produced today persist into future utility. □

Corollary 1 (Skill-Independent Policy Functions). The household's optimal policy functions $\{n^{w*}_{g_t}, n^{h*}_{g_t}, \tau^{w*}_{g_t}, \tau^{h*}_{g_t}, cc^*_{g_t}, a^*_{g_t+1}\}$ are independent of the child skill levels $(\log \theta_{cog,g_t}, \log \theta_{ncog,g_t})$.

Proof. From Proposition 1, the skill term $\sum_D \Gamma_{D,g_t} \log \theta_{D,g_t}$ is additively separable and common across choices, so it cancels in the argmax and the optimal policy depends only on $\bar{\mathcal{V}}^{s2}_{q,r}(\Omega^{s2}_{-\theta})$, not on child skills. The same holds for the Stage-1 work-arrangement choice, since the altruism weights are common across arrangements. □

Appendix E.2 Exogeneity of Parental Inputs

The CPF estimation in Section 4.1 addresses two distinct sources of econometric concern: measurement error in observed child skills, and potential endogeneity of parental inputs. The CPF coefficients on parental time and childcare enter the structural model as fixed parameters and carry directly into the counterfactual exercises of Section 6, so it is important that they capture the causal returns. Parental inputs (τ^w, τ^h, cc) are tied to labor supply decisions through the household constraints. I therefore instrument all four inputs with UK tax-and-benefit policy variation to recover the causal effects.

Instrument construction. I construct simulated instruments from external UK tax-and-benefit policy variation, each capturing a distinct policy channel. For each MCS household-wave, I use hourly wages and observed family structure into the FORTAX microsimulator (Shephard, 2009) under counterfactual configurations, so that variation across households comes from the policy regime and household characteristics (children, ages, education). The wage predictions come from four log-wage Mincer regressions estimated on the sample of married and cohabiting couples with children, separately by gender and by paid employment versus self-employment status. I control for education, age, and age squared.

1. Z^{NI}_w, Z^{NI}_h: the paid employment versus self-employment National Insurance contribution differential for the wife and husband, capturing the Class 1 versus Class 2/4 wedge that shifts

sectoral choice.[62]

2. Z^{cc}: the Work Tax Credit (WTC) childcare element value at benchmark hours, shifting the effective price of formal childcare;

3. Z^{WTC}_{2003}: the 2003 Working Tax Credit reform interacted with the number of dependent children, exploiting the response structure of the WTC childcare expansion;

4. Z^{WFTC}_{1999}: the 1999 Working Families Tax Credit expansion interacted with low-income eligibility and dependent children, capturing the extensive-margin labor supply incentive;

5. Z^{χ_3}: the household-level log-log curvature of the FORTAX net-income schedule, capturing rate-schedule progressivity;

6. Z^{k30}_h: the discrete jump in household disposable income at the husband's 29-to-30-hour threshold, where the WTC 30-hour premium activates;

7. Z^{IT}_w, Z^{IT}_h: the wife's and husband's income tax liability at modal PE hours, shifting the after-tax return to work;

8. Z^{CB}: the child benefit plus family transfers at zero earnings, varying with the number and ages of children across policy years;

9. Z^{PTR}_h: the husband's participation tax rate (the change in household disposable income from not working to working at modal PE hours), shifting the husband's extensive margin.

Instrument validity. All instruments feed predicted wages from Mincer regressions and observed family structure into FORTAX policy rules. The variation across households reflects the policy schedule applied to predetermined education, age, and family composition rather than endogenous labor supply choices. The 1999 WFTC and 2003 WTC reform instruments are simulated differentials between the pre- and post-reform statutory schedules, so their variation reflects the legislated policy change interacted with predetermined characteristics, not within-sample exposure to the reform. The 30-hour threshold instrument Z^{k30}_h uses predicted wages at modal paid employment hours rather than actual hours, so the policy discontinuity identifies variation in the WTC premium across household types rather than households' sorting around the kink.

Moment conditions. Let $X = (\tau^w, \tau^h, cc)$ be the vector of parental inputs, $\log\theta = (\log\theta_{cog}, \log\theta_{ncog})$ the lagged latent skills, and X^c the exogenous controls (sibling indicator). The excluded instruments are the cross-domain skill proxies $Z^{skill} = (Z^{ncog}, Z^{cog})$, a second

[62] Class 1 contributions are paid by employees at 12% on earnings between the Primary and Upper Earnings Limits. Class 2 is a flat weekly contribution paid by the self-employed (£2.05/week in 2003). Class 4 is a profit-based contribution at 8% on profits above the Lower Profit Limit. For most self-employed workers in the sample earnings range, total NI liability is substantially lower than Class 1, creating a net-of-tax incentive toward self-employment.

measure of the other domain from previous waves, and the simulated FORTAX policy instruments Z^{FORTAX}.

Estimation combines two blocks of moment conditions. Let $Z^{input} = (Z^{skill}, Z^{FORTAX}, X^c, 1)$ be the instrument vector for the input equations. For each input j,

$$\mathbb{E}\left[Z_i^{input}\left(X_{ji} - \pi_j' Z_i^{input}\right)\right] = 0 \tag{E.3}$$

Let $\hat{v} = X - \hat{\Pi}' Z^{input}$ denote the four input-equation residuals, which enter the outcome equation as control-function corrections. With $Z^{outcome} = (Z^{skill}, X, X^c, \hat{v}, 1)$,

$$\mathbb{E}\left[Z_i^{outcome}\left(y_i - \beta_\theta' \log\theta_i - \beta_X' X_i - \beta_c' X_i^c - \beta_v' \hat{v}_i\right)\right] = 0 \tag{E.4}$$

Consistency. The procedure makes two standard corrections. The lagged cross-domain skill instruments correct for measurement error in the lagged skills. FORTAX instruments with a control function correct for input endogeneity, valid under the exclusion restriction $Cov(Z^{FORTAX}, \nu') = 0$. Under these conditions the stacked GMM estimator is consistent (Hansen, 1982; Wooldridge, 2002).

Appendix E.3 Estimation of Child Production Function

Table E.14 reports the OLS and joint GMM estimates. The endogeneity correction raises the parental-input coefficients relative to OLS, the expected direction given negative selection and measurement error in the imputed inputs. Across specifications, the sign pattern and the education gradient are stable. Maternal time is more productive for children of college-educated mothers in both domains, and its level is positive and significant for cognitive skills. Part-time childcare raises skills for non-college households with the college interaction offsetting it. Sibling effects are negative for cognitive and positive for non-cognitive skills. Although the childcare base and college-interaction coefficients seem large, they move together across bootstrap draws, so the net effect for children of college-educated mothers is precisely estimated. It is near zero for part-time care and $+0.15$ SD (95% CI $[0.03, 0.31]$) for cognitive and $+0.22$ (CI $[0.10, 0.34]$) for non-cognitive skills under full-time care, and it is smaller than the corresponding non-college effect throughout.

The diagnostic tests the joint significance of the first-stage residuals. For cognitive skills, the Hausman-type test yields $p = 0.055$, so input exogeneity is not rejected at the 5% level. For non-cognitive skills, $p < 0.001$, so endogeneity is detected and concentrated in part-time formal childcare ($\hat{v}_{cc^{PT}}$ significant at the 1% level; $\hat{v}_{cc^{FT}}$ is not significant at conventional levels). The parental time residuals ($\hat{v}_{\tau^w}$, $\hat{v}_{\tau^h}$) are individually insignificant in both equations, so the correction does not reject the exogeneity of parental time. The supervision constraint is slack in 81% of young-child households ($\bar{T}^{cc} = 20$), so for most, parental time is not mechanically tied to formal care.

The Hansen J rejects the overidentifying restrictions ($J = 41.3$, 7 df, $p < 0.001$). Identification of the input productivities rests on the FORTAX exclusion restriction, not on the overidentifying

Table E.14: Child Production Function: OLS vs Joint GMM

	Cognitive$_{t+1}$				Non-Cognitive$_{t+1}$			
	OLS		Joint GMM		OLS		Joint GMM	
Variable	Coef	SE	Coef	SE	Coef	SE	Coef	SE
Cognitive	0.754	0.089	0.748	0.088	0.148	0.079	0.150	0.078
Non-Cognitive	0.160	0.072	0.154	0.072	0.732	0.065	0.721	0.065
log(Wife's time)	0.055	0.066	0.317	0.124	0.024	0.060	0.201	0.117
log(Wife's time)$\times$ College	0.063	0.012	0.112	0.021	0.051	0.011	0.101	0.020
log(Husband's time)	0.111	0.153	0.060	0.289	-0.023	0.118	-0.017	0.243
log(Husband's time)$\times$ College	0.063	0.017	0.045	0.019	0.073	0.016	0.060	0.018
Childcare (PT)	0.106	0.061	1.119	0.433	0.089	0.050	1.603	0.380
Childcare (FT)	0.110	0.057	0.678	0.273	0.120	0.054	0.478	0.250
Childcare (PT) $\times$ College	-0.155	0.090	-1.123	0.427	-0.156	0.070	-1.636	0.373
Childcare (FT) $\times$ College	-0.022	0.078	-0.525	0.253	0.046	0.074	-0.263	0.234
Sibling	-0.117	0.033	-0.168	0.042	0.117	0.031	0.073	0.038
Endogeneity diagnostics (first-stage residuals)								
$\hat{v}_{\tau^w}$			-0.234	0.163			0.003	0.159
$\hat{v}_{\tau^h}$			0.037	0.331			-0.052	0.289
$\hat{v}_{cc^{PT}}$			-1.040	0.436			-1.565	0.384
$\hat{v}_{cc^{FT}}$			-0.593	0.280			-0.380	0.257
Joint test (p-value)			0.055				< 0.001	
First-stage instrument strength (joint GMM)								
Sanderson-Windmeijer F: τ^w			51.3					
Sanderson-Windmeijer F: τ^h			69.1					
Sanderson-Windmeijer F: cc^{PT}			19.0					
Sanderson-Windmeijer F: cc^{FT}			36.8					
Sanderson-Windmeijer F: z_{cog}			16.9					
Sanderson-Windmeijer F: z_{ncog}			17.1					
Kleibergen-Paap rk Wald F			8.7					
Anderson-Rubin test (p)			< 0.001					
Hansen J (all instruments)			41.3 (7 df), $p < 0.001$					
Observations			6,726					

Note: OLS columns instrument only the lagged skill measures with cross-domain lagged scores for measurement-error correction; parental inputs are treated as exogenous. Joint GMM columns stack four first-stage moment conditions for the parental inputs with the second-stage CPF moment condition, correcting simultaneously for measurement error in lagged skills and endogeneity of τ^w, τ^h, cc^{PT}, and cc^{FT} using eleven FORTAX-simulated instruments; see Appendix E. Standard errors are analytical two-step sandwich standard errors clustered on the household. The joint test reports the p-value from a Wald test that all four first-stage residuals are jointly zero; rejection indicates the endogeneity correction is active. The Sanderson-Windmeijer F statistics report first-stage strength for each of the six endogenous regressors, the four inputs and the two lagged skills, and the Kleibergen-Paap rk Wald F the joint statistic; each regressor is individually strongly identified while the joint statistic reflects a weakly identified linear combination of the system. The Anderson-Rubin test is the weak-instrument-robust test that the inputs have no effect. The Hansen J tests the overidentifying restrictions. The joint GMM column corresponds to the point estimates in Table 3.

restrictions the test evaluates. Reduced-form tax and transfer variation shifts the price and return of each arrangement and, under the exclusion restriction $\text{Cov}(Z^{FORTAX}, \nu') = 0$, reaches child skills only through the inputs.

Three facts bound the concern. First, each endogenous regressor is individually well identified. The Sanderson-Windmeijer first-stage F is between 17 and 69, so no coefficient of interest is weakly identified, even though the aggregate Kleibergen-Paap F of 8.7 reflects a weakly identified linear combination of the six-equation system. Second, the estimates are robust in sign. Controlling for household income leaves the maternal-time coefficient significant and preserves the education gradient in childcare.[63] Taken one instrument at a time, the childcare effect is positive on cognitive skills under every instrument. Third, the rejection does not implicate the instruments. The J statistic tests the overidentifying restrictions jointly with the functional form, so it cannot attribute the rejection to either, and dropping the tax instruments most exposed to an income channel still rejects ($J = 28.7$, 3 df), pointing to no single invalid instrument. Most important, the estimated technology is corroborated out of sample. Embedded in the structural model it reproduces Bernal (2008)'s finding for maternal full-time work with childcare and matches untargeted moments (Appendix J). I therefore use the full-set coefficients in the structural model and treat a parsimonious, theory-tied just-identified subset ($J = 5.0$, 2 df, $p = 0.08$) as a check on the overidentifying restrictions.

The structural estimation in Section 5 uses the joint GMM coefficients.[64] The endogeneity correction ensures that the CPF parameters entering the structural model reflect policy-relevant returns to parental inputs, so that the counterfactual exercises in Section 6 rest on policy-relevant treatment effects rather than on associations that may be confounded by unobserved household heterogeneity.

Appendix F Flexibility - Choice of Hours Grid

I embed flexibility by endowing each work arrangement with a more or less coarse hours menu. Paid employed wives choose from two points corresponding to part-time and full-time hours, while paid employed husbands choose from two points reflecting hours variation within full-time employment. Self-employed workers of both genders choose from a five-point menu spanning a wider range. This difference in menu coarseness directly encodes the scheduling flexibility advantage of self-employment and is a central modeling choice. It is disciplined by the paid

[63] Because income is not an input under the log-additive technology but captured through childcare choices, conditioning on it over-controls. The attenuated childcare coefficient yields a conservative bound.

[64] The parental-time inputs are matched from BHPS cell means, so I propagate the matching uncertainty into the standard errors with a two-step bootstrap. Each replication resamples the BHPS by household, recomputes the discretized cell means, re-maps the imputed inputs into the MCS, resamples the MCS by household, and re-estimates the control function. The analytical two-sample correction of Inoue and Solon (2010) provides a smoother alternative. The standard errors widen, modestly in the cognitive equation and more in the imputation-sensitive non-cognitive time terms, but no conclusion changes. Part-time formal childcare, the input the procedure corrects, remains significant (t = 3.2). The education interactions that carry the parental-time channels remain significant, with the father's non-cognitive interaction marginally so (t = 2.2). The education interactions are precisely estimated, so the difference in time productivity by education is identified.

employment rigidity literature, by within-person evidence from the BHPS, and by an objective k-means saturation criterion applied separately to each distribution.

Appendix F.1 Work arrangement rigidity

Wage earners cannot freely adjust weekly hours within a job (Altonji and Paxson, 1988; Aaronson and French, 2004). The constraint is sharp enough that bunching estimators detect zero paid employment response to tax kinks while recovering substantial bunching among the self-employed (Saez, 2010; Chetty et al., 2011b). Kleven and Waseem (2013) and Kleven (2016) review this asymmetry. Self-employment is the canonical residual margin of hours flexibility (Hamilton, 2000; Hurst and Pugsley, 2011; Yurdagul, 2017). I further allow nonlinearity in earnings through the hours-earnings curvature $\eta_{i,j}$ in the household income equation and through the year-over-year hours adjustment evidence below, rather than through grid cardinality alone.

Appendix F.2 Within-person variation in hours worked

From the data sample, paid employed workers exhibit essentially zero year-over-year hours adjustment within the same work arrangement. Among wives who remain paid employed in two consecutive years, 85.4% report a year-over-year change of at most 5 hours and the median change is nearly zero. Among husbands who remain paid employed, the corresponding figures are 88.9% and zero. Wives and husbands who remain self-employed adjust hours far more freely. Only 61.5 and 63.6% stay within 5 hours of the previous year, the median change is 5 hours, and the mean is 6.6 and 6.4 hours, respectively. The gap between the two work arrangements is roughly 25 percentage points in the share staying within 5 hours, and 2.5 to 3 times larger in mean hours change, and is nearly identical for wives and husbands, indicating that the asymmetry is a feature of the work arrangement rather than gender. Table F.15 reports the key moments of $|\Delta n|$ and Figure B.3 shows the corresponding density.

Appendix F.3 Grid construction and validation

I locate grid points at weighted k-means cluster centroids of the BHPS distribution within each work arrangement using household weights. For paid employed workers, I use $K = 2$, where the paid employment centroids track the modal part-time and full-time mass points of the paid employed hours distribution, which is unimodal for husbands (spike near 40) and bimodal for wives (part-time near 17 and full-time near 37).[65] For self-employed workers, I use $K = 5$ to capture the broader dispersion of self-employed hours. For home production hours, I do not adjust by

[65]The discrete K-means grid is an approximation to the continuous hours technology that converges as K increases. The K grid points minimize $\int \min_k (h^* - h_k)^2 f(h^*)\, dh^*$, so adding further grid points leaves estimated moments unchanged once the key modes are captured. Paid employment hours exhibit discrete bunching consistent with firms offering tied wage-hours packages rather than a continuous menu (Rosen, 1976; Altonji and Paxson, 1988). Wives cluster near 17 and 37 hours per week, and $K = 2$ captures these two modes without approximation loss. The self-employment grid uses $K = 5$ because self-employed workers face a more continuous hours technology, and the richer grid approximates the wider distribution.

Table F.15: Within-person stability of weekly work hours, conditional on staying in the same work arrangement in two consecutive years

			$\lvert\Delta n\rvert$ (hours/week)			
	Work arrangement	N	% $\leq$ 5h	Mean	p50	p90
Wife	Paid employed	12,765	85.4	2.6	0.0	8.0
	Self-employed	818	61.5	6.6	5.0	15.0
Husband	Paid employed	14,635	88.9	2.2	0.0	6.0
	Self-employed	2,867	63.6	6.4	5.0	15.0

Note: Sample is BHPS couple-year observations with non-missing work arrangement and weekly work hours in two consecutive years and the same work arrangement in both. $\lvert\Delta n\rvert$ is the absolute year-over-year change in usual weekly work hours. The first column reports the share of person-year pairs with a change of at most 5 hours; the next three report the mean, median, and 90th percentile of $\lvert\Delta n\rvert$. Paid employees exhibit substantially less within-person variation in hours than the self-employed, consistent with the rigid wage-hours bundles documented in Altonji and Paxson (1988) and the bunching evidence in Saez (2010).

work arrangement because this is beyond firms' scope to control workers' hours. Table F.16 reports the resulting centroids alongside empirical quantiles.

Table F.16: Discrete grid points (k-means centroids), by gender

		Quantile					k-means centroid				
Variable	Arrangement	p10	p30	p50	p70	p90	c_1	c_2	c_3	c_4	c_5
Wife											
Work hours	PE ($K = 2$)	15.0	22.0	35.0	37.0	40.0	17.6	36.7	–	–	–
Work hours	SE ($K = 5$)	10.0	20.0	30.0	40.0	54.0	15.6	34.9	47.8	58.9	76.0
Home prod	All ($K = 5$)	5.0	10.0	14.0	20.0	28.0	7.1	14.1	20.1	24.9	30.9
Husband											
Work hours	PE ($K = 2$)	35.0	37.0	39.0	40.0	46.0	32.8	41.0	–	–	–
Work hours	SE ($K = 5$)	35.0	40.0	50.0	55.0	70.0	22.7	39.0	48.3	59.4	75.9
Home prod	All ($K = 3$)	1.0	2.0	4.0	7.0	10.0	2.9	8.3	16.0	–	–

Note: Cells report empirical quantiles (p10–p90) and weighted k-means centroids (c_1–c_5) used in the structural estimation. Work hours grids are arrangement-specific: 2-point under paid employment (rigid schedules) and 5-point under self-employment (flexible hours). Home production grids are common across work arrangements within gender. Grid size follows the pooled k-means fit ladder.

Figures F.12 to F.14 show the weighted k-means centroids used by the structural grid on the marginal distributions, comparing alternative grid sizes. For paid employment, husband hours are close to normal at full-time hours with a thin right tail, hence the $K = 2$ centroids align with the two modes, while the third $K = 3$ centroid lands in the long-hours tail rather than on a modal mass point. For self-employment, the $K = 5$ centroids track the modes of both wife and husband self-employed hours. The $K = 6$ centroids nearly overlap $K = 5$, indicating that the sixth centroid has no distinct mass to summarize. For home production, the $K = 5$ centroids align with the modes of the marginal distribution for the wife and $K = 3$ for the husband.

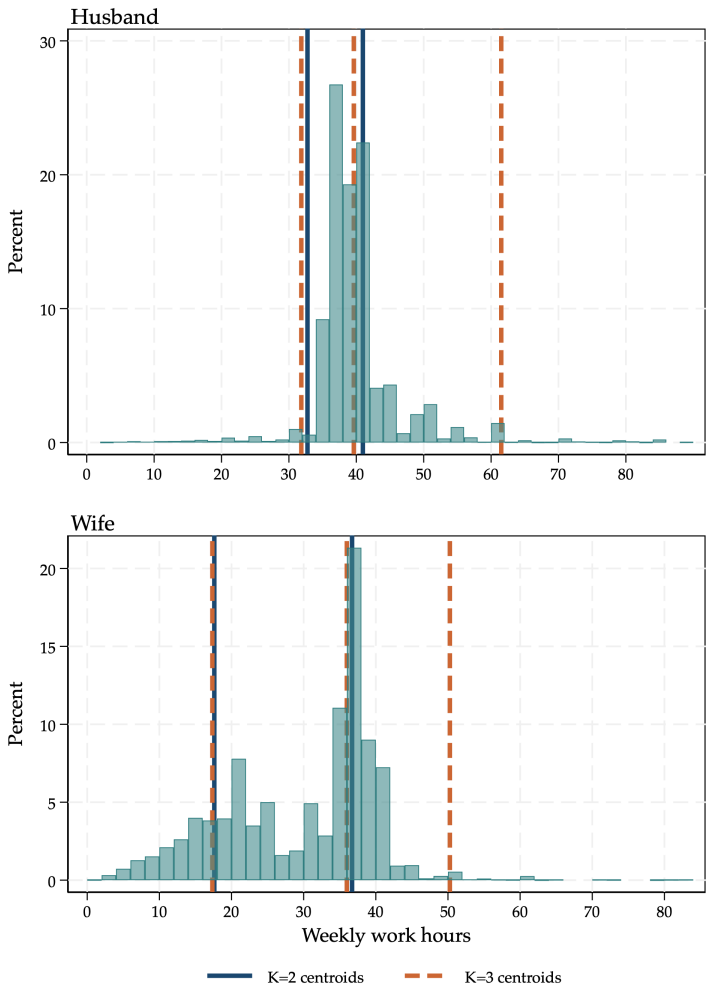


Figure F.12: Distribution of PE weekly work hours with $K = 2$ and $K = 3$ weighted k-means centroids

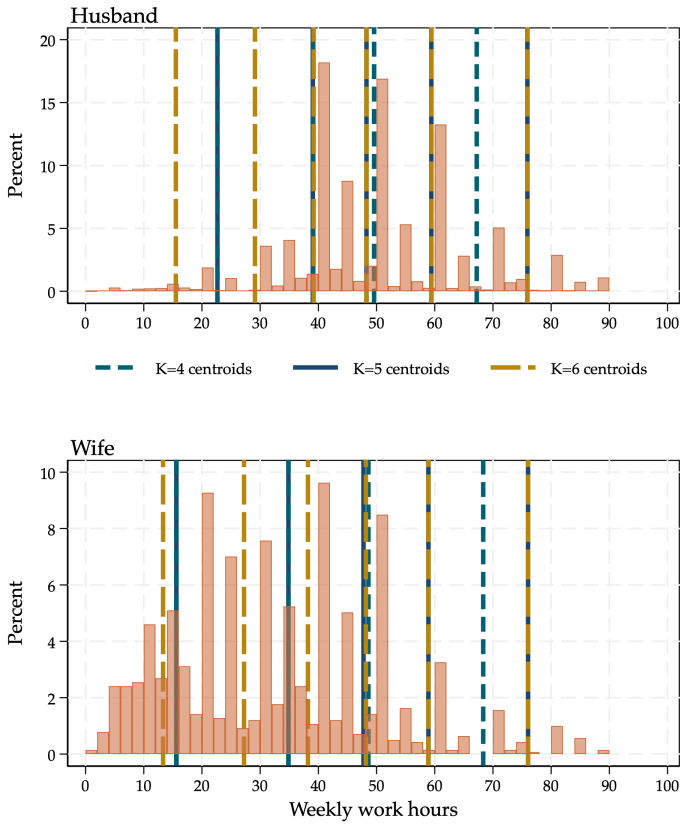


Figure F.13: Distribution of SE weekly work hours with $K = 4$, $K = 5$, and $K = 6$ weighted k-means centroids

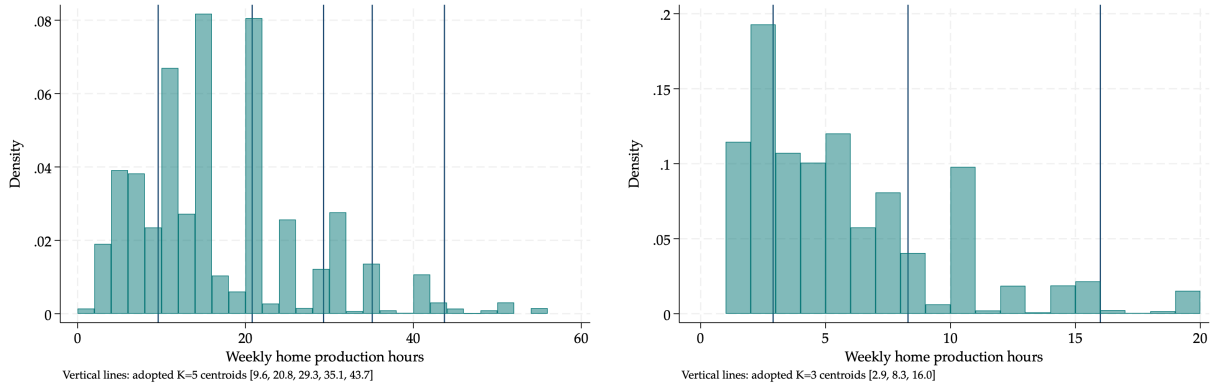


Figure F.14: Distribution of weekly home production hours, pooled across work arrangements, with k-means centroids ($K = 5$ for the wife, $K = 3$ for the husband)

Appendix F.4 Flexibility Policy Expanded Grids

The flexibility policy counterfactual expands only the paid employment work hours menu for the targeted parent, from $K = 2$ to $K = 7$. Home production is common conditional on gender and the policy leaves it unchanged, since firms set work hours but not home production. The expanded menu spans the statutory work band, between the 16-hour threshold that defines work for tax-credit purposes and the Working Time Regulations 48-hour weekly cap. It retains the two baseline paid-employment points and fills the intermediate and longer hours the rigid full-time-or-nothing market does not supply. Its spacing is not uniform but anchors on the baseline points. I construct the menu this way, rather than from the realized self-employed hours distribution, because self-employed hours are skewed toward long hours by selection into self-employment, whereas flexibility grants control over the schedule.[66]

Table F.17: Paid Employment Flexibility Expanded Grids

	Flexible Hours						
	1	2	3	4	5	6	7
Wife							
Work hours n^w (h/wk)	16.0	17.6	24.0	30.0	36.7	41.0	44.0
Husband							
Work hours n^h (h/wk)	24.0	28.0	32.8	37.0	41.0	45.0	48.0

Note: The paid employment work hours menu under the flexibility policy contains the two estimated baseline paid-employment points (wife 17.6/36.7, husband 32.8/41.0) and fills the intermediate and surrounding hours. The wife's floor (16) is the 16-hour tax-credit work threshold and the husband's ceiling (48) is the Working Time Regulations 48-hour cap; each menu spans the part-time to modestly-above-full-time range the respective parent works, within the statutory band. The menu represents the choice set flexibility provides, not the self-employed hours distribution, which is skewed toward long hours by selection into self-employment. Home production is unchanged by the policy and follows the common per-gender menu used throughout the model.

Appendix F.5 Right to Request Flexible Working: Reduced-Form Evidence

The 2003 introduction of the Right to Request Flexible Working offers a natural experiment for whether the reform changed behavior over my sample. From April 2003, employees who are parents of a child under six gained the right to request flexible working, whereas parents of older children did not. I estimate a difference-in-differences on the BHPS estimation sample, comparing mothers whose youngest child is under six (eligible) with mothers whose youngest is aged six to

[66] A grid-convergence check confirms the result is not an artifact of the discretization: the child-skill effects are stable as the hours menu is refined from five to seven to nine points. Coarser menus concentrate choices on the top rung and finer menus spread them across intermediate hours, yet the effect holds, so it reflects genuine hours choices rather than bunching at a single grid point.

eleven (ineligible), before and after the reform:

$$y_{it} = \beta\,(\text{Eligible}_{it} \times \text{Post}_t) + \gamma\,\text{Eligible}_{it} + \mathbf{X}_{it}'\delta + \eta_s + \lambda_t + \varepsilon_{it},$$

Eligibility varies with the youngest child's age, η_s are region fixed effects, λ_t are year fixed effects, and $\mathbf{X}_{it}$ includes age, age squared, college degree, and the number of children. Standard errors are clustered at the individual level. The outcomes are employment, part-time and full-time paid employment, self-employment, weekly hours, and, as the most direct first stage, whether the mother reports flexitime.

Flexitime, a shift in when rather than how many hours, does not map one-to-one to the model's hours menu, and the reform does not move it. Table F.18 shows no significant flexitime response (0.02, s.e. 0.02), matching Xue et al. (2025), who likewise find no flexitime or teleworking response to the 2014 expansion. The margin the model prices, hours, points the other way. The eligible group shifts toward part-time, a 6.6 percentage-point rise, with fewer weekly hours, but this movement fails the pre-trend. Home production shows no reform response either, so the statute moves neither the schedule nor time at home over the sample period. Xue et al. (2025) identify the response cleanly on the 2014 expansion, where women's reduced-hours uptake rises from about 3 to 10% over six years, with no response among men, an asymmetry consistent with the mother being the margin of adjustment in the model. Fathers in my sample, equally eligible, show no response (Table F.19).

The weak reduced-form response reflects how the policy works. The statute obliges an employer to consider a request but the employer can reject. Binding enforcement instead runs through indirect sex discrimination law, through the recognized childcare disparity, though an employer can still defend a refusal as proportionate.[67] The counterfactual assumes that the employer accepts the request, which provides an upper bound on the reach of the current statute.

[67] Under the Equality Act 2010 (Section 19), a requirement to work full-time or fixed hours is a provision, criterion, or practice that indirectly discriminates against women unless the employer shows it to be a proportionate means of achieving a legitimate aim.

Table F.18: Right to Request Flexible Working: Difference-in-differences, mothers

	(1) Employed	(2) Part-time	(3) Full-time	(4) Self-employed	(5) Hours	(6) Flexitime	(7) Home prod.
Eligible	-0.148***	-0.105***	-0.057***	0.012	-1.260**	0.002	0.795*
	(0.018)	(0.021)	(0.018)	(0.009)	(0.514)	(0.010)	(0.477)
Eligible × Post	0.034	0.066**	-0.026	-0.012	-0.821	0.022	0.491
	(0.025)	(0.031)	(0.027)	(0.015)	(0.704)	(0.023)	(0.600)
Age	0.074***	0.058***	0.016	0.008	0.133	0.016**	-0.755***
	(0.011)	(0.012)	(0.011)	(0.005)	(0.341)	(0.008)	(0.254)
$Age^2/100$	-0.103***	-0.081***	-0.023	-0.007	-0.219	-0.021**	1.252***
	(0.015)	(0.017)	(0.015)	(0.008)	(0.480)	(0.011)	(0.360)
College degree	0.118***	0.018	0.106***	0.005	2.490***	-0.019	-3.066***
	(0.018)	(0.021)	(0.020)	(0.010)	(0.521)	(0.013)	(0.444)
Number of children	-0.080***	0.019	-0.110***	0.018***	-3.082***	-0.020***	3.834***
	(0.011)	(0.013)	(0.011)	(0.006)	(0.322)	(0.007)	(0.302)
Observations	11929	11234	11234	11929	7862	8072	11246

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Difference-in-differences estimates for mothers with youngest child aged 0–11 in the BHPS. Eligible indicates a youngest child under six, treated after the April 2003 reform. Controls are age, age squared, college degree, and number of children, with region and year fixed effects; the post main effect is absorbed by the year effects. Standard errors clustered by individual in parentheses. The Eligible × Post coefficient is the reform effect. Flexitime is measured only from 1999 onwards.

Table F.19: Right to Request Flexible Working: Difference-in-differences, fathers

	(1) Part-time	(2) Full-time	(3) Self-employed	(4) Hours	(5) Flexitime	(6) Home prod.
Eligible	0.003	-0.003	0.015	-0.365	0.010	0.216
	(0.006)	(0.006)	(0.016)	(0.298)	(0.008)	(0.207)
Eligible × Post	-0.003	0.003	0.018	0.526	0.016	0.016
	(0.013)	(0.013)	(0.024)	(0.466)	(0.019)	(0.303)
Age	-0.007	0.007	0.000	0.171	0.005	-0.088
	(0.004)	(0.004)	(0.009)	(0.143)	(0.005)	(0.098)
$Age^2/100$	0.010*	-0.010*	0.008	-0.274	-0.005	0.178
	(0.006)	(0.006)	(0.012)	(0.193)	(0.006)	(0.130)
College degree	0.015**	-0.015**	-0.111***	-1.750***	0.061***	0.100
	(0.008)	(0.008)	(0.017)	(0.328)	(0.013)	(0.211)
Number of children	0.002	-0.002	0.033***	0.049	-0.004	0.296**
	(0.004)	(0.004)	(0.011)	(0.187)	(0.006)	(0.138)
Observations	9746	9746	11933	9746	8466	11279

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Difference-in-differences estimates for fathers with youngest child aged 0–11 in the BHPS. Eligible indicates a youngest child under six, treated after the April 2003 reform. Controls are age, age squared, college degree, and number of children, with region and year fixed effects; the post main effect is absorbed by the year effects. Standard errors clustered by individual in parentheses. The Eligible × Post coefficient is the reform effect. Flexitime is measured only from 1999 onwards. Employment is omitted because I assume fathers always work.

Appendix G Model Timeline and Computational Details

Appendix G.1 Intratemporal Timeline

The model periods fall into three stages: Stage A (pre-birth), Stage B (young child, ages 1–5), and Stage C (older child, age 7+). Transitions are marked by t^b (the period of birth) and t^B (the period in which the youngest child turns 6).[68]

1. **Stage A**: $t \leq \min\{t^b, T\}$. The household draws work arrangement-specific preference shocks $\boldsymbol{\xi}_t$ and chooses work arrangement q. Income shocks ε_t are drawn (jointly for both spouses if the wife works, husband only otherwise). Given state-specific shocks $\boldsymbol{\zeta}_t$, the household chooses assets, hours, and home production. The wife's human capital evolves, and a child arrives with probability p_t^w estimated from the data as a function of the wife's age, education, number of children, and both spouses' work arrangements (Table I.20); if a child is born, the household transitions to Stage B.

2. **Stage B**: $t^b < t < t^B$. The same sequence as Stage A, with the addition that the household also chooses whether to send the child to formal childcare, determining child-related expenditures. The household remains in Stage B until the youngest child turns 6, then transitions to Stage C.

3. **Stage C**: $t^B \leq t \leq T$. Mirrors Stage A in structure with an older child present; the choice set is unchanged but continuation values reflect the absence of future young-child periods.

Appendix G.2 Model Solution

1. *State space reduction.* Starting from observed individual characteristics from the data, I first enumerate all reachable state space combinations. This reduces the effective state space and substantially eases computation.

2. *Parental education.* Since parental education is a time-invariant binary variable (1 if college-educated, 0 otherwise), I solve the model separately for each of the four possible combinations of parental education levels.

3. *Backwards induction and terminal period setup.* I solve the model by backwards induction, beginning at Stage 2 of the terminal period T. The state space at this stage consists of the mother's human capital (h_T^w), current assets (a_T), number of children (k_T), child's age (g_T), next-period assets (a_{T+1}), income shocks ($\boldsymbol{\varepsilon}_T$), and the choices carried over from the intratemporal Stage 1. Income shocks are discretized via Gaussian quadrature applied to each parent's income process, using the corresponding variance-covariance matrix to obtain nodes and weights. The action space varies with child presence: when no child is present

[68] Households may start in Stage A or B depending on initial conditions. Once the child surpasses age 5, the household enters Stage C. Depending on the arrival of subsequent children, the youngest child may replace the previous one as the focal child.

(either not yet born or after age 5, as described in Section 3), parents allocate time between labor, leisure, and home production; when a child is between ages 1 and 5, they additionally choose whether to use childcare.

4. *Value function computation.* For each point in the state space, I compute the current-period utility and the discounted expected continuation value. I construct the ex-ante value function separately for the cases with and without a child, then combine them using the probability of having a child (p_T^w).[69] The taste shocks over time preferences are then integrated out to obtain the ex-ante value function.

5. *Human capital accumulation.* Depending on the wife's work arrangement choice and hours worked, her human capital evolves deterministically through the experience law of motion (Equation 10).

6. *Integration over income and discrete choices.* For each income-shock realization, I compute the Stage-2 ex-ante value function over next-period assets and childcare choices, integrating out the state-specific preference shocks $\boldsymbol{\zeta}_T$. I then integrate out the income shocks using the Gaussian quadrature weights. Households therefore observe the income shock before choosing hours, home production, savings, and childcare.

7. *Work arrangement choice.* At Stage 1, I compute the ex-ante value function over work arrangement choices and integrate out the work-arrangement-specific preference shocks $\boldsymbol{\xi}_T$.

8. *Iteration.* The ex-ante value function obtained at Step 7 serves as the expected continuation value for period $T-1$. Steps 3–7 are then repeated, iterating backwards until the initial period is reached.

Appendix G.3 Model Simulation

Since parental education is time-invariant, there are four possible combinations of spousal education levels: (college, college), (college, no college), (no college, college), and (no college, no college). The simulation then proceeds by initializing each woman at her observed starting age and simulating forward through the life cycle.

1. *Model selection.* At the beginning of each period, I select the model solution corresponding to the couple's education combination.

2. *Stage 1: Work arrangement choice.* I extract the Stage 1 solution, draw work-arrangement-specific preference shocks, and assign the household the work arrangement that yields the highest utility.

[69] At the terminal period, the fertility probability is zero by assumption, since the fertile age range is 21–45.

3. *Stage 2: Hours and savings.* I draw the household's income shock and condition the Stage-2 choice on the realized value, then I extract the Stage 2 solution, draw the relevant state-specific preference shocks, and select the combination of hours worked and next-period assets that maximizes the ex-post value function. If there is a young child, the action space includes childcare choice. The chosen next-period assets a_{t+1} carry forward as the asset state entering the following period.

4. *Human capital update.* Given the chosen work arrangement and hours, I update the wife's human capital according to the accumulation process.

5. *Fertility.* The household draws a fertility shock. If the realization falls below the period-specific fertility probability, a new child is born.

6. *Child skill initialization.* When a new child is born, I draw her initial joint skill vector $(\log \theta_{cog,1}, \log \theta_{ncog,1})$ from $\mathcal{N}(\mathbf{0}, \widehat{\Sigma}_0)$ using the Cholesky decomposition $\widehat{\Sigma}_0 = LL'$, where L is estimated from wave-1 MCS data (Appendix G.6).

7. *Child skill update.* An existing child's cognitive and non-cognitive skills evolve through the production function given the period's parental time and childcare inputs. By log-additivity this does not affect the second-stage behavioral moments, but it generates the age-7 skill paths used in the counterfactuals.

8. *Moment computation.* Steps 1–7 are repeated for all individuals and simulation draws. Household choices and characteristics are stored locally, and the simulated moments are computed as described in Table 2.

Appendix G.4 Model Estimation

The model is estimated by Simulated Method of Moments (SMM). The procedure is as follows.

1. *Model solution.* Given an initial parameter vector, I solve the model following the steps in Appendix G.2.

2. *Simulation.* Using the model solution, I simulate the life cycle forward and compute the simulated moments as described in Appendix G.3.

3. *Minimization.* I compute the weighted distance between the data moments and the simulated moments and evaluate the SMM objective function. I then minimize the objective function over the parameter space to obtain the parameter vector that best aligns the simulated moments with their empirical counterparts.

Appendix G.5 Weighting Matrix and Standard Errors

The weighting matrix is diagonal with elements equal to the inverse bootstrap variance of each data moment, scaled by a dataset-specific constant R_d that adjusts for differences in sample size and measurement precision across BHPS and MCS:[70]

$$W = \text{diag}\left(\frac{1}{\widehat{\text{Var}}(\mathcal{M}_d) \cdot R_d}\right). \tag{G.5}$$

The SMM estimator satisfies (Duffie and Singleton, 1993)

$$\sqrt{N}(\hat{\Theta} - \Theta_0) \xrightarrow{d} \mathcal{N}(0,\, V), \tag{G.6}$$

where Θ_0 is the true parameter vector and

$$V = \left(1 + \tfrac{1}{S}\right)(G'WG)^{-1}G'W\Omega WG\,(G'WG)^{-1}, \tag{G.7}$$

with $G \equiv \partial\mathcal{M}(\Theta_0)/\partial\Theta' \in \mathbb{R}^{M\times K}$ the Jacobian of the simulated moments, $\Omega \equiv \text{Var}(\sqrt{N}\,\hat{\mathcal{M}})$ the asymptotic variance of the data moments, and $(1 + 1/S)$ accounting for simulation noise. I approximate G by central finite differences with step size $h_k \propto |\hat{\Theta}_k|$, fixing the random seed across evaluations to eliminate simulation noise. I estimate Ω by nonparametric bootstrap over the data moments. Standard errors are $\sqrt{\text{diag}(\hat{V})}$ where

$$\hat{V} = \left(1 + \tfrac{1}{S}\right)(\hat{G}'W\hat{G})^{-1}\hat{G}'W\hat{\Omega}W\hat{G}\,(\hat{G}'W\hat{G})^{-1}. \tag{G.8}$$

Appendix G.6 Initial Distribution of Child's Skills

To assess whether initial skills depend on family characteristics, I regress initial skill measures on family income, siblings, low birth weight, and parental education following Garcia-Vazquez (2023). The income coefficients are not statistically significant ($p = 0.98$ for cognitive, $p = 0.71$ for non-cognitive), and the raw correlations confirm the same pattern (Moschini, 2023). Since the level of initial skills is not separately identified from pre-natal investment, I normalize the initial distribution to be mean-zero and common across households: $(\log\theta_{cog,1}, \log\theta_{ncog,1}) \sim \mathcal{N}(\mathbf{0}, \widehat{\Sigma}_0)$. The estimated variances are $\widehat{\sigma}^2_{cog} = 0.199$ and $\widehat{\sigma}^2_{ncog} = 0.229$, with an implied cross-domain correlation of $\widehat{\rho}_{cog,ncog,1} = -0.12$, consistent with near-orthogonality of cognitive and non-cognitive skills at infancy.

In the first wave ($g_t = 1$), I observe three measures for each skill domain $D \in \{cog, ncog\}$. The

[70] The optimal weighting matrix is the inverse of the full variance–covariance matrix of the moments (McFadden, 1989; Pakes and Pollard, 1989; Duffie and Singleton, 1993). The diagonal inverse-variance approximation is a numerically stable, consistent alternative (Gourieroux and Monfort, 1996; Attanasio, Low, and Sánchez-Marcos, 2008).

measurement system for the latent skill $\theta_{D,1}$ is:

$$\widetilde{\theta}^m_{D,1} = \mu^m_{\theta_D,1} + \alpha^m_{\theta_D,1} \log \theta_{D,1} + \varepsilon^m_{\theta_D,1}, \quad m \in \{1,2,3\}, \tag{G.9}$$

where measurement errors are uncorrelated across measures. Normalizing one reference measure ($\mu^1_{\theta_D,1} = 0$, $\alpha^1_{\theta_D,1} = 1$), the off-diagonal covariances satisfy:

$$s_{12} = \alpha^2_{\theta_D,1}\sigma^2_{\theta_D,1}, \quad s_{13} = \alpha^3_{\theta_D,1}\sigma^2_{\theta_D,1}, \quad s_{23} = \alpha^2_{\theta_D,1}\alpha^3_{\theta_D,1}\sigma^2_{\theta_D,1}, \tag{G.10}$$

yielding the closed-form estimator:

$$\widehat{\sigma}^2_{\theta_D,1} = \frac{s_{12}s_{13}}{s_{23}}. \tag{G.11}$$

For cognitive skills, the three measures are the gross motor, social, and fine motor subscales of the Denver Developmental Screening Test. For non-cognitive skills, they are the Mood and Adaptability subscales of the Carey Infant Temperament Scale and the sleep-time regularity item from the Regularity subscale. The cross-domain covariance $\widehat{\mathrm{cov}}(\log \theta_{cog,1}, \log \theta_{ncog,1})$ is estimated from the covariance between the reference measures under independence of measurement errors across domains.

In the simulation, initial skills are drawn using the Cholesky factor L of $\widehat{\Sigma}_0 = LL'$: for each (household, simulation draw, child) triple, two standard normals (z_1, z_2) are drawn and $\log \theta_{cog,1} = L_{11}z_1$, $\log \theta_{ncog,1} = L_{21}z_1 + L_{22}z_2$. A fixed seed is used so that initial skill draws are identical across counterfactual scenarios, enabling paired comparisons.

Appendix H Identification of the Structural Parameters

This appendix establishes identification of the structural parameters estimated by SMM: household preferences, parental income, and the wife's human capital. It uses the first-order conditions of a continuous relaxation of the household problem to show which variation identifies which parameter, and then maps that variation to the moments the estimator targets. I suppress time subscripts for clarity.

Assets are measured with error: $\tilde{a} = a + \epsilon^a$, where ϵ^a has mean zero and is mean-independent of household choices and states. Child expenditures are also measured with error: $\tilde{M} = M + \epsilon^M$. Consumption is not observed directly; I construct it from the budget constraint (Equation 5) as $\tilde{C} = (1+r)\tilde{a} + \mathcal{T} - \tilde{M} - \tilde{a}'$, using observed assets, next-period assets, child expenditure, and net income, so that $\tilde{C} = C + \epsilon^C$, where

$$\epsilon^C = (1+r)\epsilon^a - \epsilon^{a'} - \epsilon^M.$$

Each component error is mean zero and mean-independent of choices and states, so the same holds for ϵ^C.

Proposition 2. Under the following conditions, the preference parameters and the hours-earnings curvature parameters $\eta_{i,j}$ are identified.

(i) **Tax system regularity.** Conditional on positive hours, $\mathcal{T}(\cdot)$ is differentiable in earnings with $\frac{\partial \mathcal{T}}{\partial Y_j^i} \neq 0$ for $i \in \{w, h\}$, $j \in \{PE, SE\}$.[a]

(ii) **Observability.** Given $n^i > 0$, the joint distribution of $\left(\tilde{C},\, n^i,\, l^i,\, \tau^i,\, \frac{\partial \mathcal{T}}{\partial n^i}\right)$ is observed for $i \in \{w, h\}$. Given $k_t > 0$ and $0 < \tau^i < \bar{T}$, the distribution of (τ^i, l^i) is observed.

(iii) **Measurement error independence.** $(\epsilon^a, \epsilon^{a'}, \epsilon^M)$ are mutually independent and independent of choices, states, and time.

(iv) **Positive consumption.** $\mathbb{E}[\tilde{C} \mid n^i > 0] > 0$ for $i \in \{w, h\}$.

(v) **Work arrangement variation.** Both paid employment and self-employment are observed for each spouse, and the wife is observed in paid employment, self-employment, and non-working. The young-child indicator $\mathbb{1}[ychild = 1]$ and childcare indicators $\mathbb{1}[cc = PT]$, $\mathbb{1}[cc = FT]$ each take both values in the data conditional on each work arrangement.

[a] This follows Gayle and Shephard (2019) and ensures that the tax system does not fully offset marginal changes in earnings.

Appendix H.1 Identification Under Fixed Work Arrangements

Identifying the base leisure preference α_l^w separately from the arrangement-leisure interaction $\alpha_{PE,1}^w$ requires a household configuration that removes the interaction from the first-order conditions. A non-working wife supplies no market labor, so her leisure is valued at α_l^w. Her intratemporal conditions identify α_l^w only relative to her home-production weight. The level comes from the participation margin. Without a young child, flow utility carries no arrangement-specific intercept, so the value of not working differs from the value of working only through consumption, the continuation value of experience, and leisure, which is valued at α_l^w when she does not work and at $\alpha_l^w + \alpha_{j,1}^w$ when she works in arrangement j. The hours margin of working wives identifies the composite $\alpha_l^w + \alpha_{j,1}^w$. Given the composite, the shock scale σ_ξ, and the terminal value ψ_{hc}, the participation rate of wives without a young child identifies α_l^w.

I use two household configurations to isolate the intratemporal identification of core preference and technology parameters. The first is $q = 3$ (husband paid employed, wife not working), which anchors the wife's leisure preference α_l^w without work arrangement-leisure interactions, through her home-production allocation and her participation margin. The second is $q = 1$ (both spouses paid employed), which pins down the composite $\alpha_l^w + \alpha_{PE,1}^w$ through the consumption-tax link and thereby identifies the work arrangement-leisure interaction $\alpha_{PE,1}^w$ as a deviation from the non-working anchor.

Anchor: Non-Working Wife ($q \in \{3, 6\}$)

When the wife does not work ($n^w = 0$, $d^w_{NW} = 1$), no work arrangement-leisure interaction enters her utility. The time constraint reduces to $l^w + \tau^w = \bar{T}$, and the relevant FOCs are:

$$\frac{\partial \mathcal{L}}{\partial l^w} : \quad \frac{\alpha^w_l}{l^w} = \lambda^{TC_w} \tag{H.12}$$

$$\frac{\partial \mathcal{L}}{\partial \tau^w} : \quad \frac{\alpha^w_\tau + \alpha^w_{\tau,k} \mathbb{1}[ychild] + \Psi^w \mathbb{1}[ychild]}{\tau^w} = \lambda^{TC_w} \tag{H.13}$$

where $\Psi^w = \beta \sum_D \Gamma_{D,g_t+1}(\gamma_{D,3} + \gamma_{D,4}\,\theta^w)$ collects the child production function returns to maternal time, weighted by the shadow value Γ_{D,g_t+1} of next-period skills from Proposition 1. Equating (H.12) and (H.13) in the no-child subsample ($\mathbb{1}[ychild] = 0$, so Ψ^w drops out):

$$\frac{\alpha^w_l}{\alpha^w_\tau} = \frac{l^w_{NW}}{\tau^w_{NW}} \tag{H.14}$$

This identifies the ratio $\alpha^w_l / \alpha^w_\tau$ where home production is interior. The base level α^w_τ is identified from home production in the no-young-child cells, where, through the discrete choice over the home-production grid, a more negative α^w_τ relative to α^w_l shifts mass toward lower home production, and it is separated from the young-child interaction $\alpha^w_{\tau,k}$ by the contrast with the young-child allocations.

Level: Working Wife

The Stage 2 Lagrangian, conditional on $q = 1$ (both in paid employment), assets a', and childcare choice cc, is:

$$\begin{aligned}
\mathcal{L} = {} & \log C + (\alpha^w_l + \alpha^w_{PE,1}) \log l^w + \alpha^h_l \log l^h \\
& + [\alpha^w_\tau + \alpha^w_{\tau,n}\, n^w/\bar{n} + \alpha^w_{\tau,k} \mathbb{1}[ychild]] \log \tau^w + [\alpha^h_\tau + \alpha^h_{\tau,n}\, n^h/\bar{n} + \alpha^h_{\tau,k} \mathbb{1}[ychild]] \log \tau^h \\
& + \beta \sum_D \Gamma_{D,g_t+1} \Big[\log A_D + \gamma_{D,1} \log \theta_D + \gamma_{D,2} \log \theta_{D'} + \gamma_{D,3} \log \tau^w + \gamma_{D,4} \log \tau^w \cdot \theta^w \\
& \qquad + \gamma_{D,5} \log \tau^h + \gamma_{D,6} \log \tau^h \cdot \theta^h + \gamma_{D,7} cc^{PT} + \gamma_{D,8} cc^{FT} \\
& \qquad + \gamma_{D,9} cc^{PT} \theta^w + \gamma_{D,10} cc^{FT} \theta^w + \gamma_{D,11} sib \Big] \\
& + \beta \mathbb{E} \bar{V}(a', hc^{w\prime}) \\
& + \lambda^{TC_w}[\bar{T} - l^w - \tau^w - n^w] + \lambda^{TC_h}[\bar{T} - l^h - \tau^h - n^h] \\
& + \lambda^{BC}[(1+r)a + \mathcal{T}(Y^w_{PE}, Y^h_{PE}, \mathbf{x}) - M - C - a']
\end{aligned}$$

where $\alpha^w_{PE,1}$ is the paid employment-leisure interaction and earnings $Y^i_{PE} = y^i_{PE}(n^i)^{\eta_{i,PE}}$ are substituted directly into the net-income function $\mathcal{T}(\cdot)$. Childcare and work arrangement-child interaction terms that do not affect continuous FOCs conditional on (q, cc) are absorbed into additive constants.

Taking first-order conditions:

$$\frac{\partial \mathcal{L}}{\partial C}: \quad \frac{1}{C} = \lambda^{BC} \tag{H.15}$$

$$\frac{\partial \mathcal{L}}{\partial l^w}: \quad \frac{\alpha_l^w + \alpha_{PE,1}^w}{l^w} = \lambda^{TC_w} \tag{H.16}$$

$$\frac{\partial \mathcal{L}}{\partial l^h}: \quad \frac{\alpha_l^h}{l^h} = \lambda^{TC_h} \tag{H.17}$$

$$\frac{\partial \mathcal{L}}{\partial n^w}: \quad \lambda^{TC_w} = \lambda^{BC} \frac{\partial \mathcal{T}}{\partial n^w} \tag{H.18}$$

$$\frac{\partial \mathcal{L}}{\partial n^h}: \quad \lambda^{TC_h} = \lambda^{BC} \frac{\partial \mathcal{T}}{\partial n^h} \tag{H.19}$$

$$\frac{\partial \mathcal{L}}{\partial \tau^w}: \quad \frac{\alpha_\tau^w + \alpha_{\tau,n}^w \, n^w/\bar{n} + \alpha_{\tau,k}^w \mathbb{1}[ychild] + \Psi^w \mathbb{1}[ychild]}{\tau^w} = \lambda^{TC_w} \tag{H.20}$$

$$\frac{\partial \mathcal{L}}{\partial \tau^h}: \quad \frac{\alpha_\tau^h + \alpha_{\tau,n}^h \, n^h/\bar{n} + \alpha_{\tau,k}^h \mathbb{1}[ychild] + \Psi^h \mathbb{1}[ychild]}{\tau^h} = \lambda^{TC_h} \tag{H.21}$$

where Ψ^w is as defined in the non-working anchor block and $\Psi^h = \beta \sum_D \Gamma_{D,g_t+1}(\gamma_{D,5} + \gamma_{D,6}\,\theta^h)$ is the analogous term for paternal time. With the CPF parameters known from external estimation, Γ_{D,g_t+1} is a known linear function of $(\alpha_{cog}, \alpha_{ncog})$.

The marginal net-of-tax return to the wife's hours follows from Equation 6 and the income technology by the chain rule:

$$\frac{\partial \mathcal{T}}{\partial n^w} = \chi_2(\mathbf{x})\,\chi_3(\mathbf{x})\,(Y_{PE}^w + Y_{PE}^h)^{\chi_3(\mathbf{x})-1} \cdot \eta_{w,PE}\, y_{PE}^w\, (n^w)^{\eta_{w,PE}-1} \tag{H.22}$$

The tax parameters (χ_1, χ_2, χ_3) are known. Both y_{PE}^w and n^w are observed. An analogous expression holds for $\partial \mathcal{T}/\partial n^h$. Hours also enter utility through the workload term, contributing $\alpha_{\tau,n}^i(\log \tau^i)/\bar{n}$ to (H.18)–(H.19), and, for the wife, the continuation value through the human-capital law of motion in Equation 10, which adds the discounted marginal value of the experience from an extra hour. The sequential argument holds these two channels fixed: the workload interaction $\alpha_{\tau,n}^i$ is identified from the within-person slope of home production in own hours, and the human-capital parameters from wage growth and re-entry earnings, so conditional on these the displayed conditions identify the baseline leisure and home-production weights.

1. **Consumption path.** Combining the FOC for a' with (H.15) and the Envelope Theorem yields the Euler equation $1/C = \beta(1+r)\,\mathbb{E}[1/C']$. With β and r preset, the Euler equation governs how the marginal utility of consumption grows across periods. Its level is fixed by the budget constraint together with the borrowing and terminal conditions.

2. **Hours-earnings curvature $\eta_{i,j}$.** The income technology gives $\log Y_j^i = \log y_j^i + \eta_{i,j} \log n^i$. Weekly earnings Y_j^i and hours n^i are both observed. The curvature is identified from two implications of the model instead. First, observed log hourly income is $\log y_j^i + (\eta_{i,j} -$

1) $\log n^i$, so, given the income equation, the level and dispersion of hourly income within an arrangement move with $\eta_{i,j}$ in proportion to the mean and dispersion of log hours. Second, by (H.22) the return to a marginal hour is $\eta_{i,j}$ times the average return, so, given the leisure weights, $\eta_{i,j}$ governs how much mass households place on the long relative to the short points of the hours menu. The mean and dispersion of log hours by arrangement and the hourly-income moments therefore identify $\eta_{i,j}$ separately for paid and self-employment, where the curvature reflects the degree of hours flexibility documented in Appendix F.

3. **Husband's leisure preference α_l^h.** Since husband PE is the reference ($\alpha_{PE,1}^h = 0$), the husband's leisure FOC (H.17) contains α_l^h alone. Combining (H.15), (H.17), and (H.19) at an interior solution:

$$\alpha_l^h = \frac{\mathbb{E}\left[l^h \frac{\partial \mathcal{T}}{\partial n^h} \,\middle|\, n^h > 0 \right]}{\mathbb{E}[\tilde{C} \mid n^h > 0]}$$

The condition is linear in consumption, $\alpha_l^h C = l^h \partial \mathcal{T} / \partial n^h$, so taking expectations over working households and using $\mathbb{E}[\tilde{C}] = \mathbb{E}[C]$ under the mean-zero measurement error gives the moment above. The household-varying l^h stays inside the expectation, and dividing by $\mathbb{E}[\tilde{C}]$ rather than by $\tilde{C}$ avoids the bias in $\mathbb{E}[1/\tilde{C}]$. Condition (i) ensures the numerator is nonzero; condition (iv) ensures the denominator is positive. Both l^h and the marginal tax rate are computable from data and the known tax system.

4. **Wife's leisure preference α_l^w and paid employment-leisure interaction $\alpha_{PE,1}^w$.** Combining (H.15), (H.16), and (H.18) at an interior solution:

$$\alpha_l^w + \alpha_{PE,1}^w = \frac{\mathbb{E}\left[l^w \frac{\partial \mathcal{T}}{\partial n^w} \,\middle|\, n^w > 0,\, q = 1 \right]}{\mathbb{E}[\tilde{C} \mid n^w > 0,\, q = 1]} \tag{H.23}$$

This identifies the composite $(\alpha_l^w + \alpha_{PE,1}^w)$ at the level, through the consumption-tax link, and the participation margin of wives without a young child pins α_l^w, leaving $\alpha_{PE,1}^w$ as the residual. For home production, the paid employed wife's τ / l ratio from (H.16) and (H.20),

$$\frac{\tau_{PE}^w}{l_{PE}^w} = \frac{\alpha_\tau^w + \alpha_{\tau,n}^w \, n^w / \bar{n}}{\alpha_l^w + \alpha_{PE,1}^w}$$

is interior only when a young child is present; without one, home production is at its lower bound and the ratio does not hold with equality. The home-production weights are therefore identified from the young-child working allocations (Step 5): that ratio pins the composite $\alpha_\tau^w + \alpha_{\tau,n}^w \, n^w / \bar{n} + \alpha_{\tau,k}^w + \Psi^w$, the within-person slope of home production in own hours (Step 6) separates the workload interaction $\alpha_{\tau,n}^w$, the education variation separates the child-skill return Ψ^w, and the base level α_τ^w is pinned by the no-young-child home production as above.

5. **Home production and child skill preferences** $(\alpha_\tau^h, \alpha_{\tau,k}^w, \alpha_{\tau,k}^h, \alpha_{cog}, \alpha_{ncog}, \alpha_{cog}^{ed}, \alpha_{ncog}^{ed})$**.** The husband's home-production weights are identified symmetrically. The no-young-child home-

production level identifies the base weight α^h_τ, his within-person hours slope identifies the workload interaction $\alpha^h_{\tau,n}$, and the young-child interaction $\alpha^h_{\tau,k}$ is identified in the young-child system below, given α^h_l from Step 3.

The remaining parameters $(\alpha^w_{\tau,k}, \alpha^h_{\tau,k}, \alpha_{cog}, \alpha_{ncog}, \alpha^{ed}_{cog}, \alpha^{ed}_{ncog})$ are identified from the young-child subsample ($\mathbb{1}[ychild] = 1$). The τ/l ratio of a spouse working in arrangement j becomes:

$$\frac{\tau^i}{l^i} = \frac{\alpha^i_\tau + \alpha^i_{\tau,n}\, n^i/\bar{n} + \alpha^i_{\tau,k} + \beta \sum_D \Gamma_{D,g_t+1}(\gamma^i_D + \gamma^i_{D,\theta}\, \theta^i)}{\alpha^i_l + \alpha^i_{j,1}}$$

where I use shorthand $\gamma^w_D = \gamma_{D,3}$, $\gamma^w_{D,\theta} = \gamma_{D,4}$, $\gamma^h_D = \gamma_{D,5}$, $\gamma^h_{D,\theta} = \gamma_{D,6}$. The shadow value Γ_{D,g_t+1} appears because today's time investment produces next-period skills, which are valued at the Proposition 1 coefficient rather than the flow weight α_D alone. Because the skill weights differ by maternal education, so does the shadow value, and I write Γ^e_D for its value at $\theta^w = e$. Evaluating the wife's condition at both levels of her education and the husband's condition at each combination of the spouses' education gives me six equations:

$$\begin{aligned}
\text{Wife, } \theta^w = e: \quad & \alpha^w_{\tau,k} + \beta \sum_D \Gamma^e_D\, (\gamma_{D,3} + \gamma_{D,4}\, e) = R^w_e, \qquad e \in \{0,1\} \\
\text{Husb., } (\theta^w, \theta^h) = (e, e'): \quad & \alpha^h_{\tau,k} + \beta \sum_D \Gamma^e_D\, (\gamma_{D,5} + \gamma_{D,6}\, e') = R^h_{e,e'}, \qquad e, e' \in \{0,1\}
\end{aligned}$$

where each right-hand side equals $(\alpha^i_l + \alpha^i_{j,1}) \cdot \mathbb{E}[\tau^i/l^i \mid ychild = 1, \theta^w, \theta^h] - \alpha^i_\tau - \alpha^i_{\tau,n}\, n^i/\bar{n}$, which is computable from data and the parameters identified above. The system is linear in the six unknowns $(\alpha^w_{\tau,k}, \alpha^h_{\tau,k}, \Gamma^0_{cog}, \Gamma^0_{ncog}, \Gamma^1_{cog}, \Gamma^1_{ncog})$. The recursion of Proposition 1 makes $(\Gamma^e_{cog}, \Gamma^e_{ncog})$ a known invertible linear map of $(\alpha_{cog} + \alpha^{ed}_{cog}\, e,\ \alpha_{ncog} + \alpha^{ed}_{ncog}\, e)$, with coefficients built from the CPF persistence parameters, so the four shadow values deliver the two skill weights and their education interactions.

6. **Workload interactions** $(\alpha^w_{\tau,n}, \alpha^h_{\tau,n})$**.** With the workload interaction, the weight on the wife's home production is $\alpha^w_\tau + \alpha^w_{\tau,n}\, n^w/\bar{n}$, so the working wife's τ/l ratio in Step 4 identifies the composite weight evaluated at the observed hours point, and one additional moment is required to separate $\alpha^w_{\tau,n}$ from the level. That moment is the within-person slope of home production in own market hours. With $\alpha^w_{\tau,n} = 0$, an additional hour of work lowers home production only through the time constraint from the τ/l ratio. The husband's $\alpha^h_{\tau,n}$ is identified symmetrically from his within-person slope.

Appendix H.2 Identification With All Work Arrangements

The baseline identifies $(\alpha^w_l, \alpha^w_{PE,1}, \alpha^h_l, \alpha^w_\tau, \alpha^h_\tau, \alpha^w_{\tau,n}, \alpha^h_{\tau,n}, \alpha^w_{\tau,k}, \alpha^h_{\tau,k}, \alpha_{cog}, \alpha_{ncog}, \alpha^{ed}_{cog}, \alpha^{ed}_{ncog}, \eta_{i,j})$ from intratemporal FOCs under two fixed work arrangements ($q \in \{1,3\}$) and the participation margin

without a young child. The intratemporal arguments carry through conditional on any work arrangement q, since the work arrangement-specific utility terms enter the FOCs only through additive shifts in the leisure condition.

The wife's self-employment-leisure interaction $\alpha^w_{SE,1}$ is identified by evaluating the composite (H.23) at $d^w_{SE} = 1$ (wife in self-employment):

$$\alpha^w_l + \alpha^w_{SE,1} = \frac{\mathbb{E}\left[l^w \, \frac{\partial \mathcal{T}}{\partial n^w} \,\middle|\, n^w > 0,\, d^w_{SE} = 1 \right]}{\mathbb{E}[\tilde{C} \mid n^w > 0,\, d^w_{SE} = 1]}$$

Since α^w_l is identified from the baseline, $\alpha^w_{SE,1}$ follows by differencing. A parallel argument identifies $\alpha^h_{SE,1}$ from the husband's FOCs at $d^h_{SE} = 1$, with husband PE as reference ($\alpha^h_{PE,1} = 0$).

The remaining parameters, $\alpha^w_{j,2}$, $\alpha^w_{j,3,PT}$, $\alpha^w_{j,3,FT}$, $\alpha^{w,ed}_{j,3,PT}$, $\alpha^{w,ed}_{j,3,FT}$, $\alpha^h_{SE,2}$, $\alpha_{cc,PT}$, $\alpha_{cc,FT}$, $\alpha^{ed}_{cc,PT}$, $\alpha^{ed}_{cc,FT}$, are additively separable in the flow utility and do not affect the continuous intratemporal FOCs conditional on (q, cc). They shift the expected value $\mathcal{V}^{s2}_q$ and therefore the Stage 1 work arrangement choice probability:

$$p(d_q = 1 \mid \Omega^{s1}) = \frac{\exp(\mathcal{V}^{s2}_q / \sigma_\xi)}{\sum_{q'} \exp(\mathcal{V}^{s2}_{q'} / \sigma_\xi)}$$

Since all parameters entering $\mathcal{V}^{s2}_q$ except the target are identified from the baseline and the work arrangement-leisure interactions above, the remaining parameters are identified sequentially:

1. **Work arrangement-child interactions $\alpha^w_{j,2}$ for $j \in \{PE, SE\}$.** These shift the value of wife work arrangement j when a young child is present. Variation in wife's work arrangement shares across young-child and non-young-child states, conditional on all other identified parameters, identifies $\alpha^w_{PE,2}$ and $\alpha^w_{SE,2}$. The wife's non-working state ($j = NW$) is the reference ($\alpha^w_{NW,2} = 0$).

2. **Husband's self-employment-child interaction $\alpha^h_{SE,2}$.** Variation in husband work arrangement shares across young-child states identifies $\alpha^h_{SE,2}$, with husband PE as reference ($\alpha^h_{PE,2} = 0$).

3. **Childcare base parameters $\alpha_{cc,PT}, \alpha_{cc,FT}$.** These enter the value function when a young child is present, with $cc = 0$ as reference. Conditional on all other parameters, variation in childcare take-up rates across non-college households with young children identifies both.

4. **Childcare-education interactions $\alpha^{ed}_{cc,PT}, \alpha^{ed}_{cc,FT}$.** These shift the value of childcare for college-educated mothers. The child production function, estimated outside the structural model, implies substantially different skill returns to childcare by maternal education (the college interaction $\gamma_{D,9}$ offsets the part-time childcare effect, leaving a net return near zero for children of college-educated mothers). The utility education interactions capture preference heterogeneity in childcare valuation net of the CPF channel. They are identified from the differential childcare take-up rates between college and non-college mothers, conditional on work arrangement. Since the model is solved separately by θ^w, the Bellman equation

generates education-specific choice probabilities, and the education-split childcare moments (PT/FT take-up by work arrangement and education) provide the identifying variation.

5. **Childcare-work arrangement interactions $\alpha^{w}_{j,3,PT}, \alpha^{w}_{j,3,FT}$ and their education interactions $\alpha^{w,ed}_{j,3,PT}, \alpha^{w,ed}_{j,3,FT}$ for $j \in \{PE, SE\}$.** These shift the value of childcare conditional on wife's work arrangement, and let that shift differ by education. Variation in the joint distribution of (q, cc) by education identifies them, with the non-working baseline ($j = NW$) excluded from the sum. The education interactions capture that college paid- and self-employed mothers have a steeper childcare education gradient than non-working mothers.

6. **Scale parameters σ_ξ, σ_ζ.** The preference shock scales govern the dispersion of choices around the deterministic optimum. They are identified from the smoothness of observed choice probabilities conditional on state variables: larger σ implies more uniform choice frequencies for a given spread of $\mathcal{V}^{s2}_q$.

7. **Terminal value ψ_{hc}.** The terminal value parameter disciplines the life-cycle savings and human capital investment profiles through Equation 2. It is identified from the wife's labor-force attachment beyond what current earnings, the wage return to experience, and leisure rationalize.

Interior solutions and corners. The identification arguments use first-order conditions that assume interior choices of hours, home production, and consumption. Home production enters utility and the child-skill technology only through $\log \tau^i$, so its coefficient is the net weight $A_i = \alpha^i_\tau + \alpha^i_{\tau,n}(n^i/\bar{n}) + \alpha^i_{\tau,k} + \Psi^i$, where $\Psi^i > 0$ is the child-skill return to parental time. When $A_i > 0$, the logarithm makes both ends unattractive ($\log \tau^i \to -\infty$ as $\tau^i \downarrow 0$, and $\log l^i \to -\infty$ as $l^i \downarrow 0$), so home production is interior. This holds whenever the child-skill return outweighs the net disutility of home time, which is the case in the young-child cells at the estimates. Without a young child the child-skill terms drop out, the net weight is negative, and home production falls to its lower bound.

Corners can happen through three channels. Non-participation ($n^w = 0$) is a separate work-arrangement state with its own value function, so it does not violate the interior conditions for working wives. A binding borrowing constraint replaces the Euler equation with $a' = \underline{a}$ but leaves the intratemporal leisure-hours-home-production ratios, and hence does not affect the preference identification. The supervision constraint binds only in a subset of childcare-arrangement cells, and the identification conditions hold in the complement where it is slack. The discrete childcare choice is itself a corner among three options, and its parameters are identified from the choice probabilities rather than a first-order condition.

Appendix H.3 Identification of Income and Human Capital

Wage levels and the college return come from mean log hourly income by arrangement and education, and the husband's age profile is pinned down from mean income by age. Dispersion

comes from the within-arrangement standard deviation of hourly income, net of the hours term $(\eta_{i,j} - 1) \log n$. For wives, the dispersion of wage growth among the continuously employed isolates the transitory shock, and the within-couple income correlation identifies ρ. Because wife's human capital captured through experience is unobserved, the returns to experience and the accumulation curvature ν are identified from wage growth, fast at low experience and slower as the stock rises, and from the growth gap between full-time and lower-hours stayers. The depreciation rate is identified from the within-person wage change of wives returning from non-employment, by education. Initial experience and the two ability types are identified from how the entry wage rises with age at entry and from the persistent dispersion of wages.

Appendix I Externally Estimated Parameters

Table I.20 shows the fertility probability. Fertility follows a nonlinear life-cycle pattern and college education is associated with a higher likelihood of childbirth. The parity effects are large. A first child raises the probability of a further birth, but the probability declines sharply at higher parities. Conditional on these, fertility shifts with work arrangement: paid-employed wives have a lower birth probability than non-working wives, self-employed wives sit closer to non-working wives, and self-employed husbands have a higher probability, consistent with the flexible arrangement being more compatible with childbearing.

Table I.20: Fertility Probability

	Estimate	SE
Probability of Child		
Constant	-15.277	1.027
Age	1.054	0.068
$\text{Age}^2/100$	-1.901	0.109
College	0.255	0.063
1 kid	0.614	0.071
2 kid	-1.235	0.097
Work arrangement		
Wife PE (vs. NW)	-0.351	0.082
Wife SE (vs. NW)	-0.220	0.178
Husband SE (vs. PE)	0.387	0.082

Note: Two-year birth probability estimated through a logit.

Table I.21 reports the estimated tax-and-transfer parameters and Figure I.15 shows the FORTAX fit. The lump-sum term χ_1 is the transfer a household receives at zero earnings and the main channel of redistribution. It is largest for households with a non-working mother, who qualifies for out-of-work support. It reflects Child Benefit and the child element of the tax credits, which scale with family size. Above this floor the schedule is close to proportional, with χ_3 near one. Households thus retain a roughly constant share of additional gross earnings and redistribution is

concentrated at low income. The slope χ_2 and the curvature χ_3 are not separately interpretable, since a lower slope and a higher curvature trade off within the power form and describe the same schedule. I therefore read the fitted net-income function as a whole. Its variation across households is dominated by the mother's employment and by family size.

In fact, paid and self-employed workers face the same income tax schedule but a different National Insurance and social-insurance regime. Employees pay Class 1 contributions on earnings above the primary threshold, at a main rate of about 10 to 11% over the sample period, and their employer pays a further secondary contribution. The self-employed instead pay a flat-rate weekly Class 2 contribution together with Class 4 contributions on annual profits, at a lower main rate of about 7 to 8% and with no employer contribution (HM Revenue & Customs, 2024).[71] A self-employed worker therefore retains more of a marginal pound of earnings, a feature of a system that taxes work arrangements differently (Mirrlees et al., 2011). The self-employed also have more limited access to contributory benefits, including Statutory Sick Pay and contribution-based Jobseeker's Allowance. Self-employment income can be zero or negative when a business makes a loss, which is bottom-coded at zero in the data. The lower National Insurance burden is why the fitted net-income schedule sits slightly above paid employment at low incomes, while the lump-sum term χ_1 captures the difference in contributory-benefit access. The two schedules are otherwise close. The net tax-and-benefit wedge between the arrangements is modest relative to the pre-tax earnings gap that separates them.



Figure I.15: FORTAX fit

[71] I use this variation for the child production function estimates.

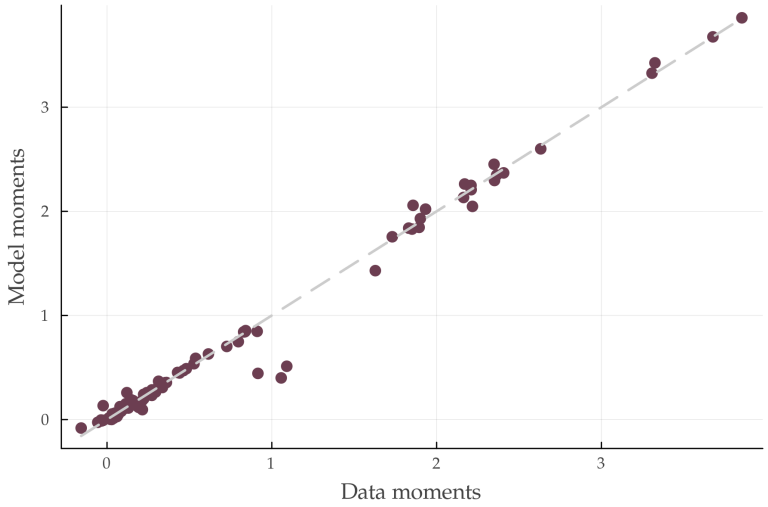
Table I.21: FORTAX Parameters

	Number of Children											
	0			1			2			3		
	χ_1	χ_2	χ_3	χ_1	χ_2	χ_3	χ_1	χ_2	χ_3	χ_1	χ_2	χ_3
Youngest child (age = 0)												
Husband PE												
Wife PE	1.51	1.23	0.92									
Wife SE	-3.32	1.21	0.93									
Wife NW	44.32	0.57	1.01									
Husband SE												
Wife PE	10.06	1.11	0.94									
Wife SE	14.21	1.19	0.92									
Wife NW	32.89	0.84	0.96									
Age = 1												
Husband PE												
Wife PE				69.41	0.75	0.98	84.72	0.54	1.03	74.34	0.74	0.98
Wife SE				79.06	0.61	1.01	77.48	0.67	1.00	156.89	0.08	1.29
Wife NW				132.08	0.11	1.24	126.25	0.16	1.18	128.11	0.15	1.19
Husband SE												
Wife PE				72.76	0.86	0.96	112.52	0.31	1.10	128.97	0.21	1.16
Wife SE				118.26	0.18	1.18	124.18	0.42	1.05	109.68	0.33	1.10
Wife NW				121.24	0.26	1.11	127.30	0.27	1.11	112.42	0.38	1.06
Age = 3												
Husband PE												
Wife PE				69.72	0.68	1.00	70.49	0.70	0.99	81.69	0.54	1.03
Wife SE				122.86	0.12	1.23	72.64	0.77	0.98	32.76	1.86	0.85
Wife NW				122.67	0.18	1.16	128.66	0.15	1.19	137.61	0.13	1.20
Husband SE												
Wife PE				89.49	0.56	1.02	110.15	0.33	1.09	106.46	0.27	1.13
Wife SE				42.88	1.16	0.93	121.00	0.37	1.07	86.72	1.04	0.93
Wife NW				132.62	0.35	1.06	120.15	0.19	1.17	106.41	0.53	1.01
Age = 5												
Husband PE												
Wife PE				40.90	1.11	0.93	50.25	0.99	0.95	35.18	1.15	0.93
Wife SE				82.75	0.31	1.10	45.67	1.12	0.93	73.46	0.64	1.00
Wife NW				116.16	0.20	1.15	133.44	0.15	1.18	126.66	0.18	1.16
Husband SE												
Wife PE				143.49	0.13	1.22	116.79	0.28	1.11	103.98	0.49	1.04
Wife SE				-140.19	6.02	0.73	133.28	0.09	1.28	109.66	0.35	1.09
Wife NW				120.27	0.19	1.16	106.03	0.46	1.03	107.49	0.32	1.09
Age = 7+												
Husband PE												
Wife PE				57.76	0.85	0.97	66.68	0.75	0.98	45.88	0.91	0.96
Wife SE				97.13	0.41	1.06	75.80	0.63	1.01	72.94	0.66	1.00
Wife NW				126.53	0.21	1.14	134.68	0.15	1.19	124.64	0.17	1.16
Husband SE												
Wife PE				76.83	0.70	0.99	81.66	0.59	1.02	113.23	0.29	1.11
Wife SE				75.02	0.90	0.95	113.69	0.51	1.03	94.84	0.87	0.96
Wife NW				124.22	0.31	1.08	115.40	0.35	1.07	108.12	0.48	1.03

Note: PE denotes paid employment; SE denotes self-employment; NW denotes non-working. The tax-and-transfer schedule follows $\mathcal{T} = \chi_1 + \chi_2(Y^w + Y^h)^{\chi_3}$ (Equation 6), where χ_1 is the lump-sum transfer or tax component, χ_2 is the slope of the effective tax rate, and χ_3 captures the degree of progressivity. Age refers to the age of the youngest child; the panel labeled Age = 0 reports parameters for childless households. Parameters are estimated by fitting FORTAX-simulated household disposable income to this functional form separately for each combination of parental work arrangement, number of children, and child age.

Appendix J Model Fit

Figure J.16 summarizes the overall fit. Figures J.17 and J.18 and Tables J.23 to J.25 report fit on the three blocks of moments targeted in SMM: work arrangement and childcare shares, hours worked and home production by work arrangement and child presence, and income and human capital dynamics.



Note: Each point is one targeted moment; the horizontal axis is the data value and the vertical axis is the simulated value at the converged estimates. The dashed line is the 45-degree line.

Figure J.16: Overall Model Fit: Simulated vs Data Moments

Table J.26 reports untargeted moments. The model reproduces work arrangement shares split by education within one to six percentage points in the no-child and young-child cells, including the education gradient in maternal non-work with young children and in paternal self-employment. Maternal home production is somewhat under-predicted there: once the investment window closes, the model no longer attaches a child-skill return to her home time, while home production persists in the data for school-age children.

The model's implied labor supply elasticities are untargeted and line up with the quasi-experimental and structural literature. A permanent 5% increase in the wife's hourly wage raises her total hours with an uncompensated elasticity of 0.30, of which roughly half operates through participation. With a young child, the elasticity rises to 0.43 and participation accounts for about half.[72] These align with the 0.2–0.6 range that Bargain, Orsini, and Peichl (2014) estimate for married women across seventeen countries, consistent with the elasticities surveyed in Keane (2011), near the 0.18 median Marshallian hours elasticity in Attanasio et al. (2018), and match the positive, moderate responses, largest for mothers of young children, that Blundell, Duncan, and Meghir (1998) recover from UK tax reforms. The husband's own-wage elasticity is -0.03. Because

[72] Elasticities are computed by re-solving and re-simulating the model after a permanent proportional shift in the wage intercepts of both work arrangements, holding the simulation draws fixed, the standard simulation practice in discrete-choice labor supply models (Bargain, Orsini, and Peichl, 2014).

he always works, this is a pure intensive-margin response, which Bargain, Orsini, and Peichl (2014) find to be close to zero for married men. The wife's cross-wage elasticity with respect to the husband's wage is -0.30, and -0.40 with a young child, the negative income effect for women with children in Blundell, Duncan, and Meghir (1998). Home production is far less elastic than market hours, much smaller than Gelber and Mitchell (2012) estimate for single women's home production.

Table J.22: Labor Supply and Home Production Elasticities

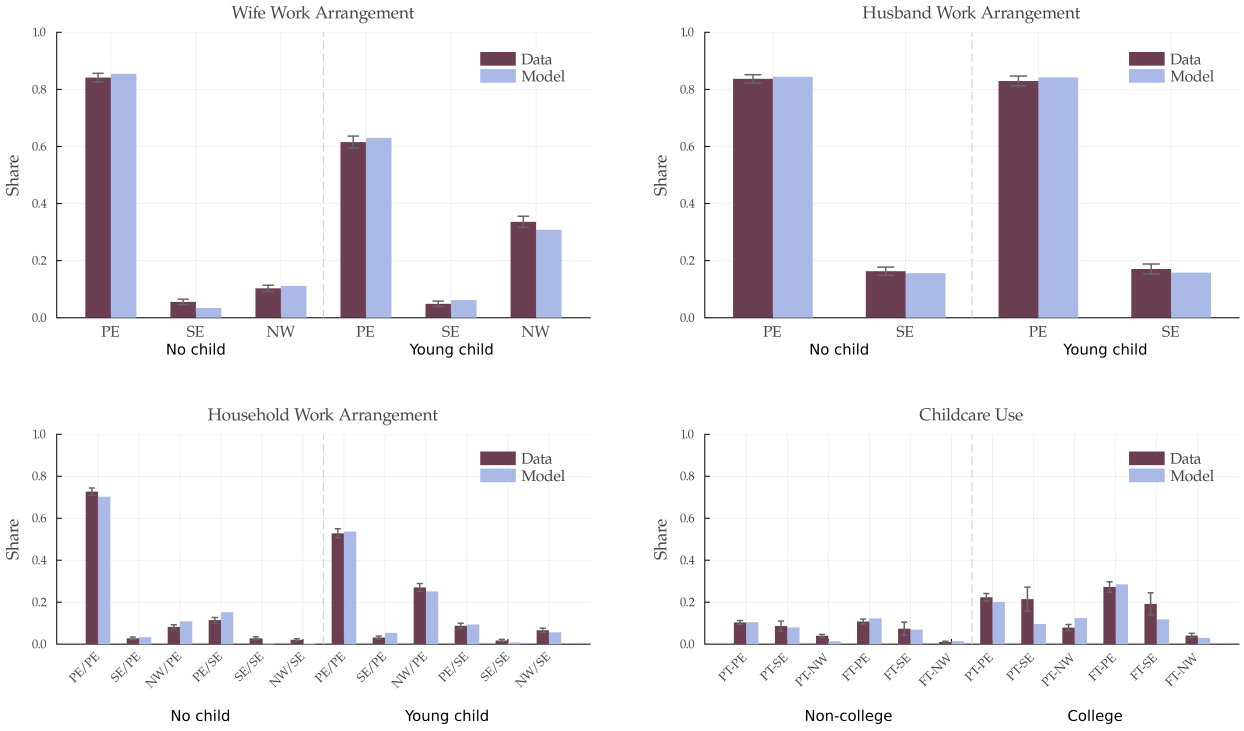
	Own-wage					Cross-wage
	Wife part.	Wife hours	Wife intensive	Wife home prod.	Husband hours	Wife hours
Pooled	$+0.161$	$+0.297$	$+0.136$	-0.038	-0.029	-0.296
Young child	$+0.225$	$+0.425$	$+0.200$	-0.033	-0.021	-0.403
Non-college	$+0.155$	$+0.281$	$+0.126$	-0.036	-0.026	-0.285
College	$+0.169$	$+0.318$	$+0.149$	-0.043	-0.033	-0.311

Note: Uncompensated (Marshallian) elasticities from a permanent 5% wage increase, re-solving and re-simulating with fixed draws. The own-wage columns raise the relevant spouse's wage and the cross-wage column reports the wife's total-hours response to a 5% increase in the husband's wage.

Table J.23: Model Fit: Hours Levels

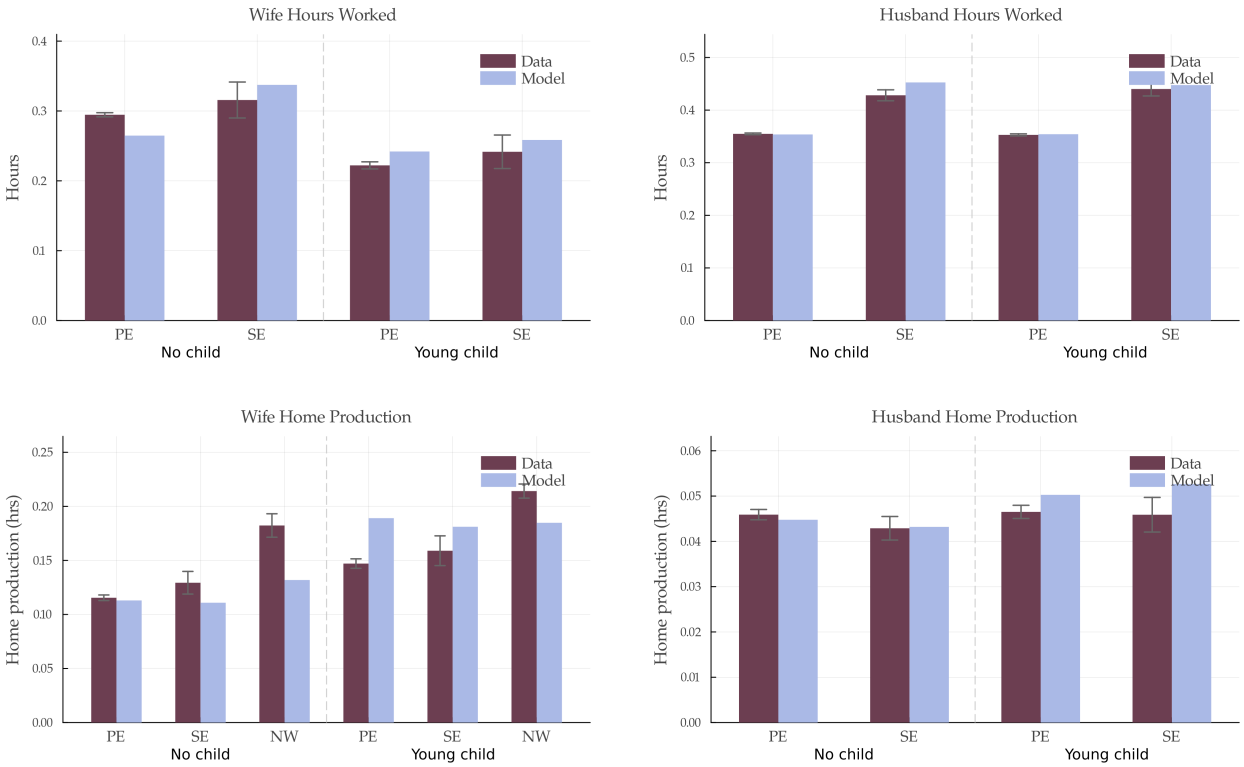
Moment	Data	Model
Wife PE log hours	3.307	3.327
Wife PE log hours dispersion	0.360	0.356
Wife SE log hours	3.325	3.426
Wife SE log hours dispersion	0.538	0.588
Husband PE log hours	3.677	3.676
Husband PE log hours dispersion	0.083	0.084
Husband SE log hours	3.852	3.859
Husband SE log hours dispersion	0.313	0.368

Note: Simulated and empirical log hours moments by work arrangement.



Note: Empirical (plum) versus simulated (periwinkle) work arrangement shares for the wife, husband, and household, and childcare use shares, by child presence; childcare is split by maternal education. Whiskers are 90% data confidence intervals.

Figure J.17: Work Arrangement and Childcare Fit



Note: Empirical (plum) versus simulated (periwinkle) mean weekly hours worked and home production hours by work arrangement and child presence. Whiskers are 90% data confidence intervals.

Figure J.18: Hours Worked and Home Production Fit

Table J.24: Model Fit: Log Hourly Wages

Moment	Data	Model
Wife PE log wage, non-college	1.851	1.829
Wife PE log wage, college	2.364	2.348
Wife PE log wage, dispersion	0.464	0.471
Husband PE log wage, non-college	2.210	2.208
Husband PE log wage, college	2.632	2.600
Husband PE log wage, dispersion	0.481	0.489
Wife SE log wage, non-college	1.628	1.430
Wife SE log wage, college	2.164	2.133
Wife SE log wage, dispersion	0.911	0.847
Husband SE log wage, non-college	1.831	1.839
Husband SE log wage, college	2.217	2.048
Husband SE log wage, dispersion	0.797	0.748
Husband PE log wage, age group 1	2.170	2.262
Husband PE log wage, age group 2	2.351	2.296
Husband PE log wage, age group 3	2.407	2.369
Husband PE log wage, age group 4	2.349	2.452
Husband SE log wage, age group 1	1.857	2.058
Husband SE log wage, age group 2	1.933	2.021
Husband SE log wage, age group 3	1.901	1.930
Husband SE log wage, age group 4	1.894	1.846
Initial wage, non-college	1.731	1.756
Initial wage, college	2.209	2.249
Initial wage age slope, non-college	-0.023	0.134
Initial wage age slope, college	0.120	0.259
Wife wage change after non-work	-0.033	-0.013
Wife wage change after employment	0.033	0.033
Initial wage dispersion, non-college	1.057	0.400
Initial wage dispersion, college	0.916	0.444
Wife wage autocovariance	0.436	0.045
Wife wage change after non-work, non-college	-0.024	-0.007
Wife wage change after non-work, college	-0.056	-0.026
Wife wage change after employment, non-college	0.030	0.025
Wife wage change after employment, college	0.042	0.043

Note: Simulated and empirical log hourly wage moments by work arrangement and education.

Table J.25: Model Fit: Human Capital Dynamics

Moment	Data	Model
Wife income growth	0.034	0.033
Initial wage, non-college	1.731	1.756
Initial wage, college	2.209	2.249
Initial wage age slope, non-college	-0.023	0.134
Initial wage age slope, college	0.120	0.259
Wife wage change after non-work	-0.033	-0.013
Wife wage change after employment	0.033	0.033
Initial wage dispersion, non-college	1.057	0.400
Initial wage dispersion, college	0.916	0.444
Wife wage autocovariance	0.436	0.045
Wife wage growth, young child	0.068	0.049
Wife wage growth, older child	0.061	0.029
Wife wage growth dispersion	1.091	0.512
Wife wage growth, full-time vs part-time	0.039	0.014
Wife wage change after non-work, non-college	-0.024	-0.007
Wife wage change after non-work, college	-0.056	-0.026
Wife wage change after employment, non-college	0.030	0.025
Wife wage change after employment, college	0.042	0.043

Note: Simulated and empirical moments for the wife's wage dynamics, including the initial wage, the post-interruption wage, wage growth, and the wage autocovariance.

Table J.26: Untargeted Moments: Model Validation

Description	Data	Model
Wife work arrangement shares by education		
PE (NC, no child)	0.833	0.809
SE (NC, no child)	0.054	0.039
NW (NC, no child)	0.113	0.152
PE (NC, young child)	0.573	0.591
SE (NC, young child)	0.043	0.064
NW (NC, young child)	0.384	0.345
PE (NC, old child)	0.754	0.763
SE (NC, old child)	0.061	0.041
NW (NC, old child)	0.185	0.196
PE (C, no child)	0.863	0.898
SE (C, no child)	0.059	0.030
NW (C, no child)	0.079	0.072
PE (C, young child)	0.709	0.693
SE (C, young child)	0.062	0.057
NW (C, young child)	0.229	0.250
PE (C, old child)	0.759	0.850
SE (C, old child)	0.074	0.036
NW (C, old child)	0.168	0.114
Husband work arrangement shares by education		
SE (NC, no child)	0.184	0.194
SE (NC, young child)	0.203	0.183
SE (NC, old child)	0.210	0.238
SE (C, no child)	0.110	0.113
SE (C, young child)	0.099	0.122
SE (C, old child)	0.137	0.156
Old-child households: hours worked		
Wife, PE	25.240	30.088
Wife, SE	31.343	39.778
Husband, PE	39.802	39.579
Husband, SE	51.241	50.376
Old-child households: home production (share of time endowment)		
Wife, PE	0.159	0.113
Wife, SE	0.167	0.110
Husband, PE	0.049	0.045
Husband, SE	0.044	0.043
Old-child households: log hourly earnings		
Wife, PE	1.893	2.058
Wife, SE	0.309	1.668
Husband, PE	2.389	2.384
Husband, SE	1.015	1.882

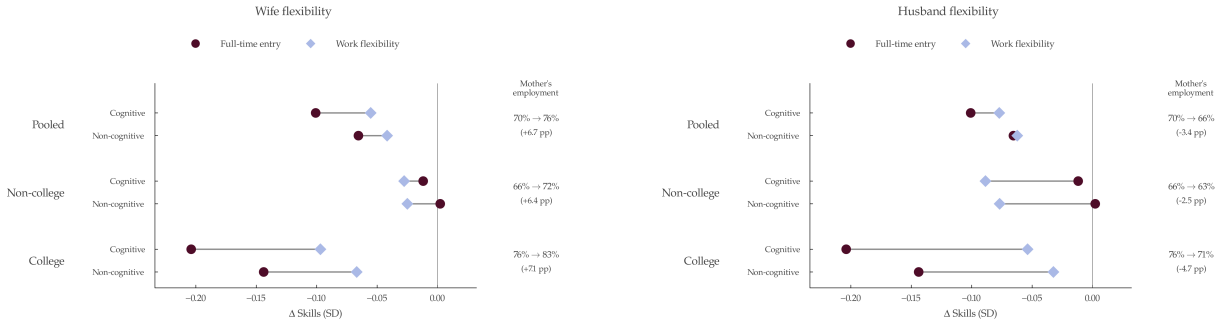
Appendix K Additional Counterfactual Results

Table K.27 lists the parameters governing each counterfactual mechanism and the related trade-offs, and Table K.30 decomposes the policy counterfactuals into the four input channels by maternal education.

Table K.27: Main Mechanisms and Tradeoff

Parameter	Governs	Tradeoff
Utility		
α_{ncog}, α_{cog}	Child skill utility	vs $\alpha^{w}_{\tau,k}$, $\alpha^{h}_{\tau,k}$: parents value child skills but bear private costs of providing the home production time
α^{PT}_{cc}, α^{FT}_{cc}	PT/FT childcare utility	Non-college educated mothers strongly averse, college educated mothers nearly indifferent
CPF technology		
$\gamma_{D,3}$, $\gamma_{D,4}$	Maternal home production	Maternal advantage in the skill technology
$\gamma_{D,5}$, $\gamma_{D,6}$	Paternal home production	Paternal home time does not substitute for released maternal time, so the father's effect on children runs through the budget rather than time
$\gamma_{D,7}$, $\gamma_{D,9}$	PT care	Strong return for non-college educated, near zero for college educated
Flexibility and household constraint		
K_{SE} vs K_{PE}	PE rigidity vs SE flexibility	Self-employment allows intermediate hours that enable home production alongside market work: *flexibility*
σ^2_{SE}	SE earnings variance	Risk channel of SE entry, distinct from the schedule channel in K_{SE}
$K_{SE} + \sigma^2_{SE}$	SE bundles flexibility and risk	Removing K_{SE} vs σ^2_{SE} yields the same ΔSE share but opposite τ^h: flexibility-driven entry selects low-hours SE ($\tau^h \uparrow$); earnings-driven entry selects high-hours SE ($\tau^h \downarrow$)
Supervision constraint	Household dynamics	Spouses cannot optimize independently: market entry by one endogenously shifts the other's home time allocation

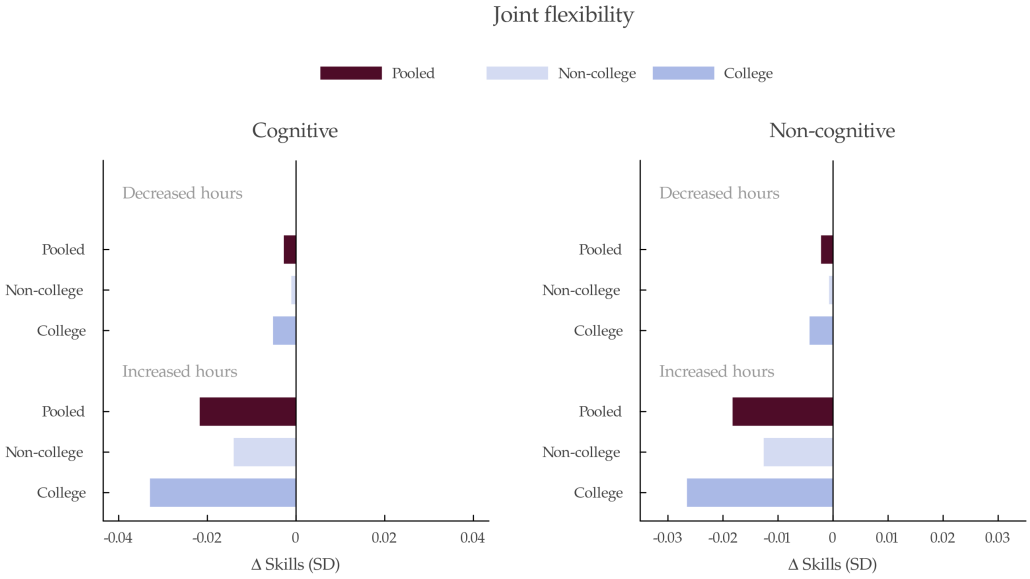
Note: $\alpha^{w}_{\tau,k}$ and $\alpha^{h}_{\tau,k}$ are the young-child interaction terms; with a young child present the flow-utility weight on $\log \tau^i$ is $\alpha^i_\tau + \alpha^i_{\tau,n}(n^i/\bar{n}) + \alpha^i_{\tau,k}$.



(a) Wife-only (b) Husband-only

Note: This figure plots the effect on a child's cognitive and non-cognitive skill at age 7 of the mother entering paid work during the child's ages 1–5, relative to remaining out of work, by maternal education. Circles are full-time entry, the causal treatment-on-treated effect in the baseline. Diamonds are flexible entry: under wife-only flexibility, the intermediate hours she chooses once her own menu widens; under husband-only flexibility, her indirect response through the household's joint allocation, since her own menu is unchanged and only the husband's wider menu shifts home production and care under the supervision constraint. The right column reports the rise in the maternal employment rate.

Figure K.19: Effect of Flexibility on Child Skills



Note: This figure plots the effects of joint flexibility on child's cognitive (left) and non-cognitive (right) skills at age 7, decomposed by the direction of the mother's cumulative hours change over the child's ages 1–5, pooled and by maternal education.

Figure K.20: Redistribution of Child Skills under Flexibility

Table K.28: Who Moves Under Each Policy, and How It Reaches the Child

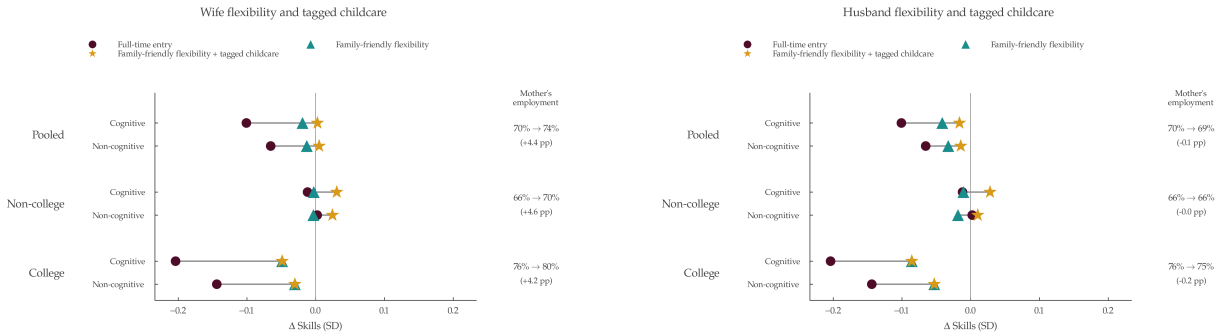
Baseline state	Share (%)	Movers (%)	Δ Market hours	Δ Home prod. (h)	Δ FT care (pp)
Panel A: Work flexibility					
Non-college mother					
Non-work	34.49	13.99	+3.56	−0.16	+0.55
PE part-time	33.41	53.89	+6.13	−0.43	+0.27
PE full-time	25.67	78.21	−11.55	+0.74	−1.04
Self-employment	6.43	58.88	−0.05	+0.06	−0.03
College mother					
Non-work	25.02	17.29	+4.69	−0.41	+2.49
PE part-time	29.33	60.72	+7.33	−0.41	+0.82
PE full-time	39.99	67.39	−9.24	+0.55	−1.88
Self-employment	5.66	60.49	−0.09	−0.13	+2.72
Panel B: Free childcare					
Non-college mother					
Non-work	34.49	0.41	+0.12	−0.29	+4.99
PE part-time	33.41	3.47	+0.63	−0.62	+8.21
PE full-time	25.67	4.39	−0.85	−0.47	+6.77
Self-employment	6.43	3.76	+0.15	−0.32	+5.61
College mother					
Non-work	25.02	2.95	+0.93	−0.85	+10.99
PE part-time	29.33	7.31	+1.35	−0.28	+12.09
PE full-time	39.99	7.11	−1.39	−0.45	+12.33
Self-employment	5.66	12.49	+0.02	−0.48	+10.24
Panel C: Flexibility and childcare combined					
Non-college mother					
Non-work	34.49	14.22	+3.63	−0.49	+5.61
PE part-time	33.41	53.88	+6.13	−1.00	+8.46
PE full-time	25.67	78.20	−11.54	+0.22	+6.06
Self-employment	6.43	59.16	−0.02	−0.40	+5.81
College mother					
Non-work	25.02	19.28	+5.26	−1.23	+13.69
PE full-time	39.99	68.00	−9.35	+0.01	+10.88
Self-employment	5.66	63.55	+0.20	−0.67	+13.73

Note: This table shows the transitions over young-child periods, baseline to each policy, by maternal education. Baseline states are non-work, paid employment part-time, paid employment full-time, and self-employment. Movers is the share of the group whose arrangement or market hours change by more than two hours. Δ Market hours, Δ Home production, and Δ FT care are the mean group changes in the wife's weekly market hours, weekly home production time, and the share using full-time formal childcare.

Table K.29: Maternal Employment Entry and Child Skills

		Pooled (SD)		Non-college (SD)		College (SD)	
Entry margin	$\Delta\tau^w$ (h/wk)	ΔCog	ΔNcog	ΔCog	ΔNcog	ΔCog	ΔNcog
Panel A: entry effect in the rigid baseline (treatment-on-treated)							
NW → PE full-time	−4.24	−0.101	−0.065	−0.012	+0.002	−0.204	−0.144
NW → PE part-time	−1.37	+0.093	+0.142	+0.140	+0.192	−0.050	−0.007
NW → self-employment	−3.02	−0.028	−0.012	−0.019	−0.004	−0.057	−0.036
Panel B: entry effect when full flexibility (RRFW) is offered							
NW → work, full-time	−2.20	−0.125	−0.097	−0.067	−0.062	−0.167	−0.123
NW → work, part-time	−1.08	−0.045	−0.034	−0.024	−0.021	−0.081	−0.054

Note: Effect on a child's skills when the mother enters work during the child's ages 1–5, relative to remaining out of work. Panel A conditions on mothers who make a clean single entry in the baseline simulation; the counterfactual holds the household's last pre-entry time and childcare inputs fixed over the spell, so the effect is causal at fixed selection. Panel B reports the same object for mothers drawn into work by the offer of the full flexibility menu, split by the hours they enter at. $\Delta\tau^w$ is the mean change in the mother's weekly home-production time.



(a) Wife-only (b) Husband-only

Note: This figure plots the effect on a child's cognitive and non-cognitive skill at age 7 of the mother entering paid work during the child's ages 1–5, relative to remaining out of work, by maternal education. Circles are full-time entry, triangles family-friendly flexibility entry, and stars the package of family-friendly flexibility with tagged childcare. The right column reports the rise in the maternal employment rate.

Figure K.21: Effects of Family-Friendly Flexibility and Tagged Childcare on Child Skills

Table K.30: CPF Channel Decomposition of Policy Skill Effects

	Non-college Mother					College Mother				
	τ^w	τ^h	cc^F	cc^I	Δ SD	τ^w	τ^h	cc^F	cc^I	Δ SD
Panel A: Work flexibility										
Wife-only										
Wife: ΔPE $+8.0$, ΔSE -1.5, ΔNW -6.5; Husb ΔSE $+1.6$						Wife: ΔPE $+8.7$, ΔSE -1.6, ΔNW -7.1; Husb ΔSE $+3.1$				
Δ Input	-0.019	$+0.010$	-0.001	$+0.005$		-0.023	$+0.001$	$+0.014$	$+0.014$	
Cog (SD)	**-0.031**	$+0.001$	$+0.000$	$+0.005$	-0.025	**-0.043**	$+0.000$	$+0.000$	$+0.004$	-0.039
Ncog (SD)	**-0.024**	$+0.000$	$+0.000$	$+0.004$	-0.019	**-0.035**	$+0.000$	$+0.000$	$+0.004$	-0.031
Joint										
Wife: ΔPE $+5.0$, ΔSE -1.3, ΔNW -3.7; Husb ΔSE -5.0						Wife: ΔPE $+4.1$, ΔSE -1.2, ΔNW -2.9; Husb ΔSE -5.3				
Δ Input	-0.018	$+0.011$	-0.000	$+0.010$		-0.017	$+0.010$	$+0.012$	$+0.020$	
Cog (SD)	**-0.021**	$+0.002$	-0.001	$+0.010$	-0.009	**-0.030**	$+0.002$	-0.001	$+0.006$	-0.022
Ncog (SD)	**-0.016**	$+0.000$	-0.001	$+0.009$	-0.008	**-0.024**	$+0.000$	-0.001	$+0.007$	-0.018
Husband-only										
Wife: ΔPE -2.6, ΔSE $+0.1$, ΔNW $+2.4$; Husb ΔSE -6.2						Wife: ΔPE -4.9, ΔSE $+0.3$, ΔNW $+4.6$; Husb ΔSE -7.9				
Δ Input	-0.005	$+0.001$	-0.001	$+0.006$		$+0.004$	$+0.011$	-0.003	$+0.003$	
Cog (SD)	$+0.002$	$+0.001$	$+0.000$	**$+0.006$**	$+0.010$	**$+0.014$**	$+0.002$	$+0.000$	$+0.001$	$+0.017$
Ncog (SD)	$+0.001$	$+0.000$	-0.000	**$+0.005$**	$+0.007$	**$+0.011$**	$+0.001$	$+0.001$	$+0.001$	$+0.013$
Panel B: Free formal childcare										
Work-conditional										
Wife: ΔPE $+0.5$, ΔSE -0.0, ΔNW -0.4; Husb ΔSE $+0.1$						Wife: ΔPE $+2.0$, ΔSE -0.2, ΔNW -1.8; Husb ΔSE $+1.2$				
Δ Input	-0.024	-0.005	-0.002	$+0.047$		-0.022	-0.006	$+0.017$	$+0.092$	
Cog (SD)	-0.014	-0.001	$+0.001$	**$+0.055$**	$+0.041$	-0.017	-0.001	$+0.001$	**$+0.031$**	$+0.014$
Ncog (SD)	-0.012	-0.000	$+0.001$	**$+0.046$**	$+0.035$	-0.014	-0.000	$+0.001$	**$+0.032$**	$+0.018$
Means-tested										
Wife: ΔPE -0.1, ΔSE $+0.0$, ΔNW $+0.1$; Husb ΔSE $+0.2$						Wife: ΔPE $+0.3$, ΔSE -0.1, ΔNW -0.2; Husb ΔSE $+1.5$				
Δ Input	-0.016	-0.004	-0.002	$+0.037$		-0.023	-0.009	$+0.037$	$+0.062$	
Cog (SD)	-0.007	-0.000	$+0.001$	**$+0.030$**	$+0.023$	-0.017	-0.001	-0.000	**$+0.018$**	-0.001
Ncog (SD)	-0.006	-0.000	$+0.001$	**$+0.025$**	$+0.020$	-0.014	-0.000	-0.001	**$+0.018$**	$+0.003$
Universal										
Wife: ΔPE $+0.1$, ΔSE $+0.0$, ΔNW -0.1; Husb ΔSE $+0.1$						Wife: ΔPE $+0.8$, ΔSE -0.2, ΔNW -0.6; Husb ΔSE $+1.1$				
Δ Input	-0.030	-0.008	-0.003	$+0.065$		-0.033	-0.012	$+0.042$	$+0.118$	
Cog (SD)	-0.016	-0.001	$+0.001$	**$+0.065$**	$+0.050$	-0.025	-0.001	$+0.001$	**$+0.037$**	$+0.012$
Ncog (SD)	-0.013	-0.000	$+0.001$	**$+0.055$**	$+0.042$	-0.021	-0.001	-0.000	**$+0.039$**	$+0.017$

Note: Shapley decomposition of the change in mean log skills at age 7 into the four CPF input channels; contributions sum exactly to the total. Panel A grants the RRFW flexible-hours menu to the wife, both spouses, or the husband. Panel B provides free formal childcare that is work-conditional (working families below £100,000), means-tested (bottom third of net income, no work condition), or universal. Input rows report mean changes over young-child periods; the dominant channel in each row is bold.

Appendix L Consumption Equivalent Variation

I measure the policy's effect on household welfare through the consumption equivalent variation (CEV), following De Nardi, Pashchenko, and Porapakkarm (2025). I compare each household with its hypothetical self under the policy counterfactual, defining the CEV as the constant proportional adjustment to its baseline consumption path that leaves it indifferent between the baseline and the policy. Let V_0 denote a household's expected discounted lifetime value,

$$V_0 = \mathbb{E} \sum_{t=1}^{\bar{t}} \beta^{t-1} \mathcal{U}_t,$$

where $\bar{t}$ is the number of model periods over the household's horizon, β the discount factor, and $\mathcal{U}_t$ the flow utility of Equation 1, in which consumption enters as $\log C_t$. Let V_0^{BS} and V_0^{CF} be the value under the baseline and under the policy counterfactual. The CEV λ is the consumption fraction that solves

$$\mathbb{E} \sum_{t=1}^{\bar{t}} \beta^{t-1} \mathcal{U}\Big((1+\lambda) C_t^{BS}, \cdot\Big) = V_0^{CF}.$$

Each model period spans two years, so its flow utility sums two annual sub-years discounted at the standard annual rate $\beta_a = 0.98$, with $\beta = \beta_a^2$ the period discount. Consumption $\log C_t$ therefore enters the period flow with coefficient $(1+\beta_a)$, and scaling the baseline consumption path by $(1+\lambda)$ raises lifetime value by $(1+\beta_a)\log(1+\lambda)\sum_t \beta^{t-1}$. Solving,

$$\lambda = \exp\left(\frac{V_0^{CF} - V_0^{BS}}{(1+\beta_a)}\text{PV}\right) - 1, \qquad \text{PV} = \sum_{t=1}^{\bar{t}} \beta^{t-1} = \frac{1-\beta^{\bar{t}}}{1-\beta}, \tag{L.24}$$

the exact welfare gain in units of lifetime consumption. I compute λ for each simulated household and average over households with a young-child period by maternal education. I then convert to pounds by multiplying by baseline household income. Because I scale λ by baseline income rather than by baseline consumption, these pound figures are income-scaled welfare values. I apply the same income base to the policy costs.

Flow utility is additively separable in the child's skills and its remaining arguments, so the lifetime value change decomposes exactly into a child-skill and a parental component,

$$\Delta V_0 \equiv V_0^{CF} - V_0^{BS} = \Delta V_0^{\text{skill}} + \Delta V_0^{\text{par}}.$$

The child component is the discounted value of the policy-induced skill path,

$$\Delta V_0^{\text{skill}} = \mathbb{E} \sum_{t=1}^{\bar{t}} \beta^{t-1} \left(\alpha_{cog}\, \Delta \log \theta_{cog,t} + \alpha_{ncog}\, \Delta \log \theta_{ncog,t}\right),$$

where skills are fixed after the investment window (age 7) and ΔV_0^{par} is the value of the changes in consumption, leisure, home production, and work-arrangement amenities. Because this uses the

household's own valuation of skills, α_{cog} and α_{ncog}, rather than an external price, I allocate the exact total CEV by the value shares,

$$\lambda^{\text{skill}} = \frac{\Delta V_0^{\text{skill}}}{\Delta V_0}\,\lambda, \qquad \lambda^{\text{par}} = \frac{\Delta V_0^{\text{par}}}{\Delta V_0}\,\lambda, \qquad \lambda^{\text{skill}} + \lambda^{\text{par}} = \lambda, \tag{L.25}$$

The two components add to the total by construction. The child-skill component λ^{skill} is the part of the total consumption-equivalent attributable to the skill change under this value-share allocation. Note that it is not the consumption-equivalent of the skill change in isolation, $\exp(\Delta V_0^{\text{skill}}/[(1+\beta_a)\,\text{PV}]) - 1$, because the CEV is nonlinear in ΔV_0.

Appendix M Discussion on the Unitary Household Framework

I assume that the household model is unitary such that spouses have equal weights ($\mu = 0.5$) and pool resources. It therefore determines aggregate time allocations and child outcomes, but not the intra-household split of welfare. To gauge how much this assumption drives the results, I re-solve the flexibility counterfactual under alternative Pareto weights μ on the wife, holding the remaining parameters fixed. Formally, the estimated flow utility decomposes as $u = u^{\text{sh}} + u^w + u^h$. The shared component u^{sh} collects consumption, the child-skill returns, and the base childcare cost, while u^w and u^h collect each spouse's private leisure, home-production, and work-arrangement terms. The weighted objective is $u^\mu = u^{\text{sh}} + 2\mu\,u^w + 2(1-\mu)\,u^h$, so the shared consumption, child-skill, savings, and terminal-value components enter unweighted and μ reweights only the spouses' private time and amenity terms. At $\mu = 0.5$, $2\mu = 2(1-\mu) = 1$ and $u^\mu = u$, so the $\mu = 0.5$ row reproduces the estimated baseline exactly.

The equal-weight baseline is consistent with evidence that the wife's weight is approximately 0.5 before childbirth, though it shifts as couples raise young children (Low et al., 2018; Verriest, 2024). Its level also varies with the couple's education. Gayle and Shephard (2019) estimate the weight by education. They find that it rises with the wife's schooling and falls with the husband's, from about 0.32 (college husband, non-college wife) to 0.58 (non-college husband, college wife) and averaging just below equal weighting (Table M.31). In a dynamic setting, Reynoso (2024) finds the same gradient. UK estimates place the wife's share in the same range and moving in the same direction with her relative earnings (Blundell, Chiappori, and Meghir, 2005; Blundell et al., 2007; Chiappori and Meghir, 2015).

I re-solve the flexibility counterfactual under each weight, holding the remaining parameters fixed. I assign each couple its education-specific weight from Table M.31, uniformly over $\mu \in [0.35, 0.55]$. Table M.32 reports the treatment-on-treated entry effects by the couple's education. The heterogeneity is invariant to the Pareto weights. A mother's full-time entry lowers her child's skill when she is college-educated, in every matching cell, and the college loss is essentially independent of the husband's education. For a non-college mother the effect is close to zero regardless of her bargaining weight, so full-time entry is roughly child-neutral rather than beneficial. Reweighting

Table M.31: Wife's Pareto weight by education

	Wife	
Husband	Non-college	College
Non-college	0.47	0.58
College	0.32	0.44

Note: Wife's Pareto weight μ by the couple's education, from Gayle and Shephard (2019) (Table II), collapsing "some college" into non-college and averaging the underlying cells. The unitary baseline is equal weighting, $\mu = 0.5$.

the spouses' private time and amenity terms shifts the magnitudes but leaves this contrast intact: college children bear the cost of the mother's full-time entry and non-college children do not. This check speaks to the entry-effect heterogeneity that drives the results rather than to the flexibility policy's employment and welfare effects.

Table M.32: Flexibility entry effects (ToT) by the couple's education

Wife × husband	Wife's weight μ	n	Cognitive	Non-cognitive
College × college	0.44	604	−0.148	−0.107
College × non-college	0.58	747	−0.210	−0.130
Non-college × college	0.32	472	+0.022	+0.045
Non-college × non-college	0.47	1060	−0.016	−0.003

Note: Treatment-on-treated effect on a child's cognitive and non-cognitive skill at age 7 (SD) of the mother entering paid full-time work during the child's ages 1–5, by the couple's education, re-solving the model with each couple assigned its education-specific Pareto weight μ on the wife from Gayle and Shephard (2019) (Table M.31).

Appendix M.1 Heterogeneity by Ability Type

Table M.33 splits the flexibility and childcare responses by the cross of maternal education and the two ability types. Within each education group, the two types, which differ only in the wife's initial labor-market experience and so in her baseline wage, produce nearly identical take-up, hours changes, and child-skill effects. The heterogeneity that drives the results runs across education more than ability.

Table M.33: Who Responds to Flexibility, by Maternal Education and Ability Type

	Baseline		Work flexibility				Free childcare	
Cell	Wife hrs	Wage	Take-up	Δn^w	ΔCog	ΔNcog	ΔCog	ΔNcog
Non-college, Low	17.3	5.8	+47.3	+2.0	−0.010	−0.009	+0.051	+0.043
Non-college, High	16.9	6.1	+46.4	+2.0	−0.008	−0.007	+0.048	+0.041
College, Low	21.2	9.4	+56.2	+2.0	−0.020	−0.015	+0.010	+0.016
College, High	21.7	10.6	+57.6	+1.9	−0.021	−0.017	+0.012	+0.017

Note: Cells are the cross of maternal education (college/non-college) and the latent ability type that affects wife's initial labor-market experience. Baseline wife hours and hourly wage are averaged over young-child periods. Take-up is the share of young-child mother-periods whose work arrangement or market hours change under flexibility. Δn^w is the average change in wife market hours. ΔCog and ΔNcog are changes in the child's log skill at age 7, in standard deviations.